Multi-study factor regression biclustering

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September 20, 2026

Abstract

High-dimensional molecular feature data are often collected across multiple studies under varying experimental and technological conditions. Biclustering aims to uncover local structure by simultaneously identifying subsets of molecular features and samples that exhibit coherent, potentially overlapping patterns. However, existing biclustering methods generally do not account for inter-study heterogeneity or distinguish shared biological structure from study-specific variation. We propose the multi-study factor regression biclustering (MSFRB) model, which unifies multi-study factor analysis, factor regression, and biclustering within a Bayesian framework. Latent factors are decomposed into common components, capturing shared structure, and study-specific components, accounting for heterogeneity such as batch effects and technological differences. Biclusters are induced through sparse exchangeable shrinkage process (ESP) priors on common factor loadings and scores, enabling simultaneous selection of feature and sample subsets and automatic determination of active factors. Covariates are incorporated to adjust for observed sample characteristics. To address identifiability, we derive a sufficient condition based on unique factor support and develop a post hoc procedure to assess recovered factors. For scalable inference, we develop a novel variational inference scheme for the ESP prior and extend it to stochastic variational inference for large datasets. Simulation studies and an application to colorectal cancer gene expression data demonstrate improved recovery of shared latent structure compared with existing approaches.

Keywords: Factor analysis, biclustering, multi-study, shrinkage, spike-and-slab, variational Bayes

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1 Introduction

High-dimensional molecular high-throughput technologies play a central role in modern biomedical research. Platforms such as transcriptomics, proteomics, and metabolomics provide measurements on thousands of molecular features across biological samples, often collected in multiple studies or cohorts, with different experimental conditions. Understanding the latent biological mechanisms that jointly regulate subsets of these features remains a fundamental statistical challenge.

A recurring empirical phenomenon in molecular data is the presence of coordinated activity among subsets of molecular features such as genes, proteins, or metabolites that manifest itself only in specific subsets of samples. Biclustering methods are designed to capture such structure by simultaneously clustering rows and columns of the data matrix, enabling the identification of overlapping feature–sample associations linked to latent biological processes (Pontes et al., 2015).

Since the pioneering work of Hartigan (1972), who introduced simultaneous clustering of rows and columns of a data matrix, and Cheng and Church (2000), who popularised biclustering in gene expression analysis, a wide range of biclustering methods has been developed. Following Madeira and Oliveira (2004), these methods can be broadly classified into approaches based on constant submatrices (e.g., Hartigan, 1972; Prelić et al., 2006), additive effects (e.g., Cheng and Church, 2000; Lazzeroni and Owen, 2002), low-rank factorisations (e.g., Kluger et al., 2003), and pattern-based structures (e.g., Tanay et al., 2002; Li et al., 2009). While these approaches differ in their assumptions about bicluster geometry and signal generation, they share the goal of identifying subsets of features that exhibit coherent behaviour within subsets of samples. Reviews of the biclustering literature are provided by Madeira and Oliveira (2004), Eren et al. (2013), and Padilha and Campello (2017).

Among these approaches, the methods most directly related to the present work are biclustering models based on sparse factorisations, where biclusters are represented through latent factors and their associated loadings. In high-dimensional molecular applications, sparsity in both factor loadings and scores is particularly desirable, as it allows each latent factor to involve only a subset of features and samples, thereby inducing potentially overlapping biclusters and facilitating their biological interpretation. A seminal example is factor analysis for bicluster acquisition (FABIA, Hochreiter et al., 2010). More recent Bayesian formulations include biclustering with mixed-sparsity components (BicMix, Gao et al., 2016), which places a three-layer shrinkage prior on factor scores and loadings; spike-and-slab biclustering models such as Denitto et al. (2017) and Moran et al. (2021), with the latter also incorporating an Indian Buffet Process (IBP) prior to automatically infer the number of biclusters; and empirical Bayes matrix factorization (EBMF, Wang and Stephens, 2021), where the degree of sparsity is learned from the data.

While these methods have proved useful in single-study settings, they do not explicitly account for heterogeneity across multiple studies, nor do they incorporate observed covariate effects within a unified biclustering framework. Among them, BicMix (Gao et al., 2016) is unique in explicitly accommodating broad confounding effects by allowing latent factors to be either sparse or dense through a combination of spike-and-slab and three parameter Beta priors on factor scores and loadings. Dense factors can then ab-

sorb broad sources of variation such as batch effects, whereas sparse factors capture more localised structure. In contrast, Moran et al. (2021) focus on recovering interpretable biclustering structure and automatically inferring the number of biclusters, whereas Wang and Stephens (2021) emphasise flexible empirical Bayes learning of sparsity patterns directly from the data. Nevertheless, none of these approaches explicitly distinguish shared biological structure from study-specific variation. Besides batch effects, such heterogeneity may arise from differences in experimental platforms and protocols, patient populations, sample collection procedures, or study design (see, e.g., Liang et al., 2026).

In this paper, we propose a multi-study factor regression biclustering (MSFRB) model, a unified framework that simultaneously accounts for covariate effects, distinguishes between shared and study-specific sources of variation, and recovers latent bicluster structure within a Bayesian sparse factor analysis setting. To model study-specific variation, we build on the multi-study factor analysis (MSFA) framework introduced by De Vito et al. (2019) in a frequentist setting and later extended to high-dimensional Bayesian inference (De Vito et al., 2021; Hansen et al., 2025). MSFA assumes that some latent factors are shared across studies, while others are specific to individual studies. This leads to a decomposition into common and study-specific factors, where the latter account for study-dependent variability arising from batch effects, measurement noise, and other sources of heterogeneity. To account for observed sample characteristics such as age and sex, we further extend the model by incorporating a design matrix of known covariates alongside the common and study-specific latent factors and their loadings. This extension is inspired by the multi-study factor regression (MSFR) model of De Vito and Avalos-Pacheco (2025) developed in the context of standard factor analysis.

To recover a bicluster structure, we impose sparsity on both the loadings and scores of the common factors, thereby identifying subsets of features that exhibit coordinated behaviour within subsets of samples. For this purpose, we employ an exchangeable shrinkage process (ESP) prior (Frühwirth-Schnatter, 2023) on the variances of factor loadings and factor scores, with additional parameters introducing several levels of sparsity (Frühwirth-Schnatter, 2023; Frühwirth-Schnatter et al., 2024). Similar in spirit to the hierarchical shrinkage constructions used in sparse factor models such as BicMix (Gao et al., 2016), the ESP prior combines element-wise sparsity, selecting individual features and samples within each factor, with global shrinkage controlling the overall degree of sparsity, and factor-level sparsity that shrinks entire factors and enables explicit factor selection and automatic determination of the number of active factors. This yields a clear distinction between active and inactive factors while allowing the sparsity structure within active factors to be learned from the data. Study-specific factor loadings are assigned similar priors, although with a simpler sparsity formulation, while the corresponding study-specific factor scores follow the standard factor analysis assumption of independent Gaussian distributions with zero mean and unit variance. Consequently, biclustering is induced only in the common component, whereas the study-specific component is primarily used to capture study-dependent heterogeneity.

Because MSFA decomposes variation into shared and study-specific components, it introduces an additional source of unidentifiable beyond the classical rotational indeterminacy of factor models. In particular, these models may suffer from information switching, whereby shared structure can be represented by study-specific factors and vice versa, leading to observationally indistinguishable parameterisations (Chandra et al., 2025; Bor-

tolato and Canale, 2026). Addressing this issue typically requires additional assumptions or constraints. For example, Chandra et al. (2025) achieve identifiability by constraining study-specific factors to lie within the shared loading subspace, thereby separating shared and study-specific covariance components. In a less restrictive approach, Bortolato and Canale (2026) impose distinct activation patterns across studies, so that each factor is uniquely determined by its study-specific support. We approach this problem differently and instead exploit the sparse support patterns induced by the biclustering formulation. In particular, when a factor possesses support that is not shared by any other common or study-specific factor, the corresponding factor can be uniquely distinguished. This observation motivates a sufficient identification condition based on unique factor support. Accordingly, we assess identifiability post-hoc rather than enforcing it during model fitting.

Inference for the proposed model is computationally challenging due to its high-dimensional latent structure and hierarchical shrinkage specification, making conventional Monte-Carlo Markov chain (MCMC) approaches impractical for the large molecular datasets that motivate our work. We therefore develop a variational inference framework for the ESP prior and derive a coordinate ascent variational inference (CAVI) algorithm for MSFRB. For very large datasets, we additionally develop a stochastic variational inference (SVI) algorithm based on subsampling, providing a scalable alternative to full-data inference.

The rest of the paper is organised as follows. Section 2 introduces the MSFRB model, building on sparse Bayesian factor analysis and its extensions to multi-study data and covariate adjustment. Section 3 describes how biclustering is incorporated through the sparse ESP prior. Identification issues arising in factor models, and in particular in multi-study factor analysis, are addressed in Section 4. An efficient variational inference (VI) scheme encompassing both CAVI and SVI is developed in Section 5. The performance of the proposed model relative to existing methods is evaluated through extensive simulation studies in Section 6 and a real-data application to a multi-study colorectal cancer gene expression dataset in Section 7. Initialisation, hyperparameter selection, detailed steps of the variational algorithms, and the identification procedure are provided in the Supplementary Material, together with an additional real-data application to multi-study colorectal cancer proteomics data. The paper concludes with a brief discussion in Section 8.

2 Multi-study factor regression biclustering

In this section, we introduce the proposed multi-study factor regression biclustering (MSFRB) model. We begin with a review of sparse Bayesian factor analysis and then show how this framework can be extended to accommodate multiple studies and observed covariates. Finally, we describe how biclustering structure can be incorporated through sparsity priors on both factor loadings and factor scores. Throughout the paper, model parameters are denoted by Greek letters and hyperparameters by Latin letters whenever possible, with the exception of the latent common factors, which follow the standard notation in the factor analysis literature.

2.1 Sparse Bayesian factor analysis

The main purpose of factor analysis is to find a lower-dimensional representation of the data by identifying latent factors that account for the observed variation. By leveraging sparsity-inducing priors, one can achieve automatic detection of the latent dimensionality, as well as a more interpretable representation of the latent factors.

Suppose we have a standard factor model:

$$\boldsymbol{Y} = \boldsymbol{\Lambda}\boldsymbol{F} + \boldsymbol{\epsilon}, \tag{2.1}$$

where $\boldsymbol{Y}$ is a $p \times n$ matrix of observed values with $j = 1, \ldots, p$ indexing variables and $i = 1, \ldots, n$ indexing observations. Further, $\boldsymbol{\Lambda}$ is a $p \times H$ factor loading matrix and $\boldsymbol{F}$ is an $H \times n$ matrix of latent factor scores, where $\boldsymbol{f}_i$ denotes the ith column of $\boldsymbol{F}$ and contains the factor scores associated with observation i. Finally, $\boldsymbol{\epsilon}$ is a $p \times n$ matrix of idiosyncratic errors following a normal distribution with variable-specific variances, $\boldsymbol{\epsilon}_i \sim \mathcal{N}_p(\mathbf{0}, \boldsymbol{\Xi})$, where $\boldsymbol{\Xi} = \operatorname{diag}(\xi_1^2, \ldots, \xi_p^2)$. The model presupposes that the variability of the data can be described by H latent factors, with $H \ll p$. In Bayesian factor analysis, it is typically assumed that the latent factors are mutually uncorrelated and have isotropic variance

$$\boldsymbol{f}_i \sim \mathcal{N}_H(\mathbf{0}, \mathbf{I}_H), \qquad \boldsymbol{f}_i \in \mathbb{R}^H \tag{2.2}$$

and that $\boldsymbol{f}_i$, $\boldsymbol{f}_l$, $\boldsymbol{\epsilon}_i$ and $\boldsymbol{\epsilon}_l$ are pairwise independent for all $i \neq l$. A common prior specification for the factor loadings is Gaussian $\lambda_{jh} \sim \mathcal{N}(0, \sigma_{jh}^2)$, for $j = 1, \ldots, p$ and $h = 1, \ldots, H$.

Sparsity is typically induced by assigning suitable priors to the factor loadings, λ_{jh}, or to their variances, σ_{jh}^2. In the context of factor analysis, sparsity often refers to *element-wise* sparsity, whereby the loadings associated with a particular factor h, represented by the column $\boldsymbol{\lambda}_h$ of the loading matrix $\boldsymbol{\Lambda}$, contain many zero or near-zero elements. In this case, only a subset of variables contributes to a given factor, which facilitates interpretation of the latent structure. A related but distinct notion is *column-wise* sparsity, which is closely linked to the automatic determination of the latent dimensionality H. Under an overfitted or infinite factor model, column-wise sparsity implies that only a subset of columns of $\boldsymbol{\Lambda}$ remain active, while the remaining columns are shrunk entirely to zero. Consequently, the number of active columns determines the effective dimension of the factor model.

There is a vast body of literature on sparse Bayesian factor analysis devoted to inducing sparsity and automatically determining the number of latent factors. A common approach is to combine element-wise and column-wise sparsity mechanisms, with the latter enabling automatic factor selection through priors that increasingly penalize higher-dimensional models. Such priors are typically assigned to variances of factor loadings, with examples including multiplicative gamma process priors (Bhattacharya and Dunson, 2011) and spike-and-slab formulations (Legramanti et al., 2020; Kowal and Canale, 2023; Frühwirth-Schnatter, 2023). Alternative approaches impose sparsity directly on the factor loadings rather than through their variances (see, e.g., Knowles and Ghahramani, 2011 and Ročková and George, 2016 who employ Indian buffet process prior).

Some models additionally incorporate global shrinkage parameters, which regulate the overall amount of shrinkage across the entire loading matrix and allow the model to learn the appropriate overall degree of sparsity from the data, or hierarchical sparsity

structures combining global, column-wise and local (element-wise) shrinkage (Gao et al., 2013; Murphy et al., 2020; Schiavon et al., 2022; Frühwirth-Schnatter et al., 2025).

2.2 Extension to multi-study framework

As mentioned in the Introduction, molecular feature data are often measured across multiple cohorts, laboratories/centres, platforms, or populations. This results in separate datasets, each of which may differ from the others in terms of the technology used, population characteristics, or laboratory conditions. Throughout the paper, we assume that these datasets correspond to different studies, indexed by $s = 1, \ldots, S$. Each study may contain a different number of observations (samples), denoted by n_s, but the same set of variables (features), indexed by $j = 1, \ldots, p$, is measured in every study. As illustrated in Figure 1, each dataset from study s has dimension $p \times n_s$ and stacked together, i.e., $\boldsymbol{Y} = [\boldsymbol{Y}_1, \boldsymbol{Y}_2, \ldots, \boldsymbol{Y}_S]$, they form a dataset $p \times n$, where $n = n_1 + \cdots + n_S$.

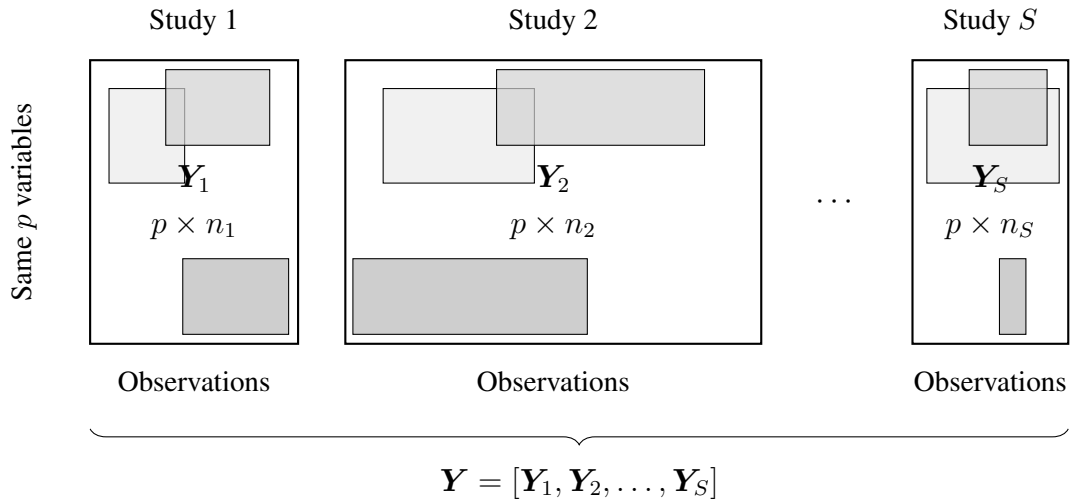


Figure 1: *Schematic representation of the multi-study data structure and bicluster configurations. The S studies $\boldsymbol{Y}_1, \ldots, \boldsymbol{Y}_S$ share the same p variables but may contain different numbers of observations n_s. Within each study, the grey blocks represent distinct biclusters sharing the same variables across studies but spanning different observations. The top two biclusters have overlapping variables, whereas the bottom biclusters are disjoint. Multiple configurations of overlapping and disjoint biclusters may occur across studies.*

To separate latent structure into components that are shared across studies and components that are specific to individual studies, multi-study factor analysis (MSFA, De Vito et al., 2019, 2021) introduces two sets of latent factors. Common latent factors, denoted by $\boldsymbol{f}_{is} \in \mathbb{R}^H$ for $i = 1, \ldots, n_s$ and $s = 1, \ldots, S$, capture variation shared across all studies, where H is the dimension of the common latent space. In addition, study-specific latent factors, denoted by $\boldsymbol{\eta}_{is} \in \mathbb{R}^{H_s}$ for $i = 1, \ldots, n_s$ and $s = 1, \ldots, S$, capture variation unique to study s, where H_s is the dimension of the corresponding latent space. Each study is associated with its own loading matrix $\boldsymbol{\Upsilon}_s \in \mathbb{R}^{p \times H_s}$, which links the study-specific factors to the observed variables. This decomposition allows information to be shared across studies through the common factors while simultaneously accounting for study-specific heterogeneity through the study-specific factors.

The resulting Bayesian MSFA model can then be written as (Liang et al., 2026):

$$\boldsymbol{y}_{is} = \boldsymbol{\Lambda}\boldsymbol{f}_{is} + \boldsymbol{\Upsilon}_s\boldsymbol{\eta}_{is} + \boldsymbol{\epsilon}_{is}, \tag{2.3}$$

where $\boldsymbol{y}_{is}$ denotes sample i from study s. Since the model explicitly decomposes variation into common and study-specific components, the idiosyncratic noise term is also allowed to vary across studies. Specifically, $\boldsymbol{\epsilon}_{is} \sim \mathcal{N}_p(\mathbf{0}, \boldsymbol{\Xi}_s)$, where $\boldsymbol{\Xi}_s = \mathrm{diag}(\xi_{1s}^2, \ldots, \xi_{ps}^2)$.

Let $\boldsymbol{F}_s \in \mathbb{R}^{H \times n_s}$ denote the matrix of common factor scores for study s, whose columns correspond to the vectors $\boldsymbol{f}_{is}$, $i = 1, \ldots, n_s$. Similarly, let $\boldsymbol{\eta}_s \in \mathbb{R}^{H_s \times n_s}$ denote the matrix of study-specific factor scores and $\boldsymbol{\epsilon}_s \in \mathbb{R}^{p \times n_s}$ the matrix of idiosyncratic errors. The model can then be written in matrix form as

$$[\boldsymbol{Y}_1, \boldsymbol{Y}_2, \ldots, \boldsymbol{Y}_S] = \boldsymbol{\Lambda}\,[\boldsymbol{F}_1, \boldsymbol{F}_2, \ldots, \boldsymbol{F}_S] + [\boldsymbol{\Upsilon}_1\boldsymbol{\eta}_1, \boldsymbol{\Upsilon}_2\boldsymbol{\eta}_2, \ldots, \boldsymbol{\Upsilon}_S\boldsymbol{\eta}_S] + [\boldsymbol{\epsilon}_1, \boldsymbol{\epsilon}_2, \ldots, \boldsymbol{\epsilon}_S]\,.$$

The above discussion provides only a concise overview of Bayesian multi-study factor models. For a more comprehensive treatment, the reader is referred to De Vito et al. (2021), Hansen et al. (2025), and Liang et al. (2026).

2.3 Multi-study factor regression

To account for the confounding effects of observed covariates, such as sex or age or ethnicity of the individuals from whom the molecular samples were obtained, we include these variables as known regressors in the model. This formulation, known as a factor regression model, was introduced in the Bayesian factor analysis context by Lucas et al. (2006), further developed in Avalos-Pacheco et al. (2022), and recently extended to the multi-study setting in (MSFR, De Vito and Avalos-Pacheco, 2025).

Let p^X denote the number of observed covariates common to all studies. The MSFR model can then be written as

$$\boldsymbol{y}_{is} = \boldsymbol{B}^X\boldsymbol{x}_{is} + \boldsymbol{\Lambda}\boldsymbol{f}_{is} + \boldsymbol{\Upsilon}_s\boldsymbol{\eta}_{is} + \boldsymbol{\epsilon}_{is},$$

where $\boldsymbol{x}_{is} \in \mathbb{R}^{p^X}$ denotes the covariate vector associated with sample i from study s, and $\boldsymbol{B}^X \in \mathbb{R}^{p \times p^X}$ is the matrix of regression coefficients.

The remaining quantities are defined as in the multi-study factor model in Eq. (2.3).

2.4 Biclustering through common factors

The multi-study factor regression model introduced above accounts for covariate effects and separates shared and study-specific sources of variation. Our primary interest, however, lies in identifying subsets of variables and observations that participate in the shared signal. To this end, we focus on the sparsity of the common component $\boldsymbol{\Lambda}\boldsymbol{F}$ and exploit the connection between sparse factor models and biclustering. Biclustering is achieved by replacing the isotropic Gaussian prior in Eq. (2.2) with a sparsity-inducing prior on the factor scores, analogous to that used for the factor loadings (see, e.g., Hochreiter et al., 2010, Gao et al., 2016, Moran et al., 2021, and Wang and Stephens, 2021). This mirrors the sparsity structure on both sides of the common signal, so that each factor is active only for a subset of variables and a subset of observations. Consequently,

each common factor defines a bicluster, allowing simultaneous clustering of variables and observations. Since a variable or observation may be active in multiple factors, the resulting biclusters are not required to be disjoint and may overlap, see Figure 1 for a visual representation of biclusters within each study where multiple configurations of such overlapping and disjoint biclusters may exist.

Various sparsity-inducing priors can be used to induce biclustering. While sparsity may be imposed directly on the factor loadings and factor scores (see, e.g., Moran et al., 2021), we instead adopt a variance decomposition approach in which the variances of the factor loadings and factor scores are expressed as products of multiple sparsity components. These components correspond to different levels of shrinkage and are each assigned their own sparsity-inducing prior. This approach is closely related to the biclustering formulation of Gao et al. (2016), which also decomposes the variance into multiple sparsity levels, although the latter adopts a hierarchical construction whereas we employ a multiplicative formulation. More specifically, for the loadings and the factors, we assume

$$\begin{aligned} \lambda_{jh} \mid \sigma^2_{jh} &\sim \mathcal{N}(0, \sigma^2_{jh}), \qquad \sigma^2_{jh} = g(\theta^{\Lambda}_h, \phi^{\Lambda}_{jh}, \kappa^{\Lambda}), \\ f_{hi} \mid \zeta^2_{hi} &\sim \mathcal{N}(0, \zeta^2_{hi}), \qquad \zeta^2_{hi} = g(\theta^{F}_h, \phi^{F}_{hi}, \kappa^{F}), \end{aligned}$$

with $g(\cdot)$ a designed hierarchical structure and where θ^{Λ}_h and θ^{F}_h control sparsity at the latent factor level, ϕ^{Λ}_{jh} and ϕ^{F}_{hi} control local (element-wise) sparsity within each factor and κ^{Λ} and κ^{F} represent global shrinkage parameters.

This hierarchy belongs to the broader class of global–local shrinkage priors (Carvalho et al., 2010; Polson and Scott, 2011; Bhadra et al., 2019), where global and local parameters jointly determine the amount of shrinkage, but extends these formulations by incorporating explicit factor-level sparsity.

3 Biclustering with sparse exchangeable shrinkage process prior

This section specifies the hierarchical prior that implements the biclustering formulation introduced in Section 2.4. The proposed prior is constructed in three stages. We first specify the global, and local (element-wise) sparsity components that take into account the within-factor sparsity in the common factor loadings and factor scores. We then introduce the exchangeable shrinkage process (ESP) prior, which induces factor-level sparsity and enables automatic determination of the effective number of latent factors. Finally, we extend the same general prior construction to the study-specific component using a simplified specification that preserves automatic factor selection while providing sufficient flexibility to capture study-specific variation. Throughout, a symmetric prior specification is adopted for the common factor loadings and factor scores, reflecting their complementary roles in defining biclusters.

3.1 Prior specification for global and local sparsity components

As discussed in Section 2.4, biclustering is achieved through sparsity in both the common factor loadings and the common factor scores. Extending this construction to

the multi-study setting, we specify the hierarchical prior jointly across all studies, with the common factor scores indexed by the study s. We decompose their variances into global, factor-level and local sparsity components:

$$\lambda_{jh} \mid \theta_h^\Lambda, \phi_{jh}^\Lambda, \kappa^\Lambda \sim \mathcal{N}(0, \theta_h^\Lambda \phi_{jh}^\Lambda \kappa^\Lambda), \quad j = 1, \ldots, p, \quad h = 1, \ldots, H,$$
$$f_{his} \mid \theta_h^F, \phi_{his}^F, \kappa^F \sim \mathcal{N}(0, \theta_h^F \phi_{his}^F \kappa^F), \quad h = 1, \ldots, H, \quad i = 1, \ldots, n_s, \quad s = 1, \ldots, S.$$

Here, θ_h^Λ and θ_h^F govern factor-level sparsity by controlling which latent factors remain active, ϕ_{jh}^Λ and ϕ_{his}^F induce local (element-wise) sparsity within the loadings and scores of factor h and κ^Λ and κ^F control global shrinkage.

For the global and local sparsity components, we adopt inverse Gamma priors commonly used in Bayesian factor analysis. To allow the degree of local sparsity to be learned from the data rather than fixed a priori, the rate parameter of each local inverse Gamma prior is itself assigned its own hyperprior at the variable or observation level:

$$\phi_{jh}^\Lambda \mid \beta_j^{\phi^\Lambda} \sim \mathcal{G}^{-1}(a^{\phi^\Lambda}, \beta_j^{\phi^\Lambda}), \qquad \beta_j^{\phi^\Lambda} \sim \mathcal{G}(c^{\phi^\Lambda}, d^{\phi^\Lambda}), \qquad \kappa^\Lambda \sim \mathcal{G}^{-1}(a^{\kappa^\Lambda}, b^{\kappa^\Lambda}),$$
$$\phi_{his}^F \mid \beta_{is}^{\phi^F} \sim \mathcal{G}^{-1}(a^{\phi^F}, \beta_{is}^{\phi^F}), \qquad \beta_{is}^{\phi^F} \sim \mathcal{G}(c^{\phi^F}, d^{\phi^F}), \qquad \kappa^F \sim \mathcal{G}^{-1}(a^{\kappa^F}, b^{\kappa^F}),$$

where $\mathcal{G}(a, b)$ denotes the gamma distribution with shape parameter a, rate parameter b, and mean a/b, and $\mathcal{G}^{-1}(a, b)$ denotes the inverse Gamma distribution with shape parameter a, rate parameter b, and mean $b/(a-1)$ for $a > 1$.

3.2 Exchangeable Shrinkage Process prior

The factor-level sparsity parameters θ_h^Λ and θ_h^F, play a central role in the proposed formulation, as they determine which latent factors remain active and which are shrunk towards zero. To model these quantities, we employ the exchangeable shrinkage process (ESP) prior developed by Frühwirth-Schnatter (2023). The ESP prior takes the form of a sequence of unordered spike-and-slab distributions and enables selection of the columns of the factor loading matrix (equivalently, the rows of the factor score matrix) corresponding to active factors.

Let H denote the maximum possible number of active latent common factors. Then the ESP prior is defined as follows. Assume $\boldsymbol{\tau} = \{\tau_h \in (0,1) : h = 1, \ldots, H\}$ to be a finite sequence of i.i.d. random probabilities. Let $\Theta = \{\theta_h\}, h = 1, \ldots, H$ be a finite sequence of column-specific shrinkage parameters assumed to be independent conditional on $\boldsymbol{\tau}$ and independent of all $\tau_l, l \neq h$ for all h. If the prior $\theta_h \mid \tau_h$ takes the following spike-and-slab form:

$$\theta_h \mid \tau_h \sim (1 - \tau_h) p_{\text{spike}}(\theta_h \mid \varphi_0) + \tau_h p_{\text{slab}}(\theta_h \mid \varphi_\theta), \tag{3.1}$$

where φ_0 and φ_θ are the hyperparameters of the spike and the slab distributions, $p_{\text{spike}}(\cdot)$ and $p_{\text{slab}}(\cdot)$ respectively, then Θ follows an ESP prior. The prior is exchangeable in the sense that it is invariant to permutations of the column indices h.

It is often assumed that each slab probability of the sequence $\tau_1, \ldots, \tau_H$ follows a beta distribution, where the first parameter depends on H. Following Frühwirth-Schnatter

(2023), we assign the slab probabilities the prior

$$\tau_h \mid \alpha^\tau \sim \mathcal{B}\left(\frac{\alpha^\tau}{H}, 1\right), \qquad h = 1, \ldots, H.$$

This corresponds to the finite Beta-process construction of Ročková and George (2016) and converges to the Indian buffet process prior as $H \to \infty$ Teh (2006).

An important property of the ESP prior is that it admits a finite generalised cumulative shrinkage process (CUSP) representation (Frühwirth-Schnatter, 2023). CUSP was introduced by Legramanti et al. (2020) in the context of sparse infinite factor models and provides a mechanism for automatic inference on the effective latent dimensionality. In the context of our formulation, it is obtained by permuting the column index h of the parameters $\theta_1, \ldots, \theta_H$ according to the decreasing slab probabilities $\tau_{(1)} > \ldots > \tau_{(H)}$. Increasing spike probabilities π_h for $h = 1, \ldots, H$, as for a CUSP prior, are obtained by defining $\pi_h = 1 - \tau_{(h)}$. Representation (3.1) allows one to choose an upper limit for the number of factors, H, and keep it fixed throughout the entire inference procedure. By performing the classification between spike and slab independently for each column, we end up with defining an effective number of active factors H^+, which is random a priori and a posteriori, and typically smaller than H.

The key properties of the ESP prior do not depend on the specific choice of the spike and slab distributions. For computational convenience and to preserve conjugacy, we assume a hierarchical structure with inverse Gamma distributions for both the spike and the slab, and Gamma hyperpriors on their rate parameters. Finally, since the concentration parameter α^τ plays a central role in determining the allocation between spike and slab, we allow it to be learned from the data by assigning it a Gamma prior $\alpha^\tau \sim \mathcal{G}(a^\tau, b^\tau)$.

Following the standard practice in biclustering literature (e.g.Gao et al., 2016; Wang and Stephens, 2021; Moran et al., 2021, we adopt a symmetric prior specification for the common factor loadings and factor scores. This symmetry is intentional as in the proposed biclustering formulation, sparsity in the loadings determines which variables participate in a latent component, while sparsity in the factor scores determines which observations participate in the corresponding component. Assigning the same global, factor-level and local shrinkage structure to both matrices therefore allows variable and observation selection to be treated on an equal footing, while permitting their respective shrinkage parameters and hyperparameters to be learned separately.

Combining the variance decomposition introduced in Section 3.1 with the ESP prior yields the following hierarchical prior for the common factor loadings:

$$\begin{aligned}
&\lambda_{jh} \,|\, \theta_h^\Lambda, \phi_{jh}^\Lambda, \kappa^\Lambda \sim \mathcal{N}(0, \theta_h^\Lambda \phi_{jh}^\Lambda \kappa^\Lambda), \quad \text{where } j = 1, \ldots, p,\ h = 1, \ldots, H, \\
&\theta_h^\Lambda \,|\, \tau_h^\Lambda \sim \tau_h^\Lambda \mathcal{G}^{-1}(a^{\theta^\Lambda}, \beta^{\theta^\Lambda}) + (1 - \tau_h^\Lambda)\mathcal{G}^{-1}(a_0^{\theta^\Lambda}, \beta_0^{\theta^\Lambda}), \qquad \tau_h^\Lambda \sim \mathcal{B}\left(\frac{\alpha^{\tau^\Lambda}}{H}, 1\right), \\
&\alpha^{\tau^\Lambda} \sim \mathcal{G}(a^{\tau^\Lambda}, b^{\tau^\Lambda}), \quad \beta^{\theta^\Lambda} \sim \mathcal{G}(c^{\theta^\Lambda}, d^{\theta^\Lambda}), \quad \beta_0^{\theta^\Lambda} \sim \mathcal{G}(c_0^{\theta^\Lambda}, d_0^{\theta^\Lambda}), \\
&\phi_{jh}^\Lambda \sim \mathcal{G}^{-1}(a^{\phi^\Lambda}, \beta_j^{\phi^\Lambda}), \qquad \beta_j^{\phi^\Lambda} \sim \mathcal{G}(c^{\phi^\Lambda}, d^{\phi^\Lambda}). \\
&\kappa^\Lambda \sim \mathcal{G}^{-1}(a^{\kappa^\Lambda}, b^{\kappa^\Lambda}),
\end{aligned} \tag{3.2}$$

An analogous prior is assigned to the common factor scores:

$$
\begin{aligned}
&f_{his} \,|\, \theta_h^F, \phi_{his}^F, \kappa^F \sim \mathcal{N}(0, \theta_h^F \phi_{his}^F \kappa^F), \text{ where } i = 1, \ldots, n_s, \ s = 1, \ldots, S, \ h = 1, \ldots, H, \\
&\theta_h^F \,|\, \tau_h^F \sim \tau_h^F \mathcal{G}^{-1}(a^{\theta^F}, \beta^{\theta^F}) + (1 - \tau_h^F)\mathcal{G}^{-1}(a_0^{\theta^F}, \beta_0^{\theta^F}), \qquad \tau_h^F \sim \mathcal{B}\left(\frac{\alpha^{\tau^F}}{H}, 1\right), \\
&\alpha^{\tau^F} \sim \mathcal{G}(a^{\tau^F}, b^{\tau^F}), \quad \beta^{\theta^F} \sim \mathcal{G}(c^{\theta^F}, d^{\theta^F}), \quad \beta_0^{\theta^F} \sim \mathcal{G}(c_0^{\theta^F}, d_0^{\theta^F}), \\
&\phi_{jh}^{\Lambda} \sim \mathcal{G}^{-1}(a^{\phi^F}, \beta_j^{\phi^F}), \qquad \beta_j^{\phi^F} \sim \mathcal{G}(c^{\phi^F}, d^{\phi^F}). \\
&\kappa^F \sim \mathcal{G}^{-1}(a^{\kappa^F}, b^{\kappa^F}),
\end{aligned}
\tag{3.3}
$$

3.3 Prior specification for the study-specific signal

Since our primary interest lies in identifying biclusters in the common signal, we do not impose sparsity on the study-specific factor scores and instead assign them a standard Gaussian prior $\boldsymbol{\eta}_{is} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{I}_{H_s})$, which is a standard choice in Bayesian factor analysis. For the study-specific factor loadings, we again employ the sparse ESP prior described in Section 3.2, but omit the global sparsity component. The global shrinkage parameter is intended to encourage an overall sparse representation by shrinking all loadings simultaneously, which is appropriate for the common component where the objective is to identify a parsimonious biclustering structure. In contrast, the study-specific component is primarily introduced to absorb study-dependent variation, such as batch and technology effects, which are generally expected to be relatively dense rather than globally sparse. We nevertheless retain the factor-level ESP prior to infer the effective number of study-specific factors automatically, together with the element-wise shrinkage prior to allow sparse study-specific patterns whenever they are supported by the data. This specification provides sufficient flexibility to model heterogeneous study-specific variation without over-constraining a component that is not of primary inferential interest.

The prior on the study-specific factor loadings is given by:

$$
\begin{aligned}
&\upsilon_{jhs} \,|\, \theta_{hs}^{\Upsilon}, \phi_{jhs}^{\Upsilon} \sim \mathcal{N}(0, \theta_{hs}^{\Upsilon} \phi_{jhs}^{\Upsilon}), \quad \text{where } j = 1, \ldots, p, \ h = 1, \ldots, H_s, \ s = 1, \ldots, S, \\
&\theta_{hs}^{\Upsilon} \,|\, \tau_{hs}^{\Upsilon} \sim \tau_{hs}^{\Upsilon} \mathcal{G}^{-1}(a^{\theta^{\Upsilon}}, \beta^{\theta^{\Upsilon}}) + (1 - \tau_{hs}^{\Upsilon})\mathcal{G}^{-1}(a_0^{\theta^{\Upsilon}}, \beta_0^{\theta^{\Upsilon}}), \qquad \tau_{hs}^{\Upsilon} \sim \mathcal{B}\left(\frac{\alpha^{\tau^{\Upsilon}}}{H}, 1\right), \\
&\alpha^{\tau^{\Upsilon}} \sim \mathcal{G}(a^{\tau^{\Upsilon}}, b^{\tau^{\Upsilon}}), \quad \beta^{\theta^{\Upsilon}} \sim \mathcal{G}(c^{\theta^{\Upsilon}}, d^{\theta^{\Upsilon}}), \quad \beta_0^{\theta^{\Upsilon}} \sim \mathcal{G}(c_0^{\theta^{\Upsilon}}, d_0^{\theta^{\Upsilon}}), \\
&\phi_{jhs}^{\Upsilon} \sim \mathcal{G}^{-1}(a^{\phi^{\Upsilon}}, \beta_{js}^{\phi^{\Upsilon}}), \qquad \beta_{js}^{\phi^{\Upsilon}} \sim \mathcal{G}(c^{\phi^{\Upsilon}}, d^{\phi^{\Upsilon}}).
\end{aligned}
\tag{3.4}
$$

Note that the column-specific variance parameters θ_{hs}^{Υ} and slab probabilities τ_{hs}^{Υ} are study-specific, whereas the corresponding hyperparameters are shared across studies. This allows information to be borrowed across studies when determining the dimensionality of the study-specific latent structure, while still permitting each study to have its own set of active factors. Similar to the common factors, the number of active study-specific factors (assigned to the slab) is denoted by H_s^+.

The performance of the proposed model depends critically on the specification of the spike-and-slab components. In particular, the spike distribution must exert substantially stronger shrinkage than the slab distribution in order to achieve effective separation

between active and inactive coefficients. If the two components are insufficiently separated, factor selection becomes unstable and inactive loadings may not be adequately shrunk towards zero. General intuition and practical guidance on calibrating the spike and slab hyperparameter in the ESP framework are provided by (Grushanina and Frühwirth-Schnatter, 2025).

The relative amount of shrinkage imposed on the common and study-specific latent components also requires careful calibration. Since the primary objective of the model is to recover the shared latent structure, the common component is assigned less aggressive factor-level shrinkage. In contrast, stronger shrinkage is imposed on the study-specific factor-level sparsity parameters, resulting in more conservative activation of study-specific factors. This reduces the tendency of study-specific factors to absorb variation that is shared across studies and helps mitigate the information-switching phenomenon discussed in Section 4. The final hyperparameter values were selected empirically based on extensive simulation experiments and are reported in Section B.2 of the Supplementary Material.

3.4 Priors for other parameters

The regression coefficients are assigned independent Gaussian priors $\beta_{jl}^X \sim N(0, q_{jl})$, where $j = 1, \ldots, p$ and $l = 1, \ldots, p^X$ denote the variable and covariate indices, respectively.

Finally, we assume a diagonal covariance matrix for the idiosyncratic errors within each study:

$$\xi_{js}^2 \sim \mathcal{G}^{-1}(a^\xi, \beta_{js}^\xi), \qquad \beta_{js}^\xi \sim \mathcal{G}(a^{\beta^\xi}, b_{js}^{\beta^\xi}).$$

The rate hyperparameters $b_{js}^{\beta^\xi}$ are assigned the data-driven values $100/R_{js}^2$, where R_{js} is the range of the data in dimension j and study s (see, for example, (Frühwirth-Schnatter, 2006) for the reasoning behind this choice).

4 Identification

In factor analysis, identifiability refers to the ability to uniquely recover the latent factors and their loadings from the observed data. Classical factor models are not identifiable because different rotations of the latent factor space yield observationally equivalent representations with identical covariance structure ((Anderson and Rubin, 1956), (Lopes and West, 2004), (Frühwirth-Schnatter et al., 2024), among many others). In multi-study factor models, this issue is further complicated by information switching, whereby signal can be equivalently represented by either shared (common) or study-specific latent factors (De Vito et al., 2019; Chandra et al., 2025; Bortolato and Canale, 2026). In this section, we discuss these sources of non-identifiability and their implications for the MS-FRB model.

4.1 Rotational indeterminacy

The rotational invariance problem inherent in factor models arises from the fact that the decomposition of the covariance structure is invariant under orthogonal transformations of the latent space. In particular, the decomposition $\boldsymbol{\Omega}_s = \boldsymbol{\Lambda}\boldsymbol{\Lambda}^\top + \boldsymbol{\Upsilon}_s\boldsymbol{\Upsilon}_s^\top + \boldsymbol{\Xi}_s$ will also hold for any semi-orthogonal matrices $\mathbf{P}$ ($\mathbf{P}\mathbf{P}^\top = \mathbf{I}$), $\mathbf{P}_s$ ($\mathbf{P}_s\mathbf{P}_s^\top = \mathbf{I}$), and $\boldsymbol{\Theta} = \boldsymbol{\Lambda}\mathbf{P}$, $\mathbf{g}_{is} = \mathbf{P}^\top \boldsymbol{f}_{is}$, $\boldsymbol{\Psi}_s = \boldsymbol{\Upsilon}_s\mathbf{P}_s$, $\mathbf{v}_{is} = \mathbf{P}_s^\top \boldsymbol{\eta}_{is}$, in which case the two models

$$\mathbf{y}_{is} = \boldsymbol{\Lambda}\boldsymbol{f}_{is} + \boldsymbol{\Upsilon}_s\boldsymbol{\eta}_{is} + \boldsymbol{\epsilon}_{is} \quad \text{and} \quad \mathbf{y}_{is} = \boldsymbol{\Theta}\mathbf{g}_{is} + \boldsymbol{\Psi}_s\mathbf{v}_{is} + \boldsymbol{\epsilon}_{is}$$

are observationally equivalent. This problem can either be addressed *a priori* via imposing restrictions on the elements of $\boldsymbol{\Lambda}$ and $\boldsymbol{\Upsilon}_s$ (see, for example, Lopes and West (2004) and Carvalho et al. (2008)), or *a posteriori* by using some type of orthogonalisation procedure, such as Procrustes or Varimax rotation (Kaiser, 1958; Poworoznek et al., 2021; McParland et al., 2014).

Let us first consider the product of the common factor loadings and factor scores, $\boldsymbol{\Lambda}\mathbf{F}$, where biclustering occurs. Both matrices are endowed with sparse structure through the sparse ESP prior, which imposes sparsity via coordinate-specific column-selection and element-wise shrinkage parameters for both factor loadings and scores. As argued in (Moran et al., 2021), generic rotations typically mix coordinates, spreading signal across multiple columns and entries, and thereby destroying the axis-aligned sparse structure favoured by the prior. Consequently, while the likelihood remains rotationally invariant, the posterior is not. Rotational indeterminacy is therefore substantially reduced, leaving only trivial ambiguities such as sign switches and permutations. A more detailed discussion of this phenomenon can be found in (Moran et al., 2021) and (Rohe and Zeng, 2023). We therefore rely on the sparsity-inducing priors to obtain effectively identifiable solutions. In Section 4.4, we further introduce a sufficient condition based on unique support that can be used to assess factor identification in practice.

It should be noted that this argument pertains only to the common factors and their loadings. Study-specific factors are expected to be less sparse, as they might capture effects inherent to an individual study, and are therefore more susceptible to rotational ambiguity. However, since the primary focus of this work is on the common factors, addressing the rotational indeterminacy of study-specific components lies beyond the scope of this paper.

4.2 Scaling

While sparsity-inducing priors substantially reduce rotational indeterminacy, another source of non-identifiability remains. When the product $\boldsymbol{\Lambda}\mathbf{F}$ is estimated, the columns of the factor loadings and scores are estimated only up to scaling. This means that $\boldsymbol{\lambda}_h^\top \boldsymbol{f}_h$ is equivalent to $(c_h\boldsymbol{\lambda}_h)^\top c_h^{-1}\boldsymbol{f}_h$ for any $c_h > 0$. In biclustering, the primary objective is to identify the nonzero elements of the matrices $\boldsymbol{\Lambda}$ and $\mathbf{F}$, as these determine the subsets of features and samples that co-vary, rather than to estimate their exact magnitudes. Since the scale of these values is not of direct interest, we follow (Moran et al., 2021) and rescale $\boldsymbol{\Lambda}$ and $\mathbf{F}$ at each iteration of the VI algorithm so that the corresponding columns

have equal norms. More specifically,

$$c_h = \left(\frac{\mathbb{E}[\|\boldsymbol{f}_h\|^2]}{\mathbb{E}[\|\boldsymbol{\lambda}_h\|^2]} \right)^{1/4}, \quad \boldsymbol{\lambda}_h \leftarrow c_h \boldsymbol{\lambda}_h, \quad \boldsymbol{f}_h \leftarrow \boldsymbol{f}_h / c_h.$$

Rescaling also improves numerical stability and leads to faster convergence by eliminating drift along scale-equivalent representations and improving the conditioning of the variational updates.

4.3 Information switching

Information switching arises because variation that is shared across studies can often be represented either through common factors or through a combination of study-specific factors, leading to multiple observationally equivalent decompositions. In the existing literature on MSFA models, information switching has been addressed differently, depending on the nature and purpose of the model. In the frequentist MSFA framework, (De Vito et al., 2019) address information switching by identifying common and study-specific factors with distinct linear subspaces. Thus, the common factors are defined as the intersection of the study-specific covariance subspaces, while the study-specific factors are defined as their complements. This requires a condition that the concatenated loading matrix $\mathbf{L} = [\boldsymbol{\Lambda}, \boldsymbol{\Upsilon}_1, \ldots, \boldsymbol{\Upsilon}_S]$ has a full column rank $r(\mathbf{L}) = H^+ + \sum_{s=1}^{S} H_s^+$, with $H^+ + \sum_{s=1}^{S} H_s^+ \leq p$, which can be rather restrictive in some settings. In the Bayesian framework, (De Vito et al., 2021) do not explicitly impose any constraints to prevent information switching between common and study-specific factors. Instead, they rely on sparsity-inducing shrinkage priors to encourage, though not guarantee, their practical separation. As they demonstrate in the discussion section, this can fail in the case of only partially shared common factors.

In subsequent years, several approaches have been proposed in the literature to address information switching. (Roy et al., 2021) address this issue by replacing the additive MSFA decomposition with a perturbation-based model, in which a single common factor structure is assumed and group differences are captured through multiplicative perturbation matrices, thereby avoiding information switching inherent in the additive structure. Unlike (Roy et al., 2021), (Grabski et al., 2023) do not depart from the additive MSFA formulation, but generalise the common-versus-study-specific decomposition by introducing a latent indicator matrix that allows factors to be shared across arbitrary subsets of studies. While their primary objective is not to address information switching, the resulting framework moves beyond the strict common-versus-study-specific dichotomy and treats the factor-sharing structure itself as an inferential target. (Chandra et al., 2025) address information switching by constraining study-specific factors to lie within the subspace spanned by the common factors, effectively modelling them as structured perturbations of the shared component and thus ensuring a unique decomposition. While this prevents information switching by model design, it can be too restrictive in case of the data showing substantial heterogeneity across studies. Finally, (Bortolato and Canale, 2026) introduce adaptive partition factor analysis (APAFA), a flexible approach in which the model learns factor activation patterns from the data at the observation level. This produces a matrix of $0/1$ activation indicators for individual observations, allowing factors to be shared across arbitrary subsets of observations and studies. Because these activation

patterns are identifiable under mild conditions, they provide an anchoring mechanism for the latent factors and thereby prevent the factor-allocation ambiguities associated with information switching.

While (Bortolato and Canale, 2026) achieve identifiability through sparse observation-level activation patterns, our setting is fundamentally different because factors are represented by biclusters, with sparsity induced simultaneously in the factor loading and score matrices. Consequently, the relevant support structure is no longer a one-dimensional activation pattern across samples, but a two-dimensional feature-sample support. This motivates an identification condition based on bicluster support rather than sample-level activation patterns.

4.4 Identification based on unique support

Motivated by the idea in (Bortolato and Canale, 2026) of using factor activation patterns for identification, we propose a sufficient identification condition tailored to the biclustering structure of the MSFRB model. In particular, the combination of element-wise sparsity in both the common factor loadings and scores, together with sparse loadings in the study-specific components, implies that non-identifiability due to information switching is highly unlikely in sufficiently sparse settings, as overlapping representations of the same signal would require substantial coincidence of supports across multiple components.

Building on this idea, we propose a post hoc identification strategy that exploits this double sparsity by recovering factor supports from posterior summaries and assessing their uniqueness. Unlike APAFA of Bortolato and Canale (2026), where identification is based on observation-level activation patterns, identification in the MSFRB model cannot generally be determined from the sample support alone. A common factor contributes to the observed data through the product of its loadings and scores, so both the features and the samples on which the factor is active determine its support. Consequently, the natural object for identification is the joint feature-sample support of the factor, rather than either support considered separately. More specifically, for each active factor, we construct its support over features and samples and define its unique support as the subset of entries not shared with any other factor. A factor is deemed identified if it has at least one element in its support that is unique. This is a sufficient condition to rule out information switching and provides a simple and interpretable diagnostic for assessing identification in practice. Below, we formulate this idea more formally.

Proposition 1. *Consider the common component of the MSFRB model:*

$$\mathbf{Y}^{(c)} = \sum_{h=1}^{H^+} \boldsymbol{\lambda}_h \boldsymbol{f}_h^\top,$$

where $\boldsymbol{\lambda}_h \in \mathbb{R}^p$ is the hth column of the common loading matrix and $\boldsymbol{f}_h \in \mathbb{R}^n$ is the corresponding common factor score vector. Here $n = \sum_{s=1}^{S} n_s$ denotes the total number of observations across all studies. For each active common factor h, define its feature support and sample support as

$$G_h = \{i \in \{1, \dots, p\} : \lambda_{jh} \neq 0\}$$

$$T_h = \{n \in \{1, \ldots, n\} : f_{hi} \neq 0\},$$

and let its total support be

$$B_h = G_h \times T_h \subseteq \{1, \ldots, p\} \times \{1, \ldots, n\}.$$

Similarly, for each active study-specific factor r in study s, let

$$G_r^s = \{j \in \{1, \ldots, p\} : \lambda_{jr}^s \neq 0\}, \qquad T_r^s = \mathcal{I}_s$$

where $\mathcal{I}_s$ is by the model design the set of all samples belonging to study s. Define the total study-specific factor support as

$$B_r^s = G_r^s \times \mathcal{I}_s.$$

For common factor h, compute its unique support as

$$U_h = B_h \setminus \left(\bigcup_{l \neq h} B_l \cup \bigcup_{s=1}^{S} \bigcup_{r=1}^{H_s^+} B_r^s \right).$$

Then if $U_h \neq \varnothing$, the hth common factor is identified with respect to information switching, in the sense that its contribution cannot be reproduced or absorbed by the remaining common factors or by any study-specific factor.

The proof of Proposition 1 is provided in the Supplementary Material, Section C.

Remark 1. *The uniqueness condition also explains why nontrivial rotations are incompatible with the recovered sparse support structure. Algebraically, the common component remains invariant under rotations of the latent factors. However, a nontrivial rotation mixes the original factor loadings and scores, thereby spreading signal across multiple factors. If (j, i) belongs to the unique support of factor h, this entry is generated exclusively by factor h in the sparse representation. A rotation that mixes factor h with other factors would generally make additional rotated factors active at this entry, destroying the uniqueness of the support. Thus, unique support acts as an anchor for a particular sparse representation of the common component, leaving only transformations that preserve the support structure, such as sign changes and column permutations.*

The details of the post hoc identification algorithm are described in the Supplementary Material, Section D.1.

5 Variational inference for multi-study factor regression biclustering

Let $\boldsymbol{\varphi}$ denote the collection of all unknown model parameters and latent variables. The goal of variational inference is to approximate the posterior distribution $p(\boldsymbol{\varphi} \mid \boldsymbol{Y})$ with some density $q^*(\boldsymbol{\varphi})$ in a class $\mathcal{Q}$. The optimal variational density $q^*(\boldsymbol{\varphi})$ is defined

as the closest to the posterior $p(\boldsymbol{\varphi} \mid \boldsymbol{Y})$ in Kullback-Leibler (KL) divergence, i.e. which satisfies the following expression:

$$\underset{q(\boldsymbol{\varphi})\in\mathcal{Q}}{\arg\min}\left\{KL(q(\boldsymbol{\varphi}) \parallel p(\boldsymbol{\varphi} \mid \boldsymbol{Y}))\right\} = \underset{q(\boldsymbol{\varphi})\in\mathcal{Q}}{\arg\min}\left\{\int_{\boldsymbol{\Phi}} q(\boldsymbol{\varphi}) \log \frac{q(\boldsymbol{\varphi})}{p(\boldsymbol{\varphi} \mid \boldsymbol{Y})} d\boldsymbol{\varphi}\right\}$$

This is equivalent to maximization of the evidence lower bound (ELBO):

$$\underset{q(\boldsymbol{\varphi})\in\mathcal{Q}}{\arg\max}\left\{ELBO(q(\boldsymbol{\varphi}))\right\} = \underset{q(\boldsymbol{\varphi})\in\mathcal{Q}}{\arg\max}\left\{\mathbb{E}_{q(\boldsymbol{\varphi})}\left[\log p(\boldsymbol{\varphi}, \boldsymbol{Y})\right] - \mathbb{E}_{q(\boldsymbol{\varphi})}\left[\log q(\boldsymbol{\varphi})\right]\right\}, \quad (5.1)$$

where all the expectations are taken with respect to $q(\boldsymbol{\varphi})$ and $p(\boldsymbol{\varphi}, \boldsymbol{Y})$ is the joint distribution of the data and model parameters, that factorises into the composition of the likelihood and prior(s) $p(\boldsymbol{\varphi}, \boldsymbol{Y}) = p(\boldsymbol{Y} \mid \boldsymbol{\varphi})p(\boldsymbol{\varphi})$.

5.1 Mean-field variational inference

A common approach to select $\mathcal{Q}$ is to use a mean-field variational family that assumes a partition of parameters into independent blocks and approximates the posterior with a product of independent densities $\mathcal{Q}^{MF} = \{q : q(\boldsymbol{\varphi}) = \prod_{m=1}^{M} q(\boldsymbol{\varphi}_m)\}$. The independent densities $q(\boldsymbol{\varphi}_m)$ are referred to as variational factors. The optimal variational factors $q^*(\boldsymbol{\varphi}_m)$ can be obtained from:

$$q_m^*(\boldsymbol{\varphi}_m) \propto \exp\left(\mathbb{E}_{q(\boldsymbol{\varphi}_{-m})}[\log p(\boldsymbol{\varphi}_m|\boldsymbol{\varphi}_{-m}, \boldsymbol{Y})]\right), \quad (5.2)$$

where $\boldsymbol{\varphi}_{-m} = (\boldsymbol{\varphi}_1, \ldots, \boldsymbol{\varphi}_{m-1}, \boldsymbol{\varphi}_{m+1}, \ldots, \boldsymbol{\varphi}_M)$ is the vector of parameters excluding the mth one ((Bishop, 2006), chapter 10).

When the conditional distribution $p(\boldsymbol{\varphi}_m|\boldsymbol{\varphi}_{-m}, \boldsymbol{Y})$ is of the exponential family, the corresponding optimal variational factors also belong to the same exponential family. Thus, the maximisation problem in Eq. (5.1) is reduced to learning the optimal variational factors $q_m^*(\boldsymbol{\varphi}_m)$. Choosing $\mathcal{Q}^{MF}$ and the prior in such a way that ensures conjugacy enables the development of a coordinate-ascent variational inference (CAVI) algorithm, which learns the optimal variational factors by iteratively updating them according to Eq. (5.2) until convergence is reached ((Bishop, 2006), chapter 10).

To implement CAVI for MSFRB as described above, we assume a mean field variational approximation where variational distributions of our parameters are independent:

$$\begin{aligned}
q(\boldsymbol{\varphi}) = q\Big(&\boldsymbol{F}, \boldsymbol{\Lambda}, \boldsymbol{\eta}, \boldsymbol{\Upsilon}, \boldsymbol{B}^X, \boldsymbol{\Xi}, \boldsymbol{\beta}^{\xi}, \boldsymbol{\Theta}^{\Lambda}, \boldsymbol{\tau}^{\Lambda}, \boldsymbol{\gamma}^{\Lambda}, \boldsymbol{\Theta}^{F}, \boldsymbol{\tau}^{F}, \boldsymbol{\gamma}^{F}, \boldsymbol{\Theta}^{\Upsilon}, \boldsymbol{\tau}^{\Upsilon}, \boldsymbol{\gamma}^{\Upsilon}, \\
&\boldsymbol{\Phi}^{\Lambda}, \boldsymbol{\beta}^{\phi^{\Lambda}}, \boldsymbol{\Phi}^{F}, \boldsymbol{\beta}^{\phi^{F}}, \boldsymbol{\Phi}^{\Upsilon}, \boldsymbol{\beta}^{\phi^{\Upsilon}}, \alpha^{\tau^{\Lambda}}, \alpha^{\tau^{F}}, \alpha^{\tau^{\Upsilon}}, \beta^{\theta^{\Lambda}}, \beta_0^{\theta^{\Lambda}}, \beta^{\theta^{F}}, \beta_0^{\theta^{F}}, \beta^{\theta^{\Upsilon}}, \beta_0^{\theta^{\Upsilon}}, \kappa^{\Lambda}, \kappa^{F}\Big) \\
= q(\alpha^{\tau^{\Lambda}})&q(\alpha^{\tau^{F}})q(\alpha^{\tau^{\Upsilon}})q(\beta^{\theta^{\Lambda}})q(\beta_0^{\theta^{\Lambda}})q(\beta^{\theta^{F}})q(\beta_0^{\theta^{F}})q(\beta^{\theta^{\Upsilon}})q(\beta_0^{\theta^{\Upsilon}})q(\kappa^{\Lambda})q(\kappa^{F}) \\
&\times \left[\prod_{j=1}^{p} q(\boldsymbol{B}_j^X) q_H(\boldsymbol{\lambda}_j) q(\beta_j^{\phi^{\Lambda}}) \prod_{h=1}^{H} q(\phi_{jh}^{\Lambda})\right] \\
&\times \left[\prod_{j=1}^{p}\prod_{s=1}^{S} q(b_{js}^{\xi}) q(\xi_{js}^2) q_{H_s}(\boldsymbol{\upsilon}_{js}) q(\beta_{js}^{\phi^{\Upsilon}}) \prod_{h=1}^{H_s} q(\phi_{jhs}^{\Upsilon})\right]
\end{aligned}$$

$$\times \left[\prod_{h=1}^{H} q(\gamma_h^\Lambda) q(\tau_h^\Lambda) q(\theta_h^\Lambda \mid \gamma_h^\Lambda) q(\gamma_h^F) q(\tau_h^F) q(\theta_h^F \mid \gamma_h^F) \right]$$
$$\times \prod_{s=1}^{S} \left[\left[\prod_{h=1}^{H_s} q(\gamma_{hs}^\Upsilon) q(\tau_{hs}^\Upsilon) q(\theta_{hs}^\Upsilon \mid \gamma_{hs}^\Upsilon) \right] \left[\prod_{i=1}^{n_s} q_{H_s}(\boldsymbol{\eta}_{is}) q_H(\boldsymbol{f}_{is}) q(\beta_{is}^{\phi^F}) \prod_{h=1}^{H} q(\phi_{his}^F) \right] \right].$$

5.2 Variational inference for the ESP prior

While for many of the model's parameters mean field variational inference is rather standard, this is not the case for the parameters involved in the ESP prior. According to our knowledge, variational inference for the ESP prior has not been described before and involves certain particularities. In this section, we describe variational inference for the general ESP prior on latent factor loadings in a basic factor model as in equation (2.1). This can be easily extended to the sparse ESP priors in the MSFRB model as described in Sections 3. The variational update steps for the sparse ESP priors on the common factor loadings and scores, as well as on the study-specific factor loadings, are presented in detail in Algorithm A.1 in the Supplementary Material.

For a basic factor model as in equation (2.1), the ESP prior on the column variance parameter of factor loadings formulated as $\theta_h \mid \tau_h \sim \tau_h \mathcal{G}^{-1}(a^\theta, b^\theta) + (1-\tau_h)\mathcal{G}^{-1}(a_0^\theta, b_0^\theta)$ can be rewritten in the following way using an auxiliary indicator γ_h:

$$\begin{aligned} &\lambda_{jh} \mid \theta_h \sim \mathcal{N}(0, \theta_h), \quad i = 1, \ldots, p, \quad h = 1, \ldots, H, \\ &\theta_h \mid \gamma_h = 1 \sim \mathcal{G}^{-1}(a^\theta, b^\theta), \quad \theta_h \mid \gamma_h = 0 \sim \mathcal{G}^{-1}(a_0^\theta, b_0^\theta), \\ &\gamma_h \sim Ber(\tau_h), \quad \tau_h \sim \mathcal{B}(\frac{\alpha^\tau}{H}, 1). \end{aligned} \tag{5.3}$$

For simplicity, we assume that a^θ, b^θ, a_0^θ, b_0^θ, and α^τ are fixed hyperparameters. Extending the algorithm to infer these quantities is straightforward through standard conjugate variational updates.

Then the mean-field variational family takes the form:

$$q(-) = \prod_{h=1}^{H} \left[q(\gamma_h) q(\tau_h) q(\theta_h|\gamma_h) \prod_{j=1}^{p} q(\lambda_{jh}) \right] \tag{5.4}$$

We choose the following variational distribution for conjugacy: $q(\gamma_h) = Ber(r_h)$, $q(\tau_h) = \mathcal{B}(\hat{a}_{\tau_h}, \hat{b}_{\tau_h})$, $q(\theta_h|\gamma_h = 1) = \mathcal{G}^{-1}(\hat{a}_{\theta_h slab}, \hat{b}_{\theta_h slab})$, $q(\theta_h|\gamma_h = 0) = \mathcal{G}^{-1}(\hat{a}_{\theta_h spike}, \hat{b}_{\theta_h spike})$, $q(\boldsymbol{\lambda}_j) = \mathcal{N}_H(\boldsymbol{\mu}_j^\lambda, \boldsymbol{\Sigma}_j^\lambda)$. The algorithm below illustrates variational updates for the respective parameters.

1. For $h = 1, \ldots, H$ update $\theta_h|\gamma_h$ from $\mathcal{G}^{-1}(\hat{a}_{\theta_h}, \hat{b}_{\theta_h})$ using

   for $\gamma_h = 1$

   $\hat{a}_{\theta_h slab} = a^\theta + \frac{p}{2}$

   $\hat{b}_{\theta_h slab} = b^\theta + \frac{1}{2}\sum_{j=1}^{p} \mathbb{E}(\lambda_{jh}^2)$

   for $\gamma_h = 0$

   $\hat{a}_{\theta_h spike} = a_0^\theta + \frac{p}{2}$

$$\hat{b}_{\theta_h spike} = b_0^\theta + \tfrac{1}{2}\sum_{j=1}^{p}\mathbb{E}(\lambda_{jh}^2)$$

where $\mathbb{E}\lambda_{jh}^2 = \left[\mu_{jh}^\lambda\right]^2 + \mathbf{\Sigma}_{j[h,h]}^\lambda$ with h and $[h,h]$ indicating the hth element of the mean vector $\boldsymbol{\mu}_j^\lambda$ and the $[h,h]$th element of the covariance matrix $\mathbf{\Sigma}_j^\Lambda$, correspondingly.

2. For $h = 1, \ldots, H$ update the responsibilities r_h in the following way:

$$s_{h\,\text{slab}} = \mathbb{E}\log\tau_h + a^\theta \log b^\theta - \log\Gamma(a^\theta) - (a^\theta + \tfrac{p}{2} + 1)\mathbb{E}_{\text{slab}}\log\theta_h - \mathbb{E}_{\text{slab}}\left(\frac{1}{\theta_h}\right)\left(b^\theta + \frac{1}{2}\sum_{j=1}^{p}\mathbb{E}(\lambda_{jh}^2)\right),$$

$$s_{h\,\text{spike}} = \mathbb{E}\log(1-\tau_h) + a_0^\theta \log b_0^\theta - \log\Gamma(a_0^\theta) - (a_0^\theta + \tfrac{p}{2} + 1)\mathbb{E}_{\text{spike}}\log\theta_h - \mathbb{E}_{\text{spike}}\left(\frac{1}{\theta_h}\right)\left(b_0^\theta + \frac{1}{2}\sum_{j=1}^{p}\mathbb{E}(\lambda_{jh}^2)\right),$$

where

$$\mathbb{E}\lambda_{jh}^2 = \left[\mu_{jh}^\lambda\right]^2 + \mathbf{\Sigma}_{j[h,h]}^\lambda,$$
$$\mathbb{E}\log\tau_h = \psi(\hat{a}_{\tau_h}) - \psi(\hat{a}_{\tau_h} + \hat{b}_{\tau_h}),$$
$$\mathbb{E}\log(1-\tau_h) = \psi(\hat{b}_{\tau_h}) - \psi(\hat{a}_{\tau_h} + \hat{b}_{\tau_h}),$$
$$\mathbb{E}(\tfrac{1}{\theta_h}) = \frac{\hat{a}_{\theta_h}}{\hat{b}_{\theta_h}},$$
$$\mathbb{E}\log\theta_h = \ln(\hat{b}_{\theta_h}) - \psi(\hat{a}_{\theta_h}),$$

where $\psi(\cdot)$ is a digamma function.

Calculate responsibility (probability that factor h is active) as

$$r_h = \frac{\exp(s_{h\,slab})}{\exp(s_{h\,slab}) + \exp(s_{h\,spike})}$$

As probabilities of a factor being active or inactive can take very small or very large values, which can lead to "Inf" or "NaN" values in the algorithm, for the sake of numerical stability we recommend to use the log-sum-exp trick, which leads to the following representation of r_h:

$$r_h = \frac{1}{1 + \exp(s_{h\,spike} - s_{h\,slab})}.$$

3. For $h = 1, \ldots, H$ update τ_h from $\mathcal{B}(\hat{a}_{\tau_h}, \hat{b}_{\tau_h})$ using

$$\hat{a}_{\tau_h} = \tfrac{\alpha^\tau}{H} + r_h$$
$$\hat{b}_{\tau_h} = 1 + (1 - r_h) = 2 - r_h$$

4. For $j = 1, \ldots, p$ update $\boldsymbol{\lambda}_j^\top$ from $\mathcal{N}_H(\boldsymbol{\mu}_j^\lambda, \boldsymbol{\Sigma}_j^\lambda)$ using

$$\boldsymbol{\Sigma}_j^\lambda = \left(\boldsymbol{\Psi}^{-1} + \mathbb{E}\left(\frac{1}{\xi_j^2}\right) \sum_{i=1}^{n} \mathbb{E}(\boldsymbol{f}_i \boldsymbol{f}_i^\top) \right)^{-1}$$

where

$$\boldsymbol{\Psi}^{-1} = \operatorname{diag}\left(\mathbb{E}\left(\tfrac{1}{\theta_1}\right), \ldots, \mathbb{E}\left(\tfrac{1}{\theta_H}\right) \right),$$
$$\mathbb{E}(\tfrac{1}{\theta_h}) = r_h \tfrac{\hat{a}_{\theta_h slab}}{\hat{b}_{\theta_h slab}} + (1 - r_h) \tfrac{\hat{a}_{\theta_h spike}}{\hat{b}_{\theta_h spike}}, h = 1, \ldots, H,$$
$$\mathbb{E}(\boldsymbol{f}_i \boldsymbol{f}_i^\top) = \boldsymbol{\mu}_i^f \left[\boldsymbol{\mu}_i^f\right]^\top + \boldsymbol{\Sigma}_i^f,$$
$$\mathbb{E}\left(\tfrac{1}{\xi_j^2}\right) = \tfrac{\hat{a}_\xi}{\hat{b}_{\xi_j}},$$

$$\boldsymbol{\mu}_j^\lambda = \boldsymbol{\Sigma}_j^\lambda \mathbb{E}\left(\frac{1}{\xi_j^2}\right) \left(\boldsymbol{\mu}_1^f, \ldots, \boldsymbol{\mu}_n^f\right) \boldsymbol{y}_j^\top.$$

5.3 CAVI algorithm for MSFRB model

The sparse ESP prior used in the biclustering algorithm is more elaborate than in Section 5.2 as in addition to the column shrinkage parameter, the variances of λ_{jh}, f_{his} and λ_{jhs} also include additional sparsity parameters. Moreover, the rate parameters of the spike and slab inverse gamma distributions, namely $\beta_0^{\theta^F}$ and β^{θ^F} in case of common factors, $\beta_0^{\theta^\Lambda}$ and β^{θ^Λ} in case of common factor loadings, and $\beta_0^{\theta^\Upsilon}$ and β^{θ^Υ} in case of study-specific factor loadings are assigned hyperpriors. Finally, the concentration parameters of the beta distribution of slab probabilities α^{τ^Λ}, α^{τ^F} and α^{τ^Υ} are also learned from data (see equations (3.2), (3.3)) and (3.4) for the full prior formulations). However, for prior choices that ensure conjugacy, these changes are easy to implement with standard enhancements of the algorithm presented in the previous section. For the inference on the spike-and-slab part of the ESP prior on all three variables λ_{jh}, f_{his} and λ_{jhs}, we utilize the same data augmentation scheme as in equation 5.3: $\gamma_h^\Lambda = 1$ if the hth column of the common factor loading matrix $\boldsymbol{\Lambda}$ is active and $\gamma_h^\Lambda = 0$ otherwise; $\gamma_h^F = 1$ if the hth row of the common factor score matrix $\boldsymbol{F}$ is active and $\gamma_h^F = 0$ otherwise; and $\gamma_{hs}^\Upsilon = 1$ if the hth column of the study s factor loading matrix $\boldsymbol{\Upsilon}_s$ is active and $\gamma_{hs}^\Upsilon = 0$ otherwise. The full details of the CAVI algorithm for the MSFRB model are presented in Algorithm A.1 in the Supplementary Material.

5.4 Stochastic variational inference

Molecular profiling data can occasionally be high dimensional not only in p but also in n_s, in which case the CAVI algorithm might become relatively slow. To overcome this bottleneck, we develop a stochastic variational inference (SVI) algorithm (Hoffman et al., 2013), which deals with high dimensionality by updating local parameters (those

parameters which updates scale with n_s e.g. study-specific or common factor scores for $i = 1, \ldots, n_s$) for only a subset of the n_s observations and then uses this subset to approximate the remaining parameters common for all individuals $i = 1, \ldots, n_s$ and $s = 1, \ldots, S$. This is done in the following way.

First, a batch proportion parameter $b \in (0, 1)$ needs to be selected, then a random sample of data in the size of $\tilde{N}_s = b \times n_s$ is drawn at each iteration of the algorithm. So, at each iteration a $p \times \tilde{N}_s$ subset of $\boldsymbol{y}_{is}$ will be taken, with randomly chosen $\tilde{N}_s$ out of n_s columns. SVI will proceed updating the local parameters $\boldsymbol{\eta}_{is}$, $\boldsymbol{f}_{is}$, ϕ^F_{his} and $\beta^{\phi^F}_{is}$ from this subset. At the next step, the local parameters updated from this batch are used for updating the global parameters, namely $\boldsymbol{\beta}^X_j$, ξ^2_{js}, $\boldsymbol{\lambda}_j$, $\boldsymbol{\upsilon}_{js}$, θ^F_h, r^F_h and κ^F.

For this purpose, to ensure the stable performance and convergence, the algorithm is reformulated as a gradient-based optimization algorithm for the global parameters. Thus, the derivative of the ELBO with respect to natural parameters takes the following form:

$$\nabla ELBO(q(\boldsymbol{\varphi}_g)) = \mathbb{E}_{q(\boldsymbol{\varphi}_{-g})}\left[\eta(\boldsymbol{Y}, \boldsymbol{\varphi})\right] - \boldsymbol{\vartheta}_g, \tag{5.5}$$

where $\boldsymbol{\varphi}$ denotes the collection of model parameters and latent variables, and $\boldsymbol{\vartheta}_g$ denotes the variational parameters associated with the gth variational factor. Setting the gradient in equation (5.5) to 0, we get $\boldsymbol{\vartheta}_g = \mathbb{E}_{q(\boldsymbol{\varphi}_{-g})}\left[\eta(\boldsymbol{Y}, \boldsymbol{\varphi})\right]$.

The expectation of the likelihood is estimated with the weight of $n_s/\tilde{N}_s$ assigned to each individual observation. At each iteration of the algorithm the global parameters are estimated as a weighted average of the global parameters updated at the current and previous iterations:

$$\boldsymbol{\vartheta}_g(t) = (1 - \rho_t)\boldsymbol{\vartheta}_g(t-1) + \rho_t \hat{\boldsymbol{\vartheta}}_g(t),$$

where ρ_t is a step size parameter such that $\sum_t \rho_t \to \infty$ and $\sum_t \rho_t^2 < \infty$ and $\boldsymbol{\vartheta}_g$ denotes the natural variational parameters associated with the gth global variational factor, with $g = 1, \ldots, G$ (in our case $G = 7$ and the set of global variational factors corresponds to the model parameters $\{\boldsymbol{B}^X_i, \xi^2_{js}, \boldsymbol{\lambda}_i, \boldsymbol{\upsilon}_{js}, \theta^F_h, r^F_h, \kappa^F\}$)

The choice of ρ_t is rather influential as it can impact the convergence of the algorithm. A typical choice, suggested in (Hoffman et al., 2013) is to set $\rho_t = (t + \tau^\rho)^{-\kappa^\rho}$, where $\kappa^\rho \in (0.5, 1]$ is the forgetting rate that controls how quickly the variational parameters change across iterations, and $\tau^\rho > 0$ is the delay, which downweights early iterations. The details of the SVI algorithm for the MSFRB model are presented in Section A.2 in the Supplementary Material.

6 Simulation Studies

To assess the performance of the proposed MSFRB model, we conducted two simulation studies. The first simulation study investigates the performance of the MSFRB model in recovering common biclusters across a range of various dimensional settings. The second study illustrates the ability of the model to correctly recover various elements of latent structure separately. The common factor loadings and factor scores are generated to produce visible block patterns. We simulate both sparse and dense common factors to assess the ability of the model to recover the underlying parameters and correctly distinguish between these two different types of factors.

In the first simulation study, we assess recovery of the true bicluster structure using the relevance, recovery and consensus measures (Prelić et al., 2006; Hochreiter et al., 2010; Moran et al., 2021). Relevance measures how well the estimated biclusters correspond to true biclusters, whereas recovery measures how well the true biclusters are represented among the estimated biclusters. The consensus score additionally penalises duplicated or spurious estimated biclusters by using an optimal one-to-one matching between true and estimated biclusters. All three measures are based on the Jaccard similarity between bicluster supports and take values in $[0, 1]$, with larger values indicating better recovery. In the second simulation study, we use two complementary metrics to compare the recovered matrices with their true (simulated) values: the relative Frobenius distance and the RV coefficient (Abdi, 2007). The relative Frobenius distance measures the reconstruction error relative to the magnitude of the true signal, with smaller values indicating better recovery. The RV coefficient measures structural similarity between two matrices, with values close to one indicating high similarity. Detailed definitions of all performance metrics are provided in Section E of the Supplementary Material. The hyperparameter values used by MSFRB in the simulation studies are reported in Table B.2 of the Supplementary Material.

We compare the performance of the proposed MSFRB model with four competing methods commonly used for the analysis of molecular profiling data. The first alternative method is the Bayesian MSFA model of (De Vito et al., 2021; Hansen et al., 2025). Like MSFRB, this model explicitly accounts for the multi-study structure of the data by decomposing variation into common and study-specific latent factors. However, it does not perform biclustering, as sparsity is imposed only on the factor loadings, whereas the latent factors are assigned standard Gaussian priors with unit variance. Consequently, the model is able to identify sparse groups of variables associated with a factor, but not sparse groups of observations. We fit the model using the variational inference implementation provided by (Hansen et al., 2025)[1].

The remaining three competing methods provide biclustering of factor loadings and scores, but do not explicitly account for the multi-study structure of the data. The first is empirical Bayes matrix factorisation (EBMF) (Wang and Stephens, 2021), which is fitted using the R package `flashier` and employs an efficient variational inference algorithm. The second is the Bayesian biclustering model of (Gao et al., 2016), fitted using the R package `BicMix`, which combines variational inference with expectation-maximisation updates. Finally, we consider spike-and-slab lasso biclustering (SSLB) (Moran et al., 2021), fitted using the `SSLB` R package, which implements a fast expectation-maximisation algorithm with variational updates.

The variational inference implementation of MSFA provided by (Hansen et al., 2025) performs best when the data are centred and standardised prior to model fitting. Therefore, each study-specific data matrix is normalised before applying this method. To ensure comparability with the other approaches, the fitted values are transformed back to the original measurement scale using the corresponding study-specific means and standard deviations before computing reconstruction errors. All reported performance metrics are

[1]Other Bayesian multi-study factor analysis models suitable for molecular profiling data, such as (Chandra et al., 2025) and (Bortolato and Canale, 2026), have also been proposed. However, these approaches are primarily aimed at covariance estimation and rely on MCMC-based inference, which can be prohibitively slow in the high-dimensional settings considered in this work.

therefore evaluated on the original data scale. Unlike the biclustering methods considered here, MSFA imposes sparsity only on the factor loadings and therefore remains susceptible to rotational indeterminacy. Consequently, direct comparison of the estimated factor loadings and factor scores with their simulated values is not meaningful. To avoid identifiability issues and ensure a fair comparison across methods, we evaluate recovery of the common latent structure through the product $\boldsymbol{\Lambda F}$, which is invariant to orthogonal rotations of the latent factors.

The remaining biclustering methods (EBMF, BicMix and SSLB) are applied directly to the pooled data matrix without additional preprocessing. Preliminary experiments based on residualising covariate and study effects prior to model fitting generally led to poorer recovery of the latent structure, suggesting that such preprocessing may remove part of the signal of interest. We therefore report results obtained from the original simulated data.

For the BicMix model of (Gao et al., 2016), implemented in the `BicMix` R package, the estimated biclustering component $\widehat{\boldsymbol{\Lambda}}\widehat{\boldsymbol{F}}$ is reconstructed using the subset of latent factors that exhibit sparsity in both the loading and factor-score directions. BicMix provides posterior inclusion probabilities for the loading and score components of each factor. We classify a component as sparse if both the loading and score inclusion probabilities exceed a predefined threshold (we set it to 0.5), and construct the estimated common component as the product of the corresponding loading and factor matrices. This ensures that only factors exhibiting simultaneous sparsity across features and samples contribute to the reconstructed biclustering structure.

Data reconstruction accuracy was evaluated on the original data scale. All reported reconstruction errors were computed by comparing the estimated and true data matrices expressed on this scale.

6.1 Simulation study 1: Bicluster recovery

This simulation study investigates the comparative performance of the MSFRB model in recovering bicluster structure common across studies. To assess the model performance with different p and n_s we consider the following settings. We set the number of studies fixed at $S = 6$. The number of features p is allowed to take values $p \in \{500, 2000\}$. For the number of samples in each study n_s we first consider the setting where $n_s \in \{200, 400\}$ for each s, and then consider the setting where n_s varies across studies taking values $n_s = \{120, 400, 100, 200, 300, 88\}$ and taking rather small values in some of the studies. For each setting, 20 replicate datasets are generated.

The data are generated as follows. The observed data matrices were generated according to the proposed MSFRB model,

$$\mathbf{Y}_s = \boldsymbol{B}^X \mathbf{X}_s + \boldsymbol{\Lambda}\mathbf{F}_s + \boldsymbol{\Upsilon}_s \boldsymbol{\eta}_s + \boldsymbol{E}_s, \qquad s = 1, \ldots, S,$$

where $\boldsymbol{B}^X$ contains the regression coefficients for the covariates, $\boldsymbol{\Lambda}$ and $\mathbf{F}_s$ denote the common factor loading and factor score matrices, $\boldsymbol{\Upsilon}_s$ and $\boldsymbol{\eta}_s$ are the study-specific loadings and scores, and $\mathbf{E}_s$ represents independent Gaussian residual errors.

For each simulated data set, we generate two covariates, namely, similar to the Section 6.1. The first covariate represents sex and is generated independently for each sample from a Bernoulli distribution, where the study-specific probabilities of being male were

chosen as $P(male) = (0.55, 0.45, 0.60, 0.50, 0.40, 0.65)$, thereby introducing moderate heterogeneity in the sex composition across studies while preserving substantial within-study variation.

The second covariate represents age and is generated from a normal distribution with standard deviation of 8 and study-specific means of $(55, 58, 52, 61, 49, 57)$ years. Generated ages were truncated to the interval $[18, 90]$ years to ensure realistic values. The two covariates were assembled into the study-specific design matrix $\mathbf{X}_s = (\mathbf{x}_{1s}, \ldots, \mathbf{x}_{n_s s})^\top$, and their effects were incorporated through the regression component $\boldsymbol{\beta}^X \mathbf{X}_s$, where each regression coefficient was independently generated from a standard normal distribution $\beta^X_{jl} \sim \mathcal{N}(0, 1)$, for $j = 1, \ldots, p$ and $l = 1, 2$.

We fix the number of common factors (and hence the number of biclusters) to $H = 10$ and the number of study-specific factors to $H_s = 2$ for each study. For each common factor h, first, the loading support $\mathcal{G}_h$ and the sample support $\mathcal{T}_h^{(s)}$ were generated. The loading supports were chosen to occupy approximately 4%, 5%, 6%, 7%, 8%, 9%, 10%, 6%, 5% and 7% of the features, respectively, while the corresponding sample supports occupied approximately 8%, 10%, 12%, 9%, 11%, 7%, 8%, 10%, 7% and 8% of the samples within each study. Three pairs of biclusters, $(1, 2)$, $(4, 5)$ and $(7, 8)$, were generated with 20%, 30% and 40% overlap, respectively, in both their feature and sample supports, whereas all remaining bicluster pairs were mutually disjoint. Conditional on the supports, the common loadings and factor scores were generated according to

$$\lambda_{jh} \sim \begin{cases} \mathcal{N}(0, \sigma_\lambda^2), & j \in \mathcal{G}_h, \\ 0, & j \notin \mathcal{G}_h, \end{cases} \qquad f_{hi}^{(s)} \sim \begin{cases} \mathcal{N}(0, \sigma_f^2), & i \in \mathcal{T}_h^{(s)}, \\ 0, & i \notin \mathcal{T}_h^{(s)}, \end{cases}$$

where $\sigma_\lambda = \sigma_f = 3$. To avoid numerically negligible signals, the generated non-zero values were required to have absolute value at least 0.1. Finally, the ordering of both features and samples was randomly permuted before constructing the observed data matrices, so that the bicluster structure was not visible from the original ordering.

The study-specific factor scores were generated independently from the standard normal distribution, $\mathcal{N}(0, 1)$. For each study, the corresponding loading vectors were generated by randomly selecting approximately 60% and 90% of the features, respectively, and sampling the associated non-zero loadings independently from $\mathcal{N}(0, 0.7^2)$ and $\mathcal{N}(0, 0.25^2)$, with the remaining loadings set to zero. The active feature supports were generated independently across studies and independently of the common bicluster supports.

Finally, independent study-specific Gaussian residual errors were generated independently as

$$\epsilon_{jis} \sim \mathcal{N}(0, \xi_{js}^2), \qquad \xi_{js}^2 \sim \mathcal{G}^{-1}(100, 39.6),$$

so that $\mathbb{E}(\xi_{js}^2) = 0.40$.

Thus, this simulation setting provides a challenging bicluster recovery problem by combining biclusters of varying sizes, three pairs of partially overlapping biclusters, heterogeneous study-specific latent structure, feature-specific residual noise, and two study-specific covariates. The difficulty is further increased by considering both balanced ($n_s = \{200, 400\}$) and highly unbalanced ($n_s = 120, 400, 100, 200, 300, 88$) study sample sizes, as well as high-dimensional settings with up to $p = 5000$ features. Together,

these characteristics create substantial ambiguity in both the feature and sample supports, making accurate recovery of the common biclusters considerably more difficult than in simulations with well-separated biclusters. Figure 2a shows the simulated observed data matrices for the setting with $n_s = 400$ observations per study and $p = 1000$ features. After jointly ordering the features and samples according to the true bicluster memberships, the ten common biclusters become clearly visible as aligned block structures across all six studies.

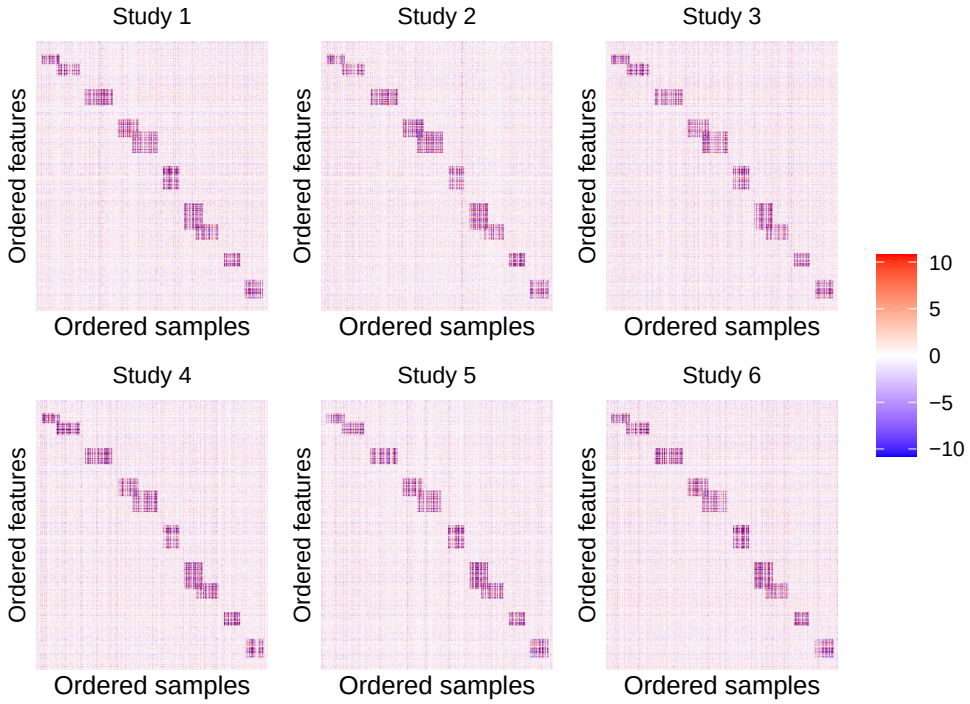
We set the maximum number of active common factors as $H = 20$ and the maximum number of study-specific factors in each study as $H_s = 10$. Model performance is assessed using the relevance, recovery, and consensus indices described in detail in Section E of the Supplementary Material. We compare the performance of the MSFRB model with three biclustering approaches, namely EBMF of (Wang and Stephens, 2021), BicMix of (Gao et al., 2016), and SSLB of (Moran et al., 2021). The MSFA model of (Hansen et al., 2025) is not included in this comparison because it is designed for latent factor estimation rather than bicluster recovery, which is the primary focus of this simulation study.

Figure 2 compares the simulated observed data matrices with the data matrices reconstructed by the MSFRB model for one representative simulation replicate. In both panels, the features and samples are ordered according to the true bicluster memberships, enabling a direct visual comparison between the underlying and recovered bicluster structures. The reconstructed data closely reproduce the common bicluster structure present in the simulated data.

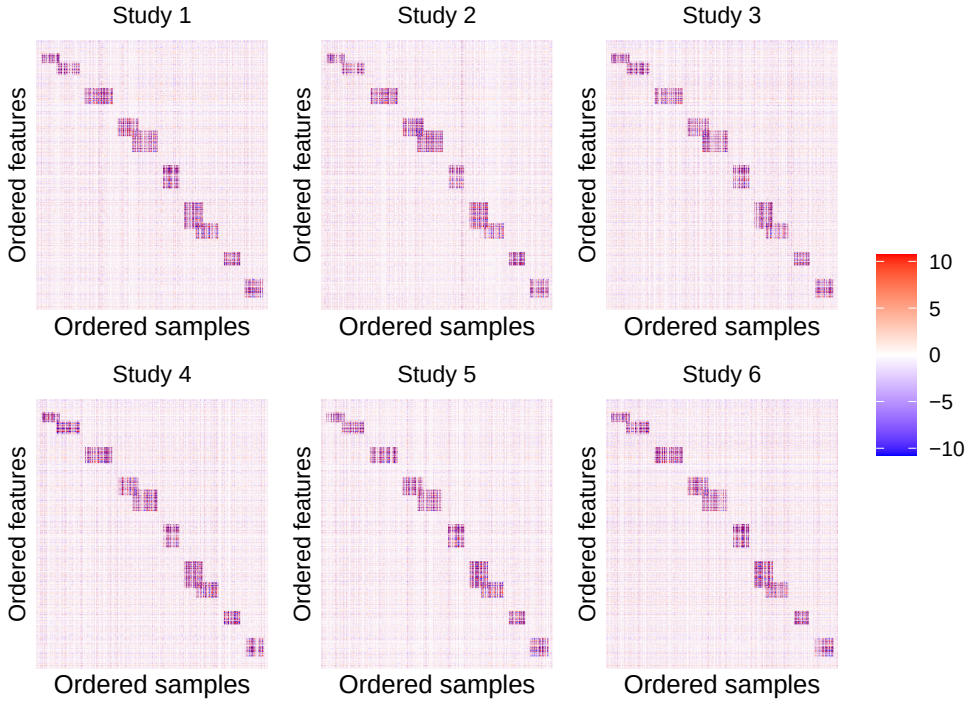
Tables 1 and 2 illustrate the recovery of the common bicluster structure of the four models across the different simulation settings. Overall, MSFRB provides the most balanced recovery of the simulated biclusters. It generally identifies a number of biclusters close to the true value of 10, while achieving the highest relevance and consensus indices across all settings. In particular, its consensus index remains between approximately 0.60 and 0.66 across the considered values of p, sample sizes, and the heterogeneous sample-size setting.

EBMF consistently achieves the highest recovery index, with values between approximately 0.68 and 0.74. However, it also consistently identifies approximately 20 biclusters, twice the true number, and its substantially lower relevance and consensus indices indicate considerable over-segmentation of the underlying bicluster structure. Thus, while the biclusters estimated by EBMF provide good coverage of the true biclusters, the individual estimated biclusters correspond less closely to the true structure. BicMix similarly tends to overestimate the number of biclusters, although with lower recovery, relevance, and consensus indices than EBMF. SSLB shows more variable behaviour across the simulation settings and generally yields lower performance in all three indices than MSFRB.

The performance of MSFRB is relatively stable with increasing dimensionality, although some slight deterioration is observed at $p = 2000$, particularly in the estimated number of biclusters and the recovery index. This effect is most pronounced for $n_s = 200$ and under heterogeneous study sizes, whereas with $n_s = 400$ MSFRB continues to recover approximately all 10 simulated biclusters even at $p = 2000$. The results therefore suggest that additional sample information improves the recovery of the number of common biclusters in higher-dimensional settings.



(a) *Simulated observed data matrices.*



(b) *Data matrices reconstructed by the MSFRB model.*

Figure 2: *Heatmaps of (a) the simulated observed data matrices and (b) the data matrices reconstructed by the MSFRB model for one randomly chosen representative replicate under the simulation setting* $n_s = 400$ *observations per study and* $p = 1000$ *features. In both panels, the features and samples are jointly ordered according to the true bicluster memberships to facilitate direct comparison. The recovered common bicluster structure closely matches the underlying simulated biclusters.*

Table 1: *Bicluster recovery performance for the common biclusters under different simulation settings for the number of samples per study n_s and number of features p. The true number of simulated biclusters is* 10*. Results are reported as mean [standard deviation] across* 20 *simulated datasets. The best value in each column and p- and n_s-setting is highlighted in bold.*

Model	n_s	p	# biclusters	Recovery	Relevance	Consensus
MSFRB	200	500	**9.95 [0.22]**	0.657 [0.028]	**0.660 [0.026]**	**0.657 [0.028]**
EBMF	200	500	19.90 [0.31]	**0.732 [0.114]**	0.372 [0.059]	0.366 [0.062]
SSLB	200	500	8.90 [3.48]	0.219 [0.060]	0.259 [0.038]	0.201 [0.042]
BicMix	200	500	17.05 [1.10]	0.520 [0.053]	0.318 [0.042]	0.306 [0.038]
MSFRB	200	1000	**8.80 [3.04]**	0.650 [0.031]	**0.665 [0.021]**	**0.650 [0.031]**
EBMF	200	1000	20.00 [0.00]	**0.676 [0.101]**	0.342 [0.049]	0.335 [0.052]
SSLB	200	1000	15.65 [5.30]	0.433 [0.058]	0.311 [0.101]	0.289 [0.086]
BicMix	200	1000	16.75 [1.21]	0.517 [0.090]	0.321 [0.055]	0.306 [0.057]
MSFRB	200	2000	**8.95 [2.16]**	0.601 [0.039]	**0.637 [0.017]**	**0.600 [0.040]**
EBMF	200	2000	20.00 [0.00]	**0.698 [0.098]**	0.352 [0.049]	0.346 [0.051]
SSLB	200	2000	19.95 [0.22]	0.592 [0.042]	0.301 [0.021]	0.297 [0.021]
BicMix	200	2000	16.45 [1.47]	0.384 [0.084]	0.241 [0.061]	0.235 [0.061]
MSFRB	400	500	**10.00 [0.00]**	0.648 [0.026]	**0.648 [0.026]**	**0.648 [0.026]**
EBMF	400	500	19.96 [0.22]	**0.716 [0.106]**	0.365 [0.053]	0.359 [0.054]
SSLB	400	500	3.35 [1.46]	0.059 [0.034]	0.159 [0.050]	0.057 [0.033]
BicMix	400	500	17.05 [1.19]	0.486 [0.057]	0.297 [0.039]	0.286 [0.041]
MSFRB	400	1000	**10.00 [0.00]**	0.642 [0.018]	**0.642 [0.018]**	**0.642 [0.018]**
EBMF	400	1000	20.00 [0.00]	**0.739 [0.091]**	0.374 [0.045]	0.369 [0.047]
SSLB	400	1000	8.40 [3.35]	0.205 [0.054]	0.254 [0.043]	0.197 [0.057]
BicMix	400	1000	16.95 [1.15]	0.433 [0.073]	0.265 [0.038]	0.254 [0.038]
MSFRB	400	2000	**9.95 [0.22]**	0.621 [0.017]	**0.624 [0.011]**	**0.621 [0.017]**
EBMF	400	2000	20.00 [0.00]	**0.688** [0.101]	0.348 [0.050]	0.341 [0.051]
SSLB	400	2000	16.30 [4.34]	0.428 [0.050]	0.293 [0.101]	0.284 [0.101]
BicMix	400	2000	15.85 [1.35]	0.377 [0.081]	0.248 [0.055]	0.238 [0.054]

6.2 Simulaton Study 2: Recovery of the underlying factor structure

This simulation study is designed to assess the ability of the MSFRB model to recover the underlying model parameters, including common and study-specific factors, their corresponding loadings, and the covariate effects. We generate data from $S = 4$ studies, with numbers of features $p \in \{500, 1000, 2000\}$ and samples $n_s \in \{200, 400\}$ per study. We consider four active common factors and two active study-specific factors in each study. The common component consists of one dense and three sparse factors, with block-structured supports for the sparse common factor loadings and scores.

The feature support is partitioned into three approximately equally sized contiguous blocks, corresponding to the three sparse common factors. The nonzero common factor loadings are generated as

$$\lambda_{jh} \sim \mathcal{N}(0, 2^2), \qquad j \in G_h, \quad h = 1, 2, 3,$$

where G_1, G_2, and G_3 denote the three feature blocks, while all loadings outside the corresponding blocks are set to 0.

The factor-score supports are defined separately within each study. The first sparse

Table 2: *Bicluster recovery performance for the common biclusters under the simulation setting with* $n_s = \{120, 400, 100, 200, 300, 88\}$. *The true number of simulated biclusters is* 10. *Results are reported as mean [standard deviation] across 20 simulated datasets. The best value in each column and* p*-setting is highlighted in bold.*

Model	p	# biclusters	Recovery	Relevance	Consensus
MSFRB	500	**9.90 [0.45]**	0.652 [0.040]	**0.658 [0.033]**	**0.649 [0.042]**
EBMF	500	19.90 [0.31]	**0.722 [0.101]**	0.366 [0.055]	0.360 [0.056]
SSLB	500	8.80 [1.94]	0.232 [0.058]	0.266 [0.038]	0.223 [0.052]
BicMix	500	16.40 [0.94]	0.471 [0.091]	0.295 [0.062]	0.287 [0.061]
MSFRB	1000	**9.75 [0.44]**	0.642 [0.031]	**0.658 [0.018]**	**0.642 [0.032]**
EBMF	1000	20.00 [0.00]	**0.728 [0.088]**	0.368 [0.044]	0.361 [0.046]
SSLB	1000	13.40 [5.26]	0.413 [0.067]	0.341 [0.086]	0.317 [0.080]
BicMix	1000	15.90 [1.07]	0.431 [0.066]	0.278 [0.047]	0.272 [0.047]
MSFRB	2000	**8.45 [2.93]**	0.597 [0.031]	**0.635 [0.020]**	**0.596 [0.031]**
EBMF	2000	20.00 [0.00]	**0.727 [0.098]**	0.365 [0.049]	0.359 [0.051]
SSLB	2000	19.95 [0.22]	0.590 [0.050]	0.301 [0.024]	0.296 [0.025]
BicMix	2000	15.30 [1.22]	0.434 [0.103]	0.287 [0.066]	0.282 [0.066]

common factor is active for the first half of the samples in each study, while the second is active for the second half. The third sparse common factor alternates across studies between being active for all samples and for the middle half of the samples. The nonzero factor scores are generated as

$$f_{hi} \sim \mathcal{N}(0, 2^2), \qquad i \in T_h, \quad h = 1, 2, 3,$$

where T_h denotes the corresponding sample support, while all remaining entries are set to 0. The dense common factor is generated by sampling its factor scores from $\mathcal{N}(0, 1.5^2)$, and its loadings from $\mathcal{N}(0, 2^2)$.

Study-specific factors are generated as $\boldsymbol{\eta}_{is}^\top \sim \mathcal{N}_2(\mathbf{0}, \mathbf{I}_2)$ for $i = 1, \ldots, n_s$ and $s = 1, \ldots, S$, with their corresponding loadings generated as $v_{jhs} \sim \mathcal{N}(0, 1)$ for all j, s, and $h = 1, 2$. The idiosyncratic variances are simulated as $\xi_{js}^2 \sim \mathcal{G}^{-1}(100, 30)$ for all j and s.

We generate two covariates, sex and age. Sex is a binary covariate that represents biological sex of an individual from which the sample is taken, and it is randomly assigned to each sample in each study as a binomial distributed variable. Age is a continuous covariate and is simulated from a normal distribution. For a visual illustration of the recovery of the individual model components, we additionally generated a lower-dimensional dataset under the same simulation design. Their corresponding true and estimated parameter matrices are presented in Section F of the Supplementary Material.

We set the maximum possible number of active common and study-specific factors to $H = 20$ and $H_s = 10$, respectively. Hyperparameters and initial values are chosen as described in Section B.1 of the Supplementary Material. We simulate 20 replicate datasets for each dimensional setting. In all replicates, in the results from the MSFRB model run, we keep only those common factors that satisfied the unique-support identification criterion.

The primary objective of this simulation study is to evaluate the extent to which the competing methods recover various aspects of the underlying latent structure. As most

Table 3: *Relative Frobenius distances and RV coefficients between the true and estimated common loading matrix $\boldsymbol{\Lambda}$, the common factor score matrix $\boldsymbol{F}$, and their product $\boldsymbol{\Lambda F}$. Values are reported as mean [standard deviation] across simulation replicates. Smaller relative Frobenius distances indicate better recovery, while RV coefficients closer to one indicate stronger agreement. The best value in each column and setting is highlighted in bold.*

Model	n_s	p	Relative Frobenius distance			RV coefficient		
			$d(\boldsymbol{\Lambda}, \widehat{\boldsymbol{\Lambda}})$	$d(\boldsymbol{F}, \widehat{\boldsymbol{F}})$	$d(\boldsymbol{\Lambda F}, \widehat{\boldsymbol{\Lambda}}\widehat{\boldsymbol{F}})$	$RV_{\boldsymbol{\Lambda}, \widehat{\boldsymbol{\Lambda}}}$	$RV_{\boldsymbol{F}, \widehat{\boldsymbol{F}}}$	$RV_{\boldsymbol{\Lambda F}, \widehat{\boldsymbol{\Lambda}}\widehat{\boldsymbol{F}}}$
MSFRB	200	500	0.052 [0.014]	0.08 [0.01]	0.056 [0.013]	**0.999**	0.999	**1.000**
MSFA	200	500	0.200 [0.098]	0.22 [0.13]	0.169 [0.016]	0.979	0.999	0.980
EBMF	200	500	**0.050 [0.055]**	**0.06 [0.07]**	**0.053 [0.017]**	0.998	**1.000**	0.998
SSLB	200	500	2.611 [1.880]	0.78 [0.22]	2.899 [2.208]	0.256	0.954	0.254
BicMix	200	500	1.941 [0.734]	1.01 [0.10]	1.190 [0.814]	0.137	0.949	0.150
MSFRB	200	1000	0.044 [0.012]	0.07 [0.01]	0.050 [0.09]	**0.999**	0.999	**1.000**
MSFA	200	1000	0.196 [0.058]	0.16 [0.09]	0.192 [0.012]	0.965	0.999	0.966
EBMF	200	1000	**0.037 [0.042]**	**0.04 [0.05]**	**0.040 [0.010]**	0.998	**0.999**	0.999
SSLB	200	1000	1.067 [0.361]	0.91 [0.13]	1.088 [0.394]	0.387	0.955	0.382
BicMix	200	1000	1.941 [0.734]	1.01 [0.10]	1.990 [0.814]	0.137	0.949	0.150
MSFRB	200	2000	0.101 [0.271	0.12 [0.23]	0.070 [0.078]	0.998	0.992	0.998
MSFA	200	2000	0.233 [0.067	0.16 [0.12]	0.229 [0.010]	0.946	0.999	0.946
EBMF	200	2000	**0.016 [0.005]**	**0.02 [0.01]**	**0.033[0.013]**	**1.000**	**1.000**	**1.000**
SSLB	200	2000	1.005 [0.367]	0.92 [0.19]	1.076 [0.402]	0.403	0.966	0.393
BicMix	200	2000	2.855 [1.491]	1.12 [0.10]	3.451 [1.824]	0.240	0.737	0.156
MSFRB	400	500	0.057 [0.020]	0.07 [0.02]	0.042 [0.008]	**0.999**	0.999	**1.000**
MSFA	400	500	0.200 [0.123]	0.21 [0.16]	0.149 [0.017]	0.980	0.999	0.981
EBMF	400	500	**0.033 [0.042]**	**0.03 [0.05]**	**0.039 [0.014]**	0.999	**1.000**	0.999
SSLB	400	500	2.976 [1.915]	0.93 [0.15]	3.334 [2.154]	0.183	0.964	0.184
BicMix	400	500	1.748 [0.466]	0.98 [0.11]	1.843 [0.481]	0.087	0.985	0.090
MSFRB	400	1000	0.046 [0.015]	0.06 [0.01]	0.043 [0.12]	**1.000**	1.000	**1.000**
MSFA	400	1000	0.166 [0.028]	0.12 [0.05]	0.171 [0.017]	0.970	0.999	0.971
EBMF	400	1000	**0.022 [0.013]**	**0.03 [0.05]**	**0.040 [0.015]**	0.999	**1.000**	0.999
SSLB	400	1000	1.364 [0.835]	0.92 [0.17]	1.464 [1.211]	0.428	0.951	0.423
BicMix	400	1000	1.940 [0.895]	1.02 [0.09]	2.059 [1.095]	0.134	0.948	0.130
MSFRB	400	2000	**0.030 [0.009]**	**0.05 [0.01]**	0.036 [0.008]	**1.000**	1.000	**1.000**
MSFA	400	2000	0.226 [0.049]	0.12 [0.09]	0.223 [0.008]	0.945	1.000	0.945
EBMF	400	2000	0.043 [0.082]	0.05 [0.11]	**0.027 [0.008]**	0.999	**1.000**	1.000
SSLB	400	2000	1.574 [0.912]	1.00 [0.18]	1.733 [1.024]	0.276	0.948	0.260
BicMix	400	2000	1.492 [0.447]	0.96 [0.08]	1.697 [0.631]	0.203	0.886	0.192

of the methods considered can be viewed as biclustering approaches, or more broadly as methods for identifying shared structure across features and samples (e.g., common pathways), we focus on recovery of the common component. Specifically, we evaluate the estimation accuracy of the matrix of common factor loadings, $\boldsymbol{\Lambda}$, the matrix of common factor scores, $\mathbf{F}$, and their product $\boldsymbol{\Lambda}\mathbf{F}$, which represents the shared signal. To assess performance, we report relative Frobenius distances and RV coefficients, which quantify discrepancies in magnitude and agreement in overall structure, respectively, between the true and estimated matrices.

For all methods, estimated factor loadings and factor scores were first aligned to the

true factors to account for sign and column permutation indeterminacies inherent to factor models. Alignment was performed by matching estimated and true factors based on the combined similarity of their loading and score vectors, followed by sign correction. The common signal was then reconstructed as the product of the aligned estimated loading and factor matrices, $\widehat{\boldsymbol{\Lambda}}\widehat{\boldsymbol{F}}$. Since BicMix allows an overcomplete representation and the simulated common signal contains both sparse and dense factors, all recovered BicMix components were considered during the matching procedure rather than restricting attention to factors classified as sparse in both dimensions. This ensures that recovery is assessed with respect to the complete common signal generated under the simulation design. The results of the simulaton are presented in Table 3.

Overall, MSFRB shows remarkably accurate recovery of the latent common structure across all considered dimensions. Although recovery of the latent structure is a by-product rather than the primary objective of the biclustering model, the relative Frobenius distances for $\boldsymbol{\Lambda}$, $\boldsymbol{F}$, and particularly their product $\boldsymbol{\Lambda F}$ remain very small, with the corresponding RV coefficients close to one. MSFRB performs only slightly worse than EBMF, a highly flexible factor model specifically designed for latent factor estimation, and becomes comparable to EBMF as the dimensionality increases, even outperforming it for the separate recovery of $\boldsymbol{\Lambda}$ and $\boldsymbol{F}$ when $n_s = 400$ and $p = 2000$.

In contrast, the other biclustering methods, SSLB and BicMix, show substantially poorer recovery of the separate latent components and their product. This may partly reflect the complexity of the simulated latent structure, which combines sparse, dense, and overlapping common factors with study-specific and covariate-driven variation, resulting in a more challenging decomposition problem than a purely sparse biclustering setting. In particular, separating the shared common component from the other sources of variation may be difficult for methods primarily designed to identify sparse bicluster structure.

Together with the bicluster recovery results of Section 6.1, these findings illustrate the flexibility of MSFRB: in addition to accurately identifying bicluster structure, it can recover the broader latent structure underlying the data.

7 Real data application

We applied the proposed MSFRB model to a multi-study colorectal cancer gene expression dataset constructed from the R-package `curatedCRCData_mRNA`. The dataset was assembled using a custom preprocessing pipeline designed to harmonise microarray-based mRNA expression studies while retaining only molecular features observed across all selected studies. The pipeline allows the user to specify the expression platform, the minimum number of observations per study, the maximum acceptable fraction of missing values before gene exclusion, and the demographic or clinical variables to retain. After filtering, studies are merged by restricting the analysis to a common set of genes with complete expression values for all observations across all included studies.

Using the pipeline, we selected the $2,000$ most variable common genes. The final dataset comprised eight colorectal cancer studies: `GSE14333_eset` (290 samples), `GSE17536_eset` (177 samples), `GSE17538.GPL570_eset` (232 samples), `GSE2109_eset` (426 samples), `GSE21815_eset` (137 samples), `GSE26682.GPL570_eset` (156 sam-

ples), GSE26682.GPL96_eset (155 samples), and GSE39582_eset (565 samples), yielding a total of $2,138$ samples. Each study contributed measurements for the same set of $2,000$ genes. Study membership was encoded by the variable study_id. Age and gender were retained as observed covariates and were included in the regression component of the model where available.

This dataset provides a natural application for MSFRB because it combines gene expressions collected across multiple independent studies conducted over different time periods and using different experimental protocols and microarray platforms. Such heterogeneity is expected to induce both biological signal shared across studies and study-specific variation due to technical, platform-related, and cohort-specific effects. The MSFRB decomposition is therefore well suited to this setting: common sparse factors are used to identify biclusters of genes and samples that represent reproducible structure across studies, while study-specific factors account for residual heterogeneity and potential batch effects. The inclusion of age and gender allows the shared latent structure to be estimated after adjustment for observed demographic variation.

For this real data analysis we used the CAVI implementation of MSFRB. The maximum number of common factors was set to be $H = 30$ and study-specific factors was set to be $H_s = 20$, with the sparse ESP prior allowing the effective number of active factors to be inferred from the data. The algorithm converged after 198 iterations. A total of six common factors were identified, all satisfying the identifiability condition based on unique support. The numbers of study-specific factors across the eight studies were $5, 8, 11, 8, 9, 3, 14$, and 9, respectively. Biclusters of genes and samples were extracted from the estimated common factors by thresholding the posterior second moments of the factor loadings and factor scores. Active factors were selected using a relative energy criterion, and gene–sample memberships were defined via factor-specific supports.

The extracted biclusters exhibit clear block-like patterns, indicating subsets of genes that are coherently up- or down-regulated within specific groups of patients. These patterns are consistent across studies and suggest that the inferred common factors capture shared sources of biological variation, while allowing for heterogeneity in the remaining samples.

We compared the performance of MSFRB with the same competing methods considered in the simulation studies, namely MSFA, EBMF, SSLB, and BicMix, using the relative Frobenius distance between the observed data and its reconstruction as a common evaluation metric. Although reconstruction accuracy is not the primary objective of the model, this metric provides a convenient way to assess how well each method captures the main structure of the data.

The results are reported in Table 4. MSFRB achieved the lowest overall reconstruction error, with a relative Frobenius distance of 0.099. EBMF performed almost identically, attaining the same overall error and comparable reconstruction accuracy across all studies. In contrast, SSLB and MSFA exhibited substantially larger reconstruction errors, while BicMix performed worse than all other methods. These results suggest that MSFRB and EBMF are better able to capture the dominant sources of variation present in the data than the competing approaches.

Table 4: *Relative Frobenius distance between observed and reconstructed data across studies.*

Model	Study 1	Study 2	Study 3	Study 4	Study 5	Study 6	Study 7	Study 8	Overall
MSFRB	0.131	0.056	0.059	0.116	0.143	0.066	0.062	0.099	0.099
MSFA	0.528	0.297	0.290	0.473	0.557	0.421	0.350	0.416	0.444
EBMF	0.133	0.056	0.060	0.115	0.150	0.069	0.062	0.098	0.099
SSLB	0.243	0.125	0.139	0.209	0.225	0.191	0.103	0.186	0.188
BicMix	0.794	0.734	0.697	0.774	0.800	0.781	0.796	0.711	0.764

8 Discussion

The paper proposes a multi-study factor regression biclustering (MSFRB) model for the comprehensive analysis of high-dimensional molecular data collected across multiple studies. The model integrates multi-study factor analysis, factor regression, and biclustering within a unified Bayesian framework, enabling simultaneous modelling of shared latent structure as well as accounting for study-specific heterogeneity and observed covariate effects. Such a combination has not been considered within a single coherent modelling framework, providing a flexible approach to integrative analysis in complex multi-study settings.

A central component of the proposed framework is the use of a sparse exchangeable shrinkage process (ESP) prior on both common factor loadings and factor scores. This prior induces sparsity at multiple levels and supports automatic factor selection, allowing the model to adapt to both sparse and moderately dense latent structures. While variational inference yields continuous approximations, the shrinkage mechanism leads to a clear practical separation between active and negligible components, which can be formalised through simple post-processing.

An additional contribution of this work is a principled approach to factor identification. Building on the structure induced by the model, we characterise active factors through their posterior energy and assess their identifiability via uniqueness of support across features and samples. This provides an interpretable criterion for distinguishing shared and study-specific components and complements the biclustering interpretation of the model. To enable scalable inference under the ESP prior, we developed a variational inference framework, including both coordinate ascent and stochastic variational algorithms. In particular, we derive mean-field variational updates for the ESP prior, enabling tractable and efficient approximation of its multi-level sparsity structure. The resulting procedures exploit the conditional structure of the model and scale well to high-dimensional settings.

Simulation studies and real data applications demonstrate that the MSFRB model accurately recovers latent structure in most settings, effectively separates shared and study-specific sources of variation, and reliably estimates covariate effects across a range of problem dimensions. These results highlight the advantages of the proposed framework over existing multi-study factor analysis and biclustering approaches.

The MSFRB model provides a flexible and principled tool for integrative analysis in multi-study settings. Several extensions can be developed within this framework. In particular, the Gaussian observation model may be relaxed to accommodate non-Gaussian response distributions, and additional structure—such as dependence or grouping among

latent factors—can be incorporated through structured shrinkage priors (in the spirit of (Schiavon et al., 2022)). These extensions can be pursued while retaining the factor regression structure and multi-study decomposition underlying the model.

Acknowledgements

The authors thank Bettina Grün for her helpful discussion and suggestions, and Antonio Canale and for the insightful comments regarding information switching.

The authors gratefully acknowledge the United Kingdom Research and Innovation Medical Research Council grants MR/W029790/1 (V.Z., L.B.), and Marmaduke Sheild Fund (L.B.). This research was supported by the NIHR Cambridge Biomedical Research Centre (NIHR203312) (L.B.). The views expressed are those of the authors and not necessarily those of the NIHR or the Department of Health and Social Care.

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Supplementary Material for: "Multi-study factor regression biclustering"

A Variational inference algorithms for multi-study factor regression biclustering

In this section, we provide the details of the CAVI and SVI algorithms outlined in the main paper.

A.1 CAVI algorithm for MSFRB

Algorithm A.1: Details of the CAVI algorithm steps for the MSFRB model

1: Set initial variational parameters

Repeat until converged

2: Update ξ_{js}^2 for $j = 1, \ldots, p$ and $s = 1, \ldots, S$ from $q^*(\xi_{js}^2) = \mathcal{G}^{-1}(\hat{a}_s^\xi, \hat{b}_{js}^\xi)$, where

$$\hat{a}_s^\xi = a_\xi + \frac{n_s}{2},$$
$$\hat{b}_{js}^\xi = \frac{\hat{a}^{b_\xi}}{\hat{b}_{js}^{b_\xi}} + \frac{1}{2}\sum_{i=1}^{n_s} \mathbb{E}\left(y_{jis} - \boldsymbol{\lambda}_j \boldsymbol{f}_{is}^\top - \boldsymbol{\lambda}_{js}\boldsymbol{\eta}_{is}^\top - \boldsymbol{B}_j^X \boldsymbol{x}_{is}\right)^2,$$

where

$$\mathbb{E}\left(y_{jis} - \boldsymbol{\lambda}_j \boldsymbol{f}_{is}^\top - \boldsymbol{\lambda}_{js}\boldsymbol{\eta}_{is}^\top - \boldsymbol{B}_j^X \boldsymbol{x}_{is}\right)^2 = y_{jis}^2 - 2y_{jis}\left(\boldsymbol{\mu}_j^\lambda \left[\boldsymbol{\mu}_{is}^f\right]^\top + \boldsymbol{\mu}_{is}^{\lambda^s}\left[\boldsymbol{\mu}_{is}^\eta\right]^\top + \boldsymbol{\mu}_j^{\beta^X}\boldsymbol{x}_{is}\right) +$$
$$2\left(\boldsymbol{\mu}_j^\lambda \left[\boldsymbol{\mu}_{is}^f\right]^\top \boldsymbol{\mu}_{is}^{\lambda^s}\left[\boldsymbol{\mu}_{is}^\eta\right]^\top + \boldsymbol{\mu}_j^\lambda \left[\boldsymbol{\mu}_{is}^f\right]^\top \boldsymbol{\mu}_j^{\beta^X}\boldsymbol{x}_{is} + \boldsymbol{\mu}_{is}^{\lambda^s}\left[\boldsymbol{\mu}_{is}^\eta\right]^\top \boldsymbol{\mu}_j^{\beta^X}\boldsymbol{x}_{is}\right) +$$
$$\boldsymbol{\mu}_j^{\beta^X}\boldsymbol{x}_{is}\left[\boldsymbol{x}_{is}\right]^\top \left[\boldsymbol{\mu}_j^{\beta^X}\right]^\top + \boldsymbol{\mu}_j^\lambda \left[\boldsymbol{\mu}_{is}^f\right]^\top \boldsymbol{\mu}_{is}^f \left[\boldsymbol{\mu}_j^\lambda\right]^\top + \boldsymbol{\mu}_{is}^{\lambda^s}\left[\boldsymbol{\mu}_{is}^\eta\right]^\top \boldsymbol{\mu}_{is}^\eta \left[\boldsymbol{\mu}_{is}^{\lambda^s}\right]^\top +$$
$$\boldsymbol{\mu}_j^\lambda \boldsymbol{\Sigma}_{is}^f \left[\boldsymbol{\mu}_j^\lambda\right]^\top + \left[\boldsymbol{\mu}_{is}^f\right]^\top \boldsymbol{\Sigma}_j^\lambda \boldsymbol{\mu}_{is}^f + \boldsymbol{\mu}_{is}^{\lambda^s}\boldsymbol{\Sigma}_{is}^\eta \left[\boldsymbol{\mu}_{is}^{\lambda^s}\right]^\top + \left[\boldsymbol{\mu}_{is}^\eta\right]^\top \boldsymbol{\Sigma}_{is}^{\lambda^s}\boldsymbol{\mu}_{is}^\eta + \mathrm{Tr}\left(\boldsymbol{\Sigma}_j^\lambda \boldsymbol{\Sigma}_{is}^f\right) + \mathrm{Tr}\left(\boldsymbol{\Sigma}_{is}^{\lambda^s}\boldsymbol{\Sigma}_{is}^\eta\right)$$

and the mean of $\mathbb{E}(\xi_{is}^{-2}) = \mathbb{E}(\frac{1}{\xi_{js}^2}) = \hat{a}_{js}^\xi / \hat{b}_{js}^\xi$.

3: Update $b_{\xi js}$ for $j = 1, \ldots, p$ and $s = 1, \ldots, S$ from $q^*(b_{\xi js}) = \mathcal{G}(\hat{a}^{b_\xi}, \hat{b}_{js}^{b_\xi})$, where

$$\hat{a}^{b_\xi} = a_g + a_\xi,$$
$$\hat{b}_{js}^{b_\xi} = b_{gjs} + \mathbb{E}(\frac{1}{\xi_{js}^2}) = b_{gjs} + \frac{\hat{a}_s^\xi}{\hat{b}_{js}^\xi},$$

where $\mathbb{E}(b_{\xi js}) = \mu_{is}^{b_\xi} = \hat{a}^{b_\xi} / \hat{b}_{js}^{b_\xi}$.

4: Update $\theta_h^\Lambda \mid \nu_h$ for $h = 1, \ldots, H$ from

- $q^*(\theta_h^\Lambda \mid \nu_h = 1) = \mathcal{G}^{-1}(\hat{a}_{\theta_h^\Lambda slab}, \hat{b}_{\theta_h^\Lambda slab})$, with

$$\hat{a}_{\theta_h^\Lambda slab} = a_\Lambda + \frac{p}{2},$$
$$\hat{b}_{\theta_h^\Lambda slab} = \frac{\hat{a}_{b_\theta}}{\hat{b}_{b_\theta}} + \frac{1}{2}\frac{\hat{a}_\kappa}{\hat{b}_\kappa}\tilde{S}_h^\Lambda,$$

- $q^*(\theta_h^\Lambda \mid \nu_h = 0) = \mathcal{G}^{-1}(\hat{a}_{\theta_h^\Lambda spike}, \hat{b}_{\theta_h^\Lambda spike})$, with

$$\hat{a}_{\theta_h^\Lambda spike} = a_0^\Lambda + \frac{p}{2},$$
$$\hat{b}_{\theta_h^\Lambda spike} = \frac{\hat{a}_{b_0^\Lambda}}{\hat{b}_{b_0^\Lambda}} + \frac{1}{2}\frac{\hat{a}_\kappa}{\hat{b}_\kappa}\tilde{S}_h^\Lambda,$$

where $\tilde{S}_h^\Lambda = \sum_{j=1}^p \left[\mathbb{E}\lambda_{jh}^2 \mathbb{E}\left(\frac{1}{\phi_{jh}^\Lambda}\right)\right] = \sum_{j=1}^p \left[\left(\left[\boldsymbol{\mu}_{jh}^\lambda\right]^2 + \boldsymbol{\Sigma}_{j[h,h]}^\lambda\right)\frac{\hat{a}_\phi}{\hat{b}_{\phi_{jh}^\Lambda}}\right]$, with h and $[h,h]$ indicating the hth element of the mean vector and the $[h,h]$th element of the covariance matrix $\boldsymbol{\Sigma}_j^\Lambda$, correspondingly.
Calculate weighted mixture of expectations for $h = 1, \ldots, H$

$$\mathbb{E}_{mix}\left(\frac{1}{\theta_h^\Lambda}\right) = r_h \frac{\hat{a}_{\theta_h^\Lambda slab}}{\hat{b}_{\theta_h^\Lambda slab}} + (1 - r_h)\frac{\hat{a}_{\theta_h^\Lambda spike}}{\hat{b}_{\theta_h^\Lambda spike}}.$$

5: Update responsibilities for $h = 1, \ldots, H$ from $q^*(\nu_h) = \mathrm{Ber}(r_h)$. First calculate

$$l_{h\,slab} = \mathbb{E}\log\tau_h^\theta + a_\Lambda \mathbb{E}\log b_\Lambda - \log\Gamma(a_\Lambda) - (a_\Lambda + \frac{p}{2} + 1)\mathbb{E}_{slab}\log\theta_h^\Lambda - \mathbb{E}_{slab}\left(\frac{1}{\theta_h^\Lambda}\right)\left(\mathbb{E}b_\Lambda + \frac{1}{2}\mathbb{E}\left(\frac{1}{\kappa}\right)\tilde{S}_h^\Lambda\right),$$

$$l_{h\,spike} = \mathbb{E}\log(1 - \tau_h^\theta) + a_0^\Lambda \mathbb{E}\log b_0^\Lambda - \log\Gamma(a_0^\Lambda) - (a_0^\Lambda + \frac{p}{2} + 1)\mathbb{E}_{spike}\log\theta_h^\Lambda - \mathbb{E}_{spike}\left(\frac{1}{\theta_h^\Lambda}\right)\left(\mathbb{E}b_0^\Lambda + \frac{1}{2}\mathbb{E}\left(\frac{1}{\kappa}\right)\tilde{S}_h^\Lambda\right),$$

where

$$\mathbb{E}\log\tau_h^\theta = \psi(\hat{a}_{\tau_h^\theta}) - \psi(\hat{a}_{\tau_h^\theta} + \hat{b}_{\tau_h^\theta}),$$
$$\mathbb{E}\log(1 - \tau_h^\theta) = \psi(\hat{b}_{\tau_h^\theta}) - \psi(\hat{a}_{\tau_h^\theta} + \hat{b}_{\tau_h^\theta}),$$
$$\mathbb{E}\log b_\Lambda = \psi(\hat{a}_{b_\theta}) - \ln\hat{b}_{b_\theta},$$
$$\mathbb{E}\log b_0^\Lambda = \psi(\hat{a}_{b_0^\Lambda}) - \ln\hat{b}_{b_0^\Lambda},$$
$$\mathbb{E}_{slab}\log\theta_h^\Lambda = \ln\hat{b}_{\theta_h^\Lambda slab} - \psi(\hat{a}_{\theta_h^\Lambda slab}),$$
$$\mathbb{E}_{spike}\log\theta_h^\Lambda = \ln\hat{b}_{\theta_h^\Lambda spike} - \psi(\hat{a}_{\theta_h^\Lambda spike}),$$
$$\mathbb{E}_{slab}\left(\frac{1}{\theta_h^\Lambda}\right) = \frac{\hat{a}_{\theta_h^\Lambda slab}}{\hat{b}_{\theta_h^\Lambda slab}},$$

$$\mathbb{E}_{spike}\left(\frac{1}{\theta_h^\Lambda}\right) = \frac{\hat{a}_{\theta_h^\Lambda spike}}{\hat{b}_{\theta_h^\Lambda spike}},$$
$$\mathbb{E}\left(\frac{1}{\kappa}\right) = \frac{\hat{a}_\kappa}{\hat{b}_\kappa},$$
$$\tilde{S}_h^\Lambda = \sum_{j=1}^{p}\left[\mathbb{E}\lambda_{jh}^2\mathbb{E}\left(\frac{1}{\phi_{jh}^\Lambda}\right)\right] = \sum_{j=1}^{p}\left[\left(\left[\boldsymbol{\mu}_{jh}^\lambda\right]^2 + \boldsymbol{\Sigma}_{j[h,h]}^\lambda\right)\frac{\hat{a}_\phi}{\hat{b}_{\phi_{jh}^\Lambda}}\right].$$

Calculate the responsibility of slab r_h as

$$r_h = \frac{1}{1+\exp(l_{h\,spike} - l_{h\,slab})}.$$

6: Update τ_h^θ for $h = 1, \ldots, H$ from $q^*(\tau_h^\theta) = \mathcal{B}(\hat{a}_{\tau_h^\theta}, \hat{b}_{\tau_h^\theta})$, where

$$\hat{a}_{\tau_h^\theta} = \frac{\hat{a}_\alpha^\Lambda}{\hat{b}_\alpha^\Lambda H} + r_h,$$
$$\hat{b}_{\tau_h^\theta} = 2 - r_h.$$

7: Update α_Λ from $q^*(\alpha_\Lambda) = \mathcal{G}(\hat{a}_\alpha^\Lambda, \hat{b}_\alpha^\Lambda)$, where

$$\hat{a}_\alpha^\Lambda = a_\alpha^\Lambda + H,$$
$$\hat{b}_\alpha^\Lambda = b_\alpha^\Lambda - \frac{1}{H}\sum_{h=1}^{H}\mathbb{E}\log\tau_h^\theta,$$

with $\mathbb{E}\log\tau_h^\theta = \psi(\hat{a}_{\tau_h^\theta}) - \psi(\hat{a}_{\tau_h^\theta} + \hat{b}_{\tau_h^\theta})$.

8: Update ϕ_{jh}^Λ for $j = 1, \ldots, p$ and $h = 1, \ldots, H$ from $q^*(\phi_{jh}^\Lambda) = \mathcal{G}^{-1}(\hat{a}_\phi, \hat{b}_{\phi_{jh}^\Lambda})$, where

$$\hat{a}_\phi = a_\phi + \frac{1}{2},$$
$$\hat{b}_{\phi_{jh}^\Lambda} = \frac{\hat{a}_{\beta_\phi}}{\hat{b}_{\beta_{j\phi}}} + \frac{1}{2}\frac{\hat{a}_\kappa}{\hat{b}_\kappa}\mathbb{E}_{mix}\left(\frac{1}{\theta_h^\Lambda}\right)\mathbb{E}\lambda_{jh}^2,$$

with $\mathbb{E}_{mix}\left(\frac{1}{\theta_h^\Lambda}\right) = r_h\frac{\hat{a}_{\theta_h^\Lambda slab}}{\hat{b}_{\theta_h^\Lambda slab}} + (1-r_h)\frac{\hat{a}_{\theta_h^\Lambda spike}}{\hat{b}_{\theta_h^\Lambda spike}}$ and $\mathbb{E}\lambda_{jh}^2 = \left[\boldsymbol{\mu}_{jh}^\lambda\right]^2 + \boldsymbol{\Sigma}_{j[h,h]}^\lambda$.

9: Update κ^Λ from $q^*(\kappa) = \mathcal{G}^{-1}(\hat{a}_\kappa, \hat{b}_\kappa)$, where

$$\hat{a}_\kappa = a^\kappa + \frac{1}{2}pH,$$
$$\hat{b}_\kappa = b^\kappa + \frac{1}{2}\sum_{h=1}^{H}\left[\mathbb{E}_{mix}\left(\frac{1}{\theta_h^\Lambda}\right)\tilde{S}_h^\Lambda\right],$$

with $\mathbb{E}_{mix}\left(\frac{1}{\theta_h^\Lambda}\right) = r_h\frac{\hat{a}_{\theta_h^\Lambda slab}}{\hat{b}_{\theta_h^\Lambda slab}} + (1-r_h)\frac{\hat{a}_{\theta_h^\Lambda spike}}{\hat{b}_{\theta_h^\Lambda spike}}$ and $\tilde{S}_h^\Lambda = \sum_{j=1}^{p}\left[\left(\left[\boldsymbol{\mu}_{jh}^\lambda\right]^2 + \boldsymbol{\Sigma}_{j[h,h]}^\lambda\right)\frac{\hat{a}_\phi}{\hat{b}_{\phi_{jh}^\Lambda}}\right]$.

10: Update $\lambda_j^\top$ for $j = 1, \ldots, p$ from $q^*(\lambda_j) = \mathcal{N}_H(\boldsymbol{\mu}_j^\lambda, \boldsymbol{\Sigma}_j^\lambda)$, where

$$\boldsymbol{\Sigma}_j^\lambda = \left(\boldsymbol{\Psi}_j^{-1} + \sum_{s=1}^{S} \frac{\hat{a}_s^\xi}{\hat{b}_{js}^\xi} \sum_{i=1}^{n_s} \left(\boldsymbol{\mu}_{is}^f \left[\boldsymbol{\mu}_{is}^f \right]^\top + \boldsymbol{\Sigma}_{is}^f \right) \right)^{-1},$$

$$\boldsymbol{\mu}_j^\lambda = \boldsymbol{\Sigma}_j^\lambda \sum_{s=1}^{S} \frac{\hat{a}_s^\xi}{\hat{b}_{js}^\xi} \left(\boldsymbol{\mu}_{1s}^f, \ldots, \boldsymbol{\mu}_{n_s s}^f \right) \left(\boldsymbol{y}_{is}^\top - \left(\boldsymbol{\mu}_{1s}^\eta, \ldots, \boldsymbol{\mu}_{n_s s}^\eta \right) \left[\boldsymbol{\mu}_{js}^\lambda \right]^\top - \boldsymbol{X}_s^\top \left[\boldsymbol{\mu}_j^{\beta^X} \right]^\top \right),$$

and where $\boldsymbol{\Psi}_j^{-1} = \operatorname{diag}\left(\frac{\hat{a}_\kappa}{\hat{b}_\kappa} \frac{\hat{a}_\phi}{\hat{b}_{\phi_{j1}}} \mathbb{E}_{mix}\left(\frac{1}{\theta_1}\right), \ldots, \frac{\hat{a}_\kappa}{\hat{b}_\kappa} \frac{\hat{a}_\phi}{\hat{b}_{\phi_{jH}}} \mathbb{E}_{mix}\left(\frac{1}{\theta_H}\right) \right)$ with $\mathbb{E}_{mix}\left(\frac{1}{\theta_h^\Lambda}\right) = r_h \frac{\hat{a}_{\theta_h^\Lambda slab}}{\hat{b}_{\theta_h^\Lambda slab}} + (1 - r_h) \frac{\hat{a}_{\theta_h^\Lambda spike}}{\hat{b}_{\theta_h^\Lambda spike}}$.

11: Update b_Λ from $q^*(b_\Lambda) = \mathcal{G}(\hat{a}_{b_\theta}, \hat{b}_{b_\theta})$, where

$$\hat{a}_{b_\theta} = a_1^\Lambda + a_\Lambda \sum_{h=1}^{H} r_h,$$

$$\hat{b}_{b_\theta} = b_1^\Lambda + \sum_{h=1}^{H} r_h \frac{\hat{a}_{\theta_h^\Lambda slab}}{\hat{b}_{\theta_h^\Lambda slab}}.$$

12: Update b_0^Λ from $q^*(b_0^\Lambda) = \mathcal{G}(\hat{a}_{b_0^\Lambda}, \hat{b}_{b_0^\Lambda})$, where

$$\hat{a}_{b_0^\Lambda} = a_2^\Lambda + a_0^\Lambda \sum_{h=1}^{H} (1 - r_h),$$

$$\hat{b}_{b_0^\Lambda} = b_2^\Lambda + \sum_{h=1}^{H} (1 - r_h) \frac{\hat{a}_{\theta_h^\Lambda spike}}{\hat{b}_{\theta_h^\Lambda spike}}.$$

13: Update $\beta_{j\phi}$ for $j = 1, \ldots, p$ from $q^*(\beta_{j\phi}) = \mathcal{G}(\hat{a}_{\beta_\phi}, \hat{b}_{\beta_{j\phi}})$, where

$$\hat{a}_{\beta_\phi} = c + H a_\phi,$$

$$\hat{b}_{\beta_{j\phi}} = d + \sum_{h=1}^{H} \frac{\hat{a}_\phi}{\hat{b}_{\phi_{jh}^\Lambda}}.$$

14: Update $\theta_h^\Lambda \mid \upsilon_h$ for $h = 1, \ldots, H$ from

- $q^*(\theta_h^\Lambda \mid \upsilon_h = 1) = \mathcal{G}^{-1}(\hat{a}_{\theta_h^\Lambda slab}, \hat{b}_{\theta_h^\Lambda slab})$, with

$$\hat{a}_{\theta_h^\Lambda slab} = a_F + \frac{N}{2},$$

$$\hat{b}_{\theta_h^\Lambda slab} = \frac{\hat{a}_{b_\gamma}}{\hat{b}_{b_\gamma}} + \frac{1}{2} \frac{\hat{a}_\nu}{\hat{b}_\nu} \tilde{S}_h^F,$$

- $q^*(\theta_h^\Lambda \mid \upsilon_h = 0) = \mathcal{G}^{-1}(\hat{a}_{\theta_h^\Lambda spike}, \hat{b}_{\theta_h^\Lambda spike})$, with

$$\hat{a}_{\theta_h^\Lambda spike} = a_0^F + \frac{N}{2},$$

$$\hat{b}_{\theta_h^\Lambda spike} = \frac{\hat{a}_{b_0^F}}{\hat{b}_{b_0^F}} + \frac{1}{2}\frac{\hat{a}_\nu}{\hat{b}_\nu}\tilde{S}_h^F,$$

where $\tilde{S}_h^F = \sum_{s=1}^{S}\sum_{i=1}^{n_s}\left[\mathbb{E}f_{his}^2\mathbb{E}\left(\frac{1}{\phi_{his}^F}\right)\right] = \sum_{s=1}^{S}\sum_{i=1}^{n_s}\left[\left(\left[\mu_{his}^f\right]^2 + \mathbf{\Sigma}_{ns[h,h]}^f\right)\frac{\hat{a}_\zeta}{\hat{b}_{\phi_{his}^F}}\right]$, with h and $[h,h]$ indicating the hth element of the mean vector and the $[h,h]$th element of the covariance matrix $\mathbf{\Sigma}_{is}^f$, correspondingly.

Calculate weighted mixture of expectations for $h = 1, \ldots, H$

$$\mathbb{E}_{mix}\left(\frac{1}{\theta_h^\Lambda}\right) = z_h\frac{\hat{a}_{\theta_h^\Lambda slab}}{\hat{b}_{\theta_h^\Lambda slab}} + (1 - z_h)\frac{\hat{a}_{\theta_h^\Lambda spike}}{\hat{b}_{\theta_h^\Lambda spike}}.$$

15: Update responsibilities for $h = 1, \ldots, H$ from $q^*(\upsilon_h) = \mathrm{Ber}(z_h)$. First calculate

$$m_{h\,slab} = \mathbb{E}\log\pi_h + a_F\mathbb{E}\log b_F - \log\Gamma(a_F) - (a_F + \frac{N}{2} + 1)\mathbb{E}_{slab}\log\theta_h^\Lambda - \mathbb{E}_{slab}\left(\frac{1}{\theta_h^\Lambda}\right)\left(\mathbb{E}b_F + \frac{1}{2}\mathbb{E}\left(\frac{1}{\nu}\right)\tilde{S}_h^F\right),$$

$$m_{h\,spike} = \mathbb{E}\log(1-\pi_h) + a_0^F\mathbb{E}\log b_0^F - \log\Gamma(a_0^F) - (a_0^F + \frac{N}{2} + 1)\mathbb{E}_{spike}\log\theta_h^\Lambda - \mathbb{E}_{spike}\left(\frac{1}{\theta_h^\Lambda}\right)\left(\mathbb{E}b_0^F + \frac{1}{2}\mathbb{E}\left(\frac{1}{\nu}\right)\tilde{S}_h^F\right),$$

where

$$\mathbb{E}\log\pi_h = \psi(\hat{a}_{\pi_h}) - \psi(\hat{a}_{\pi_h} + \hat{b}_{\pi_h}),$$
$$\mathbb{E}\log(1-\pi_h) = \psi(\hat{b}_{\pi_h}) - \psi(\hat{a}_{\pi_h} + \hat{b}_{\pi_h}),$$
$$\mathbb{E}\log b_F = \psi(\hat{a}_{b_\gamma}) - \ln\hat{b}_{b_\gamma},$$
$$\mathbb{E}\log b_0^F = \psi(\hat{a}_{b_0^F}) - \ln\hat{b}_{b_0^F},$$
$$\mathbb{E}_{slab}\log\theta_h^\Lambda = \ln\hat{b}_{\theta_h^\Lambda slab} - \psi(\hat{a}_{\theta_h^\Lambda slab}),$$
$$\mathbb{E}_{spike}\log\theta_h^\Lambda = \ln\hat{b}_{\theta_h^\Lambda spike} - \psi(\hat{a}_{\theta_h^\Lambda spike}),$$
$$\mathbb{E}_{slab}\left(\frac{1}{\theta_h^\Lambda}\right) = \frac{\hat{a}_{\theta_h^\Lambda slab}}{\hat{b}_{\theta_h^\Lambda slab}},$$
$$\mathbb{E}_{spike}\left(\frac{1}{\theta_h^\Lambda}\right) = \frac{\hat{a}_{\theta_h^\Lambda spike}}{\hat{b}_{\theta_h^\Lambda spike}},$$
$$\mathbb{E}\left(\frac{1}{\nu}\right) = \frac{\hat{a}_\nu}{\hat{b}_\nu},$$
$$\tilde{S}_h^F = \sum_{s=1}^{S}\sum_{i=1}^{n_s}\left[\mathbb{E}f_{his}^2\mathbb{E}\left(\frac{1}{\phi_{his}^F}\right)\right] = \sum_{s=1}^{S}\sum_{i=1}^{n_s}\left[\left(\left[\mu_{his}^f\right]^2 + \mathbf{\Sigma}_{ns[h,h]}^f\right)\frac{\hat{a}_\zeta}{\hat{b}_{\phi_{his}^F}}\right].$$

Calculate the responsibility of slab z_h as

$$z_h = \frac{1}{1 + \exp(m_{h\,spike} - m_{h\,slab})}.$$

16: Update π_h for $h = 1, \ldots, H$ from $q^*(\pi_h) = \mathcal{B}(\hat{a}_{\pi_h}, \hat{b}_{\pi_h})$, where

$$\hat{a}_{\pi_h} = \frac{\hat{a}_\alpha^F}{\hat{b}_\alpha^F H} + z_h,$$
$$\hat{b}_{\pi_h} = 2 - z_h.$$

17: Update α_γ from $q^*(\alpha_\gamma) = \mathcal{G}(\hat{a}_\alpha^F, \hat{b}_\alpha^F)$, where

$$\hat{a}_\alpha^F = a_\alpha^F + H,$$
$$\hat{b}_\alpha^F = b_\alpha^F - \frac{1}{H}\sum_{h=1}^{H} \mathbb{E}\log \pi_h,$$

with $\mathbb{E}\log \pi_h = \psi(\hat{a}_{\pi_h}) - \psi(\hat{a}_{\pi_h} + \hat{b}_{\pi_h})$.

18: Update ϕ_{his}^F for $s = 1, \ldots, S, i = 1, \ldots, n_s$ and $h = 1, \ldots, H$ from $q^*(\phi_{his}^F) = \mathcal{G}^{-1}(\hat{a}_\zeta, \hat{b}_{\phi_{his}^F})$, where

$$\hat{a}_\zeta = a_\zeta + \frac{1}{2},$$
$$\hat{b}_{\phi_{his}^F} = \frac{\hat{a}_{\beta_\zeta}}{\hat{b}_{\beta_{ns\zeta}}} + \frac{1}{2}\frac{\hat{a}_\nu}{\hat{b}_\nu}\mathbb{E}_{mix}\left(\frac{1}{\theta_h^\Lambda}\right)\mathbb{E}f_{his}^2,$$

with $\mathbb{E}_{mix}\left(\frac{1}{\theta_h^\Lambda}\right) = z_h \frac{\hat{a}_{\theta_h^\Lambda slab}}{\hat{b}_{\theta_h^\Lambda slab}} + (1 - z_h)\frac{\hat{a}_{\theta_h^\Lambda spike}}{\hat{b}_{\theta_h^\Lambda spike}}$ and $\mathbb{E}f_{his}^2 = \left[\mu_{his}^f\right]^2 + \mathbf{\Sigma}_{[h,h]ns}^f$.

19: Update κ^F from $q^*(\nu) = \mathcal{G}^{-1}(\hat{a}_\nu, \hat{b}_\nu)$, where

$$\hat{a}_\nu = c^\kappa + \frac{1}{2}HN,$$
$$\hat{b}_\nu = d^\kappa + \frac{1}{2}\sum_{h=1}^{H}\left[\mathbb{E}_{mix}\left(\frac{1}{\theta_h^\Lambda}\right)\tilde{S}_h^F\right],$$

with $\mathbb{E}_{mix}\left(\frac{1}{\theta_h^\Lambda}\right) = z_h \frac{\hat{a}_{\theta_h^\Lambda slab}}{\hat{b}_{\theta_h^\Lambda slab}} + (1 - z_h)\frac{\hat{a}_{\theta_h^\Lambda spike}}{\hat{b}_{\theta_h^\Lambda spike}}$ and $\tilde{S}_h^F = \sum_{s=1}^{S}\sum_{i=1}^{n_s}\left[\left(\left[\mu_{his}^f\right]^2 + \mathbf{\Sigma}_{ns[h,h]}^f\right)\frac{\hat{a}_\zeta}{\hat{b}_{\phi_{his}^F}}\right]$.

20: Update $\boldsymbol{f}_{is}$ for $i = 1, \ldots, n_s$ and $s = 1, \ldots, S$ from $q^*(\boldsymbol{f}_{is}) = \mathcal{N}_H(\boldsymbol{\mu}_{is}^f, \mathbf{\Sigma}_{is}^f)$, where

$$\mathbf{\Sigma}_{is}^f = \left(\mathbf{\Omega}_{is}^{-1} + \sum_{j=1}^{p}\frac{\hat{a}_s^\xi}{\hat{b}_{js}^\xi}\left(\boldsymbol{\mu}_j^\lambda\left[\boldsymbol{\mu}_j^\lambda\right]^\top + \mathbf{\Sigma}_j^\lambda\right)\right)^{-1},$$

$$\boldsymbol{\mu}_{is}^f = \mathbf{\Sigma}_{is}^f\left(\boldsymbol{\mu}_1^\lambda, \ldots, \boldsymbol{\mu}_p^\lambda\right)^\top \operatorname{diag}\left(\frac{\hat{a}_s^\xi}{\hat{b}_{1s}^\xi}, \ldots, \frac{\hat{a}_s^\xi}{\hat{b}_{ps}^\xi}\right)\left(\boldsymbol{y}_{is} - \left(\boldsymbol{\mu}_{1s}^\lambda, \ldots, \boldsymbol{\mu}_{ps}^\lambda\right)\boldsymbol{\mu}_{is}^\eta - \left(\boldsymbol{\mu}_1^{\beta^X}, \ldots, \boldsymbol{\mu}_p^{\beta^X}\right)\boldsymbol{x}_{is}\right),$$

and where $\mathbf{\Omega}_{is}^{-1} = \operatorname{diag}\left(\frac{\hat{a}_\nu}{\hat{b}_\nu}\frac{\hat{a}_\zeta}{\hat{b}_{\zeta_{1ns}}}\mathbb{E}_{mix}\left(\frac{1}{\gamma_1}\right), \ldots, \frac{\hat{a}_\nu}{\hat{b}_\nu}\frac{\hat{a}_\zeta}{\hat{b}_{\zeta_{Hns}}}\mathbb{E}_{mix}\left(\frac{1}{\gamma_H}\right)\right)$ with $\mathbb{E}_{mix}\left(\frac{1}{\theta_h^\Lambda}\right) = z_h \frac{\hat{a}_{\theta_h^\Lambda slab}}{\hat{b}_{\theta_h^\Lambda slab}} + (1 - z_h)\frac{\hat{a}_{\theta_h^\Lambda spike}}{\hat{b}_{\theta_h^\Lambda spike}}$.

21: Update b_F from $q^*(b_F) = \mathcal{G}(\hat{a}_{b_\gamma}, \hat{b}_{b_\gamma})$, where

$$\hat{a}_{b_\gamma} = a_1^F + a_\Lambda \sum_{h=1}^{H} z_h,$$
$$\hat{b}_{b_\gamma} = b_1^F + \sum_{h=1}^{H} z_h \frac{\hat{a}_{\theta_h^\Lambda slab}}{\hat{b}_{\theta_h^\Lambda slab}}.$$

22: Update b_0^F from $q^*(b_0^F) = \mathcal{G}(\hat{a}_{b_0^F}, \hat{b}_{b_0^F})$, where

$$\hat{a}_{b_0^F} = a_2^F + a_0^F \sum_{h=1}^{H} (1 - z_h),$$
$$\hat{b}_{b_0^F} = b_2^F + \sum_{h=1}^{H} (1 - z_h) \frac{\hat{a}_{\theta_h^\Lambda spike}}{\hat{b}_{\theta_h^\Lambda spike}}.$$

23: Update $\beta_{ns\zeta}$ for $i = 1, \ldots, n_s$ and $s = 1, \ldots, S$ from $q^*(\beta_{ns\zeta}) = \mathcal{G}(\hat{a}_{\beta_\zeta}, \hat{b}_{\beta_{ns\zeta}})$, where

$$\hat{a}_{\beta_\zeta} = e + H a_\zeta,$$
$$\hat{b}_{\beta_{ns\zeta}} = w + \sum_{h=1}^{H} \frac{\hat{a}_\zeta}{\hat{b}_{\phi_{his}^F}}.$$

24: Update $\theta_{hs} \mid \nu_{hs}$ for $h = 1, \ldots, H_s$ and $s = 1, \ldots, S$ from

- $q^*(\theta_{hs} \mid \nu_{hs} = 1) = \mathcal{G}^{-1}(\hat{a}_{\theta_{hs} slab}, \hat{b}_{\theta_{hs} slab})$, with

$$\hat{a}_{\theta_{hs} slab} = a_\Lambda^S + \frac{p}{2},$$
$$\hat{b}_{\theta_{hs} slab} = \frac{\hat{a}_{b_\Lambda^S}}{\hat{b}_{b_\Lambda^S}} + \frac{1}{2} \tilde{S}_{hs}^\Lambda,$$

- $q^*(\theta_{hs} \mid \nu_{hs} = 0) = \mathcal{G}^{-1}(\hat{a}_{\theta_{hs} spike}, \hat{b}_{\theta_{hs} spike})$, with

$$\hat{a}_{\theta_{hs} spike} = a_0^{S\Lambda} + \frac{p}{2},$$
$$\hat{b}_{\theta_{hs} spike} = \frac{\hat{a}_{b_0^{S\Lambda}}}{\hat{b}_{b_0^{S\Lambda}}} + \frac{1}{2} \tilde{S}_{hs}^\Lambda,$$

where $\tilde{S}_{hs}^\Lambda = \sum_{j=1}^{p} \left[\mathbb{E}\lambda_{ihs}^2 \mathbb{E}\left(\frac{1}{\phi_{jhs}}\right) \right] = \sum_{j=1}^{p} \left[\left(\left[\boldsymbol{\mu}_{jhs}^\lambda \right]^2 + \boldsymbol{\Sigma}_{js[h,h]}^\lambda \right) \frac{\hat{a}_{\phi s}}{\hat{b}_{\phi_{jhs}}} \right]$, with h and $[h, h]$ indicating the hth element of the mean vector and the $[h, h]$th element of the covariance matrix $\boldsymbol{\Sigma}_{is}^\Lambda$, correspondingly.

Calculate weighted mixture of expectations for $h = 1, \ldots, H_s$ and $s = 1, \ldots, S$

$$\mathbb{E}_{mix}\left(\frac{1}{\theta_{hs}}\right) = r_{hs} \frac{\hat{a}_{\theta_{hs} slab}}{\hat{b}_{\theta_{hs} slab}} + (1 - r_{hs}) \frac{\hat{a}_{\theta_{hs} spike}}{\hat{b}_{\theta_{hs} spike}}.$$

25: Update responsibilities for $h = 1, \ldots, H_s$ and $s = 1, \ldots, S$ from $q^*(\nu_{hs}) = \text{Ber}(r_{hs})$. First calculate

$$l_{hs\,slab} = \mathbb{E}\log\tau_{hs} + a_\Lambda^S \mathbb{E}\log b_\Lambda^S - \log\Gamma(a_\Lambda^S) - (a_\Lambda^S + \frac{p}{2} + 1)\mathbb{E}_{slab}\log\theta_{hs} - \mathbb{E}_{slab}\left(\frac{1}{\theta_{hs}}\right)\left(\mathbb{E}b_\Lambda^S + \frac{1}{2}\tilde{S}_{hs}^\Lambda\right),$$

$$l_{hs\,spike} = \mathbb{E}\log(1-\tau_{hs}) + a_0^{S\Lambda} \mathbb{E}\log b_0^{S\Lambda} - \log\Gamma(a_0^{S\Lambda}) - (a_0^{S\Lambda} + \frac{p}{2} + 1)\mathbb{E}_{spike}\log\theta_{hs} - \mathbb{E}_{spike}\left(\frac{1}{\theta_{hs}}\right)\left(\mathbb{E}b_0^{S\Lambda} + \frac{1}{2}\tilde{S}_{hs}^\Lambda\right),$$

where

$$\begin{aligned}
&\mathbb{E}\log\tau_{hs} = \psi(\hat{a}_{\tau_{hs}}) - \psi(\hat{a}_{\tau_{hs}} + \hat{b}_{\tau_{hs}}),\\
&\mathbb{E}\log(1-\tau_{hs}) = \psi(\hat{b}_{\tau_{hs}}) - \psi(\hat{a}_{\tau_{hs}} + \hat{b}_{\tau_{hs}}),\\
&\mathbb{E}\log b_\Lambda^S = \psi(\hat{a}_{b_\Lambda^S}) - \ln\hat{b}_{b_\Lambda^S},\\
&\mathbb{E}\log b_0^{S\Lambda} = \psi(\hat{a}_{b_0^{S\Lambda}}) - \ln\hat{b}_{b_0^{S\Lambda}},\\
&\mathbb{E}_{slab}\log\theta_{hs} = \ln\hat{b}_{\theta_{hs}slab} - \psi(\hat{a}_{\theta_{hs}slab}),\\
&\mathbb{E}_{spike}\log\theta_{hs} = \ln\hat{b}_{\theta_{hs}spike} - \psi(\hat{a}_{\theta_{hs}spike}),\\
&\mathbb{E}_{slab}\left(\frac{1}{\theta_{hs}}\right) = \frac{\hat{a}_{\theta_{hs}slab}}{\hat{b}_{\theta_{hs}slab}},\\
&\mathbb{E}_{spike}\left(\frac{1}{\theta_{hs}}\right) = \frac{\hat{a}_{\theta_{hs}spike}}{\hat{b}_{\theta_{hs}spike}},\\
&\tilde{S}_{hs}^\Lambda = \sum_{j=1}^p \left[\mathbb{E}\lambda_{ihs}^2 \mathbb{E}\left(\frac{1}{\phi_{jhs}}\right)\right] = \sum_{j=1}^p \left[\left(\left[\boldsymbol{\mu}_{jhs}^\lambda\right]^2 + \boldsymbol{\Sigma}_{js[h,h]}^\lambda\right)\frac{\hat{a}_{\phi s}}{\hat{b}_{\phi_{jhs}}}\right].
\end{aligned}$$

Calculate the responsibility of slab r_{hs} as

$$r_{hs} = \frac{1}{1 + \exp(l_{hs\,spike} - l_{hs\,slab})}.$$

26: Update τ_{hs} for $h = 1, \ldots, H_s$ and $s = 1, \ldots, S$ from $q^*(\tau_{hs}) = \mathcal{B}(\hat{a}_{\tau_{hs}}, \hat{b}_{\tau_{hs}})$, where

$$\begin{aligned}
\hat{a}_{\tau_{hs}} &= \frac{\hat{a}_\alpha^{S\Lambda}}{\hat{b}_\alpha^{S\Lambda} H_s} + r_{hs},\\
\hat{b}_{\tau_{hs}} &= 2 - r_{hs}.
\end{aligned}$$

27: Update α_Λ^S from $q^*(\alpha_\Lambda^S) = \mathcal{G}(\hat{a}_\alpha^{S\Lambda}, \hat{b}_\alpha^{S\Lambda})$, where

$$\begin{aligned}
\hat{a}_\alpha^{S\Lambda} &= a_\alpha^{S\Lambda} + H_s S,\\
\hat{b}_\alpha^{S\Lambda} &= b_\alpha^{S\Lambda} - \frac{1}{H_s}\sum_{s=1}^S \sum_{h=1}^{H_s} \mathbb{E}\log\tau_{hs},
\end{aligned}$$

with $\mathbb{E}\log\tau_{hs} = \psi(\hat{a}_{\tau_{hs}}) - \psi(\hat{a}_{\tau_{hs}} + \hat{b}_{\tau_{hs}})$.

28: Update ϕ_{jhs} for $j = 1, \ldots, p$, $h = 1, \ldots, H$ and $s = 1, \ldots, S$ from $q^*(\phi_{jhs}) = \mathcal{G}^{-1}(\hat{a}_{\phi s}, \hat{b}_{\phi_{jhs}})$, where

$$\hat{a}_{\phi s} = a_\phi^S + \frac{1}{2},$$
$$\hat{b}_{\phi_{jhs}} = \frac{\hat{a}_{\beta_{\phi s}}}{\hat{b}_{\beta_{\phi js}}} + \frac{1}{2}\mathbb{E}_{mix}\left(\frac{1}{\theta_{hs}}\right)\mathbb{E}\lambda_{ihs}^2,$$

with $\mathbb{E}_{mix}\left(\frac{1}{\theta_{hs}}\right) = r_{hs}\frac{\hat{a}_{\theta_{hs}slab}}{\hat{b}_{\theta_{hs}slab}} + (1 - r_{hs})\frac{\hat{a}_{\theta_{hs}spike}}{\hat{b}_{\theta_{hs}spike}}$ and $\mathbb{E}\lambda_{ihs}^2 = \left[\boldsymbol{\mu}_{jhs}^\lambda\right]^2 + \boldsymbol{\Sigma}_{j[h,h]}^\lambda$.

29: Update $\lambda_{is}^\top$ for $j = 1, \ldots, p$ and $s = 1, \ldots, S$ from $q^*(\lambda_{is}) = \mathcal{N}_{H_s}(\boldsymbol{\mu}_{is}^\lambda, \boldsymbol{\Sigma}_{is}^\lambda)$, where

$$\boldsymbol{\Sigma}_{is}^\lambda = \left(\boldsymbol{\Psi}_{is}^{-1} + \frac{\hat{a}_s^\xi}{\hat{b}_{js}^\xi}\sum_{i=1}^{n_s}\left(\boldsymbol{\mu}_{is}^\eta\left[\boldsymbol{\mu}_{is}^\eta\right]^\top + \boldsymbol{\Sigma}_{is}^\eta\right)\right)^{-1},$$
$$\boldsymbol{\mu}_{is}^\lambda = \boldsymbol{\Sigma}_{is}^\lambda\frac{\hat{a}_s^\xi}{\hat{b}_{js}^\xi}\left(\boldsymbol{\mu}_{1s}^\eta, \ldots, \boldsymbol{\mu}_{n_s s}^\eta\right)\left(\boldsymbol{y}_{is}^\top - \left(\boldsymbol{\mu}_{1s}^f, \ldots, \boldsymbol{\mu}_{n_s s}^f\right)\left[\boldsymbol{\mu}_j^\lambda\right]^\top - \boldsymbol{X}_s^\top\left[\boldsymbol{\mu}_j^{\beta^X}\right]^\top\right),$$

and where $\boldsymbol{\Psi}_{js}^{-1} = \mathrm{diag}\left(\frac{\hat{a}_{\phi s}}{\hat{b}_{\phi_{j1s}}}\mathbb{E}_{mix}\left(\frac{1}{\theta_{1s}}\right), \ldots, \frac{\hat{a}_{\phi s}}{\hat{b}_{\phi_{jH_s s}}}\mathbb{E}_{mix}\left(\frac{1}{\theta_{H_s s}}\right)\right)$ with $\mathbb{E}_{mix}\left(\frac{1}{\theta_{hs}}\right) = r_{hs}\frac{\hat{a}_{\theta_{hs}slab}}{\hat{b}_{\theta_{hs}slab}} + (1 - r_{hs})\frac{\hat{a}_{\theta_{hs}spike}}{\hat{b}_{\theta_{hs}spike}}$.

30: Update b_Λ^S from $q^*(b_\Lambda^S) = \mathcal{G}(\hat{a}_{b_\Lambda^S}, \hat{b}_{b_\Lambda^S})$, where

$$\hat{a}_{b_\Lambda^S} = a_1^{S\Lambda} + a_\Lambda^S\sum_{s=1}^{S}\sum_{h=1}^{H_s} r_{hs},$$
$$\hat{b}_{b_\Lambda^S} = b_1^{S\Lambda} + \sum_{s=1}^{S}\sum_{h=1}^{H_s} r_{hs}\frac{\hat{a}_{\theta_{hs}slab}}{\hat{b}_{\theta_{hs}slab}}.$$

31: Update $b_0^{S\Lambda}$ from $q^*(b_0^{S\Lambda}) = \mathcal{G}(\hat{a}_{b_0^{S\Lambda}}, \hat{b}_{b_0^{S\Lambda}})$, where

$$\hat{a}_{b_0^{S\Lambda}} = a_2^{S\Lambda} + a_0^{S\Lambda}\sum_{s=1}^{S}\sum_{h=1}^{H_s}(1 - r_{hs}),$$
$$\hat{b}_{b_0^{S\Lambda}} = b_2^{S\Lambda} + \sum_{s=1}^{S}\sum_{h=1}^{H_s}(1 - r_{hs})\frac{\hat{a}_{\theta_{hs}spike}}{\hat{b}_{\theta_{hs}spike}}.$$

32: Update $\beta_{\phi\{js\}}$ for $j = 1, \ldots, p$ and $s = 1, \ldots, S$ from $q^*(\beta_{\phi_{js}}) = \mathcal{G}(\hat{a}_{\beta_{\phi s}}, \hat{b}_{\beta_{\phi js}})$, where

$$\hat{a}_{\beta_{\phi s}} = c^S + H_s a_\phi^S$$
$$\hat{b}_{\beta_{\phi js}} = d^S + \sum_{h=1}^{H_s}\frac{\hat{a}_{\phi s}}{\hat{b}_{\phi_{jhs}}}.$$

33: Update η_{is} for $i = 1, \ldots, n_s$ and $s = 1, \ldots, S$ from $q^*(\eta_{is}) \sim \mathcal{N}_{H_s}(\boldsymbol{\mu}_{is}^\eta, \boldsymbol{\Sigma}_{is}^\eta)$, where

$$\boldsymbol{\Sigma}_{is}^\eta = \left(\boldsymbol{I}_{H_s} + \sum_{j=1}^{p}\frac{\hat{a}_s^\xi}{\hat{b}_{js}^\xi}\left(\boldsymbol{\mu}_{is}^\lambda\left[\boldsymbol{\mu}_{is}^\lambda\right]^\top + \boldsymbol{\Sigma}_{is}^\lambda\right)\right)^{-1},$$

$$\boldsymbol{\mu}_{is}^{\eta} = \boldsymbol{\Sigma}_{is}^{\eta} \left(\boldsymbol{\mu}_{1s}^{\lambda}, \ldots, \boldsymbol{\mu}_{ps}^{\lambda}\right)^{\top} \operatorname{diag}\left(\frac{\hat{a}_s^{\xi}}{\hat{b}_{1s}^{\xi}}, \ldots, \frac{\hat{a}_s^{\xi}}{\hat{b}_{ps}^{\xi}}\right) \left(\boldsymbol{y}_{is} - \left(\boldsymbol{\mu}_1^{\lambda}, \ldots, \boldsymbol{\mu}_p^{\lambda}\right) \boldsymbol{\mu}_{is}^{f} - \left(\boldsymbol{\mu}_1^{\beta^X}, \ldots, \boldsymbol{\mu}_p^{\beta^X}\right) \boldsymbol{x}_{is}\right).$$

34: Update $\boldsymbol{B}_j^X$ for $j = 1, \ldots, p$ from $q^*(\beta_j^X) \sim \mathcal{N}_{p_x}(\boldsymbol{\mu}_j^{\beta^X}, \boldsymbol{\Sigma}_j^{\beta^X})$, where

$$\boldsymbol{\Sigma}_j^{\beta^X} = \left(\boldsymbol{Q}_j^{-1} + \sum_{s=1}^{S} \frac{\hat{a}_s^{\xi}}{\hat{b}_{js}^{\xi}} \boldsymbol{X}_s \left[\boldsymbol{X}_s\right]^{\top}\right)^{-1},$$

$$\boldsymbol{\mu}_j^{\beta^X} = \boldsymbol{\Sigma}_j^{\beta^X} \left(\sum_{s=1}^{S} \frac{\hat{a}_s^{\xi}}{\hat{b}_{js}^{\xi}} \left(\boldsymbol{x}_{1s}, \ldots, \boldsymbol{x}_{n_s s}\right) \left(\boldsymbol{y}_{is}^{\top} - \left(\boldsymbol{\mu}_{1s}^{f}, \ldots, \boldsymbol{\mu}_{n_s s}^{f}\right) \left[\boldsymbol{\mu}_j^{\lambda}\right]^{\top} - \left(\boldsymbol{\mu}_{1s}^{\eta}, \ldots, \boldsymbol{\mu}_{n_s s}^{\eta}\right) \left[\boldsymbol{\mu}_{js}^{\lambda}\right]^{\top}\right)\right),$$

where $\boldsymbol{Q}_j = \operatorname{diag}(q_{j1}, \ldots, q_{ip_x})$.

A.2 SVI algorithm for MSFRB

The SVI algorithm requires some modifications compared to CAVI. First, the local parameters are updated only for a subset of the data. Second, the global parameters need to be reformulated in terms of natural parameters and updated as a weighted average of the value at the current and previous iterations. The variational updates for other parameters, namely, $b_{\xi js}$, r_h, τ_h^{θ}, θ_h^{Λ}, α_{Λ}, ϕ_{jh}^{Λ}, κ^{Λ}, b_{Λ}, b_0^{Λ}, $\beta_{j\phi}$, π_h, α_{γ}, b_F, b_0^F, r_{hs}, τ_{hs}, θ_{hs}, α_{Λ}^S, b_{Λ}^S and $b_0^{S\Lambda}$, are performed like in Algorithm A.1 and the updated global variables are reparameterised into their original forms according to Table A.1.

Algorithm A.2: Details of the SVI algorithm steps for the MSFRB model

1: Set initial variational parameters
2: Compute iteration-independent variational parameters:
for $s = 1, \ldots, S$ update $\hat{a}_s^{\xi} = a_{\xi} + \frac{n_s}{2}$,
update $\hat{a}^{b_{\xi}} = a_g + a_{\xi}$,
update $\hat{a}_{\theta_h^{\Lambda} slab} = a_{\Lambda} + \frac{p}{2}$,
update $\hat{a}_{\theta_h^{\Lambda} spike} = a_0^{\Lambda} + \frac{p}{2}$,
update $\hat{a}_{\phi} = a_{\phi} + \frac{1}{2}$,
update $\hat{a}_{\kappa} = a^{\kappa} + \frac{1}{2} pH$,
update $\hat{a}_{\beta_{\phi}} = c + H a_{\phi}$,
update $\hat{a}_{\theta_h^{\Lambda} slab} = a_F + \frac{N}{2}$,
update $\hat{a}_{\theta_h^{\Lambda} spike} = a_0^F + \frac{N}{2}$,
update $\hat{a}_{\zeta} = a_{\zeta} + \frac{1}{2}$,
update $\hat{a}_{\nu} = c^{\kappa} + \frac{1}{2} HN$,
update $\hat{a}_{\beta_{\zeta}} = e + H a_{\zeta}$,
update $\hat{a}_{\theta_{hs} slab} = a_{\Lambda}^S + \frac{p}{2}$,
update $\hat{a}_{\theta_{hs} spike} = a_0^{S\Lambda} + \frac{p}{2}$,
update $\hat{a}_{\phi s} = a_{\phi}^S + \frac{1}{2}$,
update $\hat{a}_{\beta_{\phi s}} = c^S + H_s a_{\phi}^S$,
update $\hat{a}_{\alpha}^{\Lambda} = a_{\alpha}^{\Lambda} + H$,
update $\hat{a}_{\alpha}^F = a_{\alpha}^F + H$,
update $\hat{a}_{\alpha}^{S\Lambda} = a_{\alpha}^{S\Lambda} + H_s S$.

Table A.1: *Reparametrisation formulas to convert natural parameters of the global variables into the conventional form and vice versa*

Variable	Natural parameter	Original parameter
ξ_{js}^2	$\eta_{js}^{\xi} = -\hat{b}_{js}^{\xi}$	$\hat{b}_{js}^{\xi} = -\eta_{js}^{\xi}$
κ^F	$\eta_{\nu} = -\hat{b}_{\nu}$	$\hat{b}_{\nu} = -\eta_{\nu}$
θ_h^{Λ}	$\eta_{h2}^{\gamma} = -\hat{b}_{\theta_h^{\Lambda}}$	$\hat{b}_{\theta_h^{\Lambda}} = -\eta_{h2}^{\gamma}$
z_h	$\eta_{z_h} = \log \frac{z_h}{1-z_h}$	$z_h = \frac{1}{1+\exp(-\eta_{z_h})}$
$\boldsymbol{\lambda}_j$	$\boldsymbol{\eta}_{j1}^{\lambda} = -\frac{1}{2}\left(\boldsymbol{\Sigma}_j^{\lambda}\right)^{-1}$ $\boldsymbol{\eta}_{j2}^{\lambda} = \left(\boldsymbol{\Sigma}_j^{\lambda}\right)^{-1}\boldsymbol{\mu}_j^{\lambda}$	$\boldsymbol{\Sigma}_j^{\lambda} = -\frac{1}{2}\left(\boldsymbol{\eta}_{j1}^{\lambda}\right)^{-1}$ $\boldsymbol{\mu}_j^{\lambda} = -\frac{1}{2}\left(\boldsymbol{\eta}_{j1}^{\lambda}\right)^{-1}\boldsymbol{\eta}_{j2}^{\lambda}$
$\boldsymbol{\lambda}_{js}$	$\boldsymbol{\eta}_{js1}^{\lambda} = -\frac{1}{2}\left(\boldsymbol{\Sigma}_{js}^{\lambda}\right)^{-1}$ $\boldsymbol{\eta}_{js2}^{\lambda} = \left(\boldsymbol{\Sigma}_{js}^{\lambda}\right)^{-1}\boldsymbol{\mu}_{js}^{\lambda}$	$\boldsymbol{\Sigma}_{js}^{\lambda} = -\frac{1}{2}\left(\boldsymbol{\eta}_{js1}^{\lambda}\right)^{-1}$ $\boldsymbol{\mu}_{js}^{\lambda} = -\frac{1}{2}\left(\boldsymbol{\eta}_{js1}^{\lambda}\right)^{-1}\boldsymbol{\eta}_{js2}^{\lambda}$
$\boldsymbol{B}_j^X$	$\boldsymbol{\eta}_{j1}^{\beta^X} = -\frac{1}{2}\left(\boldsymbol{\Sigma}_j^{\beta^X}\right)^{-1}$ $\boldsymbol{\eta}_{j2}^{\beta^X} = \left(\boldsymbol{\Sigma}_j^{\beta^X}\right)^{-1}\boldsymbol{\mu}_j^{\beta^X}$	$\boldsymbol{\Sigma}_j^{\beta^X} = -\frac{1}{2}\left(\boldsymbol{\eta}_{j1}^{\beta^X}\right)^{-1}$ $\boldsymbol{\mu}_j^{\beta^X} = -\frac{1}{2}\left(\boldsymbol{\eta}_{j1}^{\beta^X}\right)^{-1}\boldsymbol{\eta}_{j2}^{\beta^X}$

for $t = 1, \ldots$ **until convergence is reached**

3: Set the step size $\rho_t = (t + \tau^{\rho})^{-\kappa^{\rho}}$

4: For $s = 1, \ldots, S$ draw $\tilde{N}_s = \lfloor b_s \times n_s \rfloor$ data points from $1, \ldots, n_s$ without replacement and call it $\mathcal{I}_s(t)$. Set $\tilde{\boldsymbol{Y}}_s^t = \{\boldsymbol{y}_{is} \in \mathcal{I}_s(t)\}$

5: For $s = 1, \ldots, S$ and $n \in \mathcal{I}_s(t)$ calculate the updates for the local variational parameters for each observation in $\tilde{\boldsymbol{Y}}_s^t$:
update $\hat{b}_{\zeta his}$ for $h = 1, \ldots, H$ as in Step 18 of Algorithm A.1
update $\boldsymbol{\Sigma}_{is}^f$ and $\boldsymbol{\mu}_{is}^f$ as in Step 20 of Algorithm A.1
update $\hat{b}_{\beta_{ns\zeta}}$ as in Step 23 of Algorithm A.1
update $\boldsymbol{\Sigma}_{is}^{\eta}$ and $\boldsymbol{\mu}_{is}^{\eta}$ as in Step 33 of Algorithm A.1

6: Calculate the updates for global parameters using $\tilde{\boldsymbol{Y}}_s^t$ and the current step size ρ_t:
(a) For $s = 1, \ldots, S$ and $j = 1, \ldots, p$ update

$$\hat{\eta}_{js}^{\xi} = -\frac{\hat{a}^{b_{\xi}}}{\hat{b}_{js}^{b_{\xi}}} - \frac{1}{2}\frac{n_s}{\tilde{N}_s}\sum_{n \in \mathcal{I}_s(t)}\Bigg[y_{jis}^2 - 2y_{jis}\left(\boldsymbol{\mu}_j^{\lambda}\left[\boldsymbol{\mu}_{is}^f\right]^{\top} + \boldsymbol{\mu}_{is}^{\lambda^s}\left[\boldsymbol{\mu}_{is}^{\eta}\right]^{\top} + \boldsymbol{\mu}_j^{\beta^X}\boldsymbol{x}_{is}\right) +$$
$$2\left(\boldsymbol{\mu}_j^{\lambda}\left[\boldsymbol{\mu}_{is}^f\right]^{\top}\boldsymbol{\mu}_{is}^{\lambda^s}\left[\boldsymbol{\mu}_{is}^{\eta}\right]^{\top} + \boldsymbol{\mu}_j^{\lambda}\left[\boldsymbol{\mu}_{is}^f\right]^{\top}\boldsymbol{\mu}_j^{\beta^X}\boldsymbol{x}_{is} + \boldsymbol{\mu}_{is}^{\lambda^s}\left[\boldsymbol{\mu}_{is}^{\eta}\right]^{\top}\boldsymbol{\mu}_j^{\beta^X}\boldsymbol{x}_{is}\right) +$$
$$\boldsymbol{\mu}_j^{\beta^X}\boldsymbol{x}_{is}\left[\boldsymbol{x}_{is}\right]^{\top}\left[\boldsymbol{\mu}_j^{\beta^X}\right]^{\top} + \boldsymbol{\mu}_j^{\lambda}\left[\boldsymbol{\mu}_{is}^f\right]^{\top}\boldsymbol{\mu}_{is}^f\left[\boldsymbol{\mu}_j^{\lambda}\right]^{\top} + \boldsymbol{\mu}_{is}^{\lambda^s}\left[\boldsymbol{\mu}_{is}^{\eta}\right]^{\top}\boldsymbol{\mu}_{is}^{\eta}\left[\boldsymbol{\mu}_{is}^{\lambda^s}\right]^{\top} +$$
$$\boldsymbol{\mu}_j^{\lambda}\boldsymbol{\Sigma}_{is}^f\left[\boldsymbol{\mu}_j^{\lambda}\right]^{\top} + \left[\boldsymbol{\mu}_{is}^f\right]^{\top}\boldsymbol{\Sigma}_j^{\lambda}\boldsymbol{\mu}_{is}^f + \boldsymbol{\mu}_{is}^{\lambda^s}\boldsymbol{\Sigma}_{is}^{\eta}\left[\boldsymbol{\mu}_{is}^{\lambda^s}\right]^{\top} + \left[\boldsymbol{\mu}_{is}^{\eta}\right]^{\top}\boldsymbol{\Sigma}_{is}^{\lambda^s}\boldsymbol{\mu}_{is}^{\eta} + \mathrm{Tr}\left(\boldsymbol{\Sigma}_j^{\lambda}\boldsymbol{\Sigma}_{is}^f\right) + \mathrm{Tr}\left(\boldsymbol{\Sigma}_{is}^{\lambda^s}\boldsymbol{\Sigma}_{is}^{\eta}\right)\Bigg]$$

and set η_{js}^{ξ} as the weighted average of the values at this and the previous iterations

$$\eta_{js}^{\xi}(t) = (1-\rho_t)\eta_{js}^{\xi}(t-1) + \rho_t \hat{\eta}_{js}^{\xi}$$

(b) For $j = 1, \ldots, p$ update

$$\hat{\eta}_{j1}^{\lambda} = -\frac{1}{2}\left(\boldsymbol{\Psi}_j^{-1} + \sum_{s=1}^{S} \frac{\hat{a}_s^{\xi}}{\hat{b}_{js}^{\xi}} \frac{n_s}{\tilde{N}_s} \sum_{n \in \mathcal{I}_s(t)} \left(\boldsymbol{\mu}_{is}^{f} \left[\boldsymbol{\mu}_{is}^{f}\right]^{\top} + \boldsymbol{\Sigma}_{is}^{f}\right)\right),$$

$$\hat{\eta}_{j2}^{\lambda} = \sum_{s=1}^{S} \frac{\hat{a}_s^{\xi}}{\hat{b}_{js}^{\xi}} \frac{n_s}{\tilde{N}_s} \sum_{n \in \mathcal{I}_s(t)} \left(\boldsymbol{\mu}_{is}^{f} \left(\boldsymbol{y}_{is}^{\top} - \boldsymbol{\mu}_{is}^{\eta} \left[\boldsymbol{\mu}_{js}^{\lambda}\right]^{\top} - \boldsymbol{x}_{is}^{\top} \left[\boldsymbol{\mu}_{j}^{\beta^X}\right]^{\top}\right)\right),$$

where $\boldsymbol{\Psi}_j^{-1}$ is as defined in Step 10 of Algorithm A.1.
Set η_{j1}^{λ} and η_{j2}^{λ} as the weighted average of the values at this and the previous iterations

$$\eta_{j1}^{\lambda}(t) = (1-\rho_t)\eta_{j1}^{\lambda}(t-1) + \rho_t \hat{\eta}_{j1}^{\lambda},$$
$$\eta_{j2}^{\lambda}(t) = (1-\rho_t)\eta_{j2}^{\lambda}(t-1) + \rho_t \hat{\eta}_{j2}^{\lambda}.$$

(c) For $j = 1, \ldots, p$ and $s = 1, \ldots, S$ update

$$\hat{\eta}_{js1}^{\lambda} = -\frac{1}{2}\left(\boldsymbol{\Psi}_{js}^{-1} + \frac{\hat{a}_s^{\xi}}{\hat{b}_{js}^{\xi}} \frac{n_s}{\tilde{N}_s} \sum_{n \in \mathcal{I}_s(t)} \left(\boldsymbol{\mu}_{is}^{\eta} \left[\boldsymbol{\mu}_{is}^{\eta}\right]^{\top} + \boldsymbol{\Sigma}_{is}^{\eta}\right)\right),$$

$$\hat{\eta}_{js2}^{\lambda} = \frac{\hat{a}_s^{\xi}}{\hat{b}_{js}^{\xi}} \frac{n_s}{\tilde{N}_s} \sum_{n \in \mathcal{I}_s(t)} \left(\boldsymbol{\mu}_{is}^{\eta} \left(\boldsymbol{y}_{is}^{\top} - \boldsymbol{\mu}_{is}^{f} \left[\boldsymbol{\mu}_{j}^{\lambda}\right]^{\top} - \boldsymbol{x}_{is}^{\top} \left[\boldsymbol{\mu}_{j}^{\beta^X}\right]^{\top}\right)\right),$$

where $\boldsymbol{\Psi}_{js}^{-1}$ is as defined in Step 29 of Algorithm A.1.
Set η_{j1}^{λ} and η_{j2}^{λ} as the weighted average of the values at this and the previous iterations

$$\eta_{js1}^{\lambda}(t) = (1-\rho_t)\eta_{js1}^{\lambda}(t-1) + \rho_t \hat{\eta}_{js1}^{\lambda},$$
$$\eta_{js2}^{\lambda}(t) = (1-\rho_t)\eta_{js2}^{\lambda}(t-1) + \rho_t \hat{\eta}_{js2}^{\lambda}.$$

(d) For $j = 1, \ldots, p$ update

$$\hat{\eta}_{j1}^{\beta^X} = -\frac{1}{2}\left(\boldsymbol{Q}_j^{-1} + \sum_{s=1}^{S} \frac{\hat{a}_s^{\xi}}{\hat{b}_{js}^{\xi}} \frac{n_s}{\tilde{N}_s} \sum_{n \in \mathcal{I}_s(t)} \boldsymbol{x}_{is} \left[\boldsymbol{x}_{is}\right]^{\top}\right),$$

$$\hat{\eta}_{j2}^{\beta^X} = \sum_{s=1}^{S} \frac{\hat{a}_s^{\xi}}{\hat{b}_{js}^{\xi}} \frac{n_s}{\tilde{N}_s} \sum_{n \in \mathcal{I}_s(t)} \left(\boldsymbol{x}_{is} \left(\boldsymbol{y}_{is}^{\top} - \boldsymbol{\mu}_{is}^{f} \left[\boldsymbol{\mu}_{j}^{\lambda}\right]^{\top} - \boldsymbol{\mu}_{is}^{\eta} \left[\boldsymbol{\mu}_{js}^{\lambda}\right]^{\top}\right)\right),$$

where $\boldsymbol{Q}_j = \mathrm{diag}(q_{j1}, \ldots, q_{ip_x})$.
Set $\eta_{j1}^{\beta^X}$ and $\eta_{j2}^{\beta^X}$ as the weighted average of the values at this and the previous iterations

$$\eta_{j1}^{\beta^X}(t) = (1-\rho_t)\eta_{j1}^{\beta^X}(t-1) + \rho_t \hat{\eta}_{j1}^{\beta^X},$$
$$\eta_{j2}^{\beta^X}(t) = (1-\rho_t)\eta_{j2}^{\beta^X}(t-1) + \rho_t \hat{\eta}_{j2}^{\beta^X}.$$

(e) For $h = 1, \ldots, H$ update

$$\hat{\eta}_{\gamma_{hslab}} = -\frac{\hat{a}_{b_\gamma}}{\hat{b}_{b_\gamma}} - \frac{1}{2}\frac{\hat{a}_\nu}{\hat{b}_\nu}\breve{S}_h^F,$$

$$\hat{\eta}_{\gamma_{hspike}} = -\frac{\hat{a}_{b_0^F}}{\hat{b}_{b_0^F}} - \frac{1}{2}\frac{\hat{a}_\nu}{\hat{b}_\nu}\breve{S}_h^F,$$

where $\breve{S}_h^F = \sum_{s=1}^S \frac{n_s}{\tilde{N}_s} \sum_{n \in \mathcal{I}_s(t)} \left[\left(\left[\mu_{his}^f \right]^2 + \mathbf{\Sigma}_{ns[h,h]}^f \right) \frac{\hat{a}_\zeta}{\hat{b}_{\phi_{his}^F}} \right]$, with h and $[h,h]$ indicating the hth element of the mean vector and the $[h,h]$th element of the covariance matrix $\mathbf{\Sigma}_{is}^f$, correspondingly.
Set $\eta_{\gamma_{hslab}}$ and $\eta_{\gamma_{hspike}}$ as the weighted average of the values at this and the previous iterations

$$\eta_{\gamma_{hslab}}(t) = (1-\rho_t)\eta_{\gamma_{hslab}}(t-1) + \rho_t \hat{\eta}_{\gamma_{hslab}},$$
$$\eta_{\gamma_{hspike}}(t) = (1-\rho_t)\eta_{\gamma_{hspike}}(t-1) + \rho_t \hat{\eta}_{\gamma_{hspike}}.$$

(f) Update

$$\hat{\eta}_\nu = -d^\kappa - \frac{1}{2}\sum_{h=1}^H \left[\mathbb{E}_{mix}\left(\frac{1}{\theta_h^\Lambda} \right) \breve{S}_h^F \right],$$

where $\mathbb{E}_{mix}\left(\frac{1}{\theta_h^\Lambda}\right)$ is as defined in Step 19 of Algorithm A.1 and $\breve{S}_h^F = \sum_{s=1}^S \frac{n_s}{\tilde{N}_s} \sum_{n \in \mathcal{I}_s(t)} \left[\left(\left[\mu_{his}^f \right]^2 + \mathbf{\Sigma}_{ns[h,h]}^f \right.$
Set η_ν as the weighted average of the values at this and the previous iterations

$$\eta_\nu(t) = (1-\rho_t)\eta_\nu(t-1) + \rho_t \hat{\eta}_\nu.$$

(g) For $h = 1, \ldots, H$ update

$$\hat{\eta}_{z_h} = m_{hslab} - m_{hspike},$$

where

$$m_{h\,slab} = \mathbb{E}\log\pi_h + a_F \mathbb{E}\log b_F - \log\Gamma(a_F) - (a_F + \frac{N}{2} + 1)\mathbb{E}_{slab}\log\theta_h^\Lambda - \mathbb{E}_{slab}\left(\frac{1}{\theta_h^\Lambda}\right)\left(\mathbb{E}b_F + \frac{1}{2}\mathbb{E}\left(\frac{1}{\nu}\right)\breve{S}_h^F\right),$$

$$m_{h\,spike} = \mathbb{E}\log(1-\pi_h) + a_0^F \mathbb{E}\log b_0^F - \log\Gamma(a_0^F) - (a_0^F + \frac{N}{2} + 1)\mathbb{E}_{spike}\log\theta_h^\Lambda - \mathbb{E}_{spike}\left(\frac{1}{\theta_h^\Lambda}\right)\left(\mathbb{E}b_0^F + \frac{1}{2}\mathbb{E}\left(\frac{1}{\nu}\right)\breve{S}_h^F\right),$$

with $\breve{S}_h^F$ defined in the previous step (f) and other quantities defined in Step 15 of Algorithm A.1.
Set η_{z_h} as the weighted average of the values at this and the previous iterations

$$\eta_{z_h}(t) = (1-\rho_t)\eta_{z_h}(t-1) + \rho_t \hat{\eta}_{z_h}.$$

7: Update the remaining parameters which do not depend on local variables according to Algorithm A.1.

B Initial values, convergence check and hyperparameters

B.1 Initialisations

In a VI algorithm, the result can be rather sensitive to initial parameters and unfortunate starting values can lead to the algorithm being stuck at a local optimum. Therefore a careful strategy for robust initialisation of the model parameters is usually required. We initialise our algorithm in a similar way to the procedure described in Hansen et al. (2025), with the difference that at first we run the regression on the known covariates to get the initial estimates of the mean values of $\boldsymbol{B}^X$ and then proceed using the residuals from the regression. This is done according to the following step-by-step process:

1. First, the regression model $\boldsymbol{y}_{is} = \boldsymbol{B}^X \boldsymbol{x}_{is} + \boldsymbol{e}_{is}$ is solved for all $i = 1, \ldots, n_s$ and all $s = 1, \ldots, S$, which gives the initial values for $\mu_{jl}^{\beta^X} = \hat{\beta}_{jl}$ for $j = 1, \ldots, p$ and $l = 1, \ldots, p_x$. Note that the $p \times n$ dimensional data matrix $\boldsymbol{Y} = \{y_{jis}\}$ is constructed by stacking the study-specific data matrices $\boldsymbol{Y}_s$ one after another for $s = 1, \ldots, S$ and setting $n = \sum_{s=1}^{S} n_s$.

2. Initial values for $\boldsymbol{\mu}_j^{\lambda}$ and $\boldsymbol{\mu}_{is}^{f}$ are obtained from decomposing the $p \times n$ error matrix from the regression of the previous step $\boldsymbol{E} = \{e_{jis}\}$ as $\boldsymbol{E}^\top = \tilde{\boldsymbol{f}}\tilde{\boldsymbol{\Lambda}}^\top$, where the common loadings matrix $\tilde{\boldsymbol{\Lambda}}$ of dimensions $p \times H$ and factor scores $\tilde{\boldsymbol{f}}$ of dimensions $n \times H$ are estimated via an unconstrained sparse principal component analysis (PCA) (Erichson et al., 2020) available in R-package `sparsepca`, with H components Then we set $\boldsymbol{\mu}_j^{\lambda} = \tilde{\lambda}_{jh}$, where $\tilde{\lambda}_{jh}$ are elements (j, h) of $\tilde{\boldsymbol{\Lambda}}$, and $\boldsymbol{\mu}_{is}^{f} = \tilde{f}_{ish}$, where $\tilde{f}_{ish}$ refers to the element of the stacked matrix of common factor scores $\tilde{\boldsymbol{f}}$ corresponding to sample i of subject s and factor h.

3. As the next step, to define the study-specific residuals $\tilde{\boldsymbol{E}}_s$ as

$$\tilde{\boldsymbol{E}}_s^\top = \boldsymbol{E}_s^\top \left(\boldsymbol{I}_p - \tilde{\boldsymbol{\Lambda}} \left(\tilde{\boldsymbol{\Lambda}}^\top \tilde{\boldsymbol{\Lambda}} \right)^+ \tilde{\boldsymbol{\Lambda}}^\top \right),$$

   where $\boldsymbol{E}_s$ is the $p \times n_s$ matrix of regression residuals with the indices $1, \ldots, n_s$ corresponding to subject s and the superscript "+" indicates the Moore-Penrose inverse. These are the sparse PCA residuals proposed by Camacho et al. (2020), which subtract the common part of the variability from the data.

4. Now the initial values for $\boldsymbol{\mu}_{js}^{\Upsilon}$ and $\boldsymbol{\mu}_{is}^{\eta}$ are obtained from decomposing the $p \times n_s$ study-specific error matrix $\tilde{\boldsymbol{E}}_s$ as $\tilde{\boldsymbol{E}}_s^\top = \tilde{\boldsymbol{\eta}}_s \tilde{\boldsymbol{\Upsilon}}_s^\top$, with the constrained that $\mathrm{Cov}(\tilde{\boldsymbol{\eta}}_s) = \boldsymbol{I}_{H_s}$. The study-specific loadings matrix $\tilde{\boldsymbol{\Upsilon}}_s$ of dimensions $p \times H_s$ and factor scores $\tilde{\boldsymbol{\eta}}_s$ of dimensions $n_s \times H_s$ are estimated via an unconstrained sparse principal component analysis (PCA) (Erichson et al., 2020) available in R-package `sparsepca` with H_s components and are appropriately scaled. Then we set $\boldsymbol{\mu}_{js}^{\Upsilon} = \tilde{\upsilon}_{jhs}$, where $\tilde{\upsilon}_{jhs}$ is element (j, h) of $\tilde{\boldsymbol{\Upsilon}}_s$, and $\boldsymbol{\mu}_{is}^{\eta} = \tilde{\eta}_{his}$, where $\tilde{\eta}_{his}$ refers to the (i, h) element of $\tilde{\boldsymbol{\eta}}_s$.

5. Initial values for the parameters of study-specific errors ξ_{js}^2 are determined in the following way. For $s = 1, \ldots, S$, set $\hat{a}_{js}^{\xi} = a^{\xi} + n_s/2$, then let $(\tilde{\xi}_{1s}^2, \ldots, \tilde{\xi}_{ps}^2) = \mathrm{diag}(\tilde{\boldsymbol{E}}_s \tilde{\boldsymbol{E}}_s^{\top} - \tilde{\boldsymbol{\Lambda}}_s \tilde{\boldsymbol{\Lambda}}_s^{\top})$ and set $\hat{b}_{js}^{\xi} = (\hat{a}_{js}^{\xi} - 1) \times \tilde{\xi}_{js}^2$. Set $\hat{b}_{js}^{\beta^{\xi}} = b_{js}^{\beta^{\xi}} + \frac{(\hat{a}_{js}^{\xi} - 1)}{\hat{b}_{js}^{\xi}}$.

6. Initial values for $\boldsymbol{\Sigma}_j^{\beta^X}$ are defined by setting $\boldsymbol{\Sigma}_j^{\beta^X} = \left(\boldsymbol{Q}_j^{-1} + \sum_{s=1}^{S} \frac{(\hat{a}_{js}^{\xi} - 1)}{\hat{b}_{js}^{\xi}} \boldsymbol{X}_s \boldsymbol{X}_s^{\top} \right)^{-1}$, where $\boldsymbol{Q}_j = \mathrm{diag}(q_{j1}, \ldots, q_{jp_X})$ and $\boldsymbol{X}_s$ is the $p_X \times n_s$ matrix of covariates for subject s.

7. Concentration parameters for slab probabilities are initialised as a mean of the prior distribution. As the prior distribution is gamma, the initial values for the slab concentration parameters are the following: for common loadings matrix $\alpha^{\tau^{\Lambda}} = a^{\tau^{\Lambda}}/b^{\tau^{\Lambda}}$, for the common factors $\alpha^{\tau^F} = a^{\tau^F}/b^{\tau^F}$, and for study-specific factor loadings $\alpha^{\tau^{\Upsilon}} = a^{\tau^{\Upsilon}}/b^{\tau^{\Upsilon}}$.

8. Initial values for the global sparsity parameters κ^{Λ} and κ^F are set equal for the prior hyperparameters, namely $\hat{a}^{\kappa^{\Lambda}} = a^{\kappa^{\Lambda}}$, $\hat{b}^{\kappa^{\Lambda}} = b^{\kappa^{\Lambda}}$, $\hat{a}^{\kappa^F} = a^{\kappa^F}$ and $\hat{b}^{\kappa^F} = b^{\kappa^F}$.

9. Initial values for the element-wise sparsity parameters ϕ_{jh}^{Λ}, ϕ_{his}^{F} and ϕ_{jhs}^{Υ} are set equal to prior hyperparameters, with the rate parameters being equal to its prior mean, namely: $\hat{a}_j^{\beta^{\phi^{\Lambda}}} = c^{\phi^{\Lambda}}$, $\hat{b}_j^{\beta^{\phi^{\Lambda}}} = d^{\phi^{\Lambda}}$, $\hat{a}_{jh}^{\phi^{\Lambda}} = a^{\phi^{\Lambda}}$, $\hat{b}_{jh}^{\phi^{\Lambda}} = c^{\phi^{\Lambda}}/d^{\phi^{\Lambda}}$, $\hat{a}_{is}^{\beta^{\phi^F}} = c^{\phi^F}$, $\hat{b}_{is}^{\beta^{\phi^F}} = d^{\phi^F}$, $\hat{a}_{his}^{\phi^F} = c^{\phi^{\Lambda}}$, $\hat{b}_{his}^{\phi^F} = c^{\phi^F}/d^{\phi^F}$, and $\hat{a}_{js}^{\beta^{\phi^{\Upsilon}}} = c^{\phi^{\Upsilon}}$, $\hat{b}_{js}^{\beta^{\phi^{\Upsilon}}} = d^{\phi^{\Upsilon}}$, $\hat{a}_{jhs}^{\phi^{\Upsilon}} = a^{\phi^{\Upsilon}}$, $\hat{b}_{jhs}^{\phi^{\Upsilon}} = c^{\phi^{\Upsilon}}/d^{\phi^{\Upsilon}}$.

10. Initial values of the spike and the slab parameters of θ_h^{Λ} are set equal to prior hyperparameters, with the rate parameter being equal to its prior mean, namely: $\hat{a}^{\beta_0^{\theta^{\Lambda}}} = c_0^{\theta^{\Lambda}}$, $\hat{b}^{\beta_0^{\theta^{\Lambda}}} = d_0^{\theta^{\Lambda}}$, $\hat{a}_{\theta_h^{\Lambda} spike} = a_0^{\theta^{\Lambda}}$, $\hat{b}_{\theta_h^{\Lambda} spike} = c_0^{\theta^{\Lambda}}/d_0^{\theta^{\Lambda}}$, and $\hat{a}^{\beta^{\theta^{\Lambda}}} = c^{\theta^{\Lambda}}$, $\hat{b}^{\beta^{\theta^{\Lambda}}} = d^{\theta^{\Lambda}}$, $\hat{a}_{\theta_h^{\Lambda} slab} = a^{\theta^{\Lambda}}$, $\hat{b}_{\theta_h^{\Lambda} slab} = c^{\theta^{\Lambda}}/d^{\theta^{\Lambda}}$.

11. Initial values of the spike and the slab parameters of θ_h^{F} are set equal to prior hyperparameters, with the rate parameter being equal to its prior mean, namely: $\hat{a}^{\beta_0^{\theta^F}} = c_0^{\theta^F}$, $\hat{b}^{\beta_0^{\theta^F}} = d_0^{\theta^F}$, $\hat{a}_{\theta_h^F spike} = a_0^{\theta^F}$, $\hat{b}_{\theta_h^F spike} = c_0^{\theta^F}/d_0^{\theta^F}$, and $\hat{a}^{\beta^{\theta^F}} = c^{\theta^F}$, $\hat{b}^{\beta^{\theta^F}} = d^{\theta^F}$, $\hat{a}_{\theta_h^F slab} = a^{\theta^F}$, $\hat{b}_{\theta_h^F slab} = c^{\theta^F}/d^{\theta^F}$.

12. Initial values of the spike and the slab parameters of θ_{hs}^{Υ} are set equal to prior hyperparameters, with the rate parameter being equal to its prior mean, namely: $\hat{a}^{\beta_0^{\theta^{\Upsilon}}} = c_0^{\theta^{\Upsilon}}$, $\hat{b}^{\beta_0^{\theta^{\Upsilon}}} = d_0^{\theta^{\Upsilon}}$, $\hat{a}_{\theta_{hs}^{\Upsilon} spike} = a_0^{\theta^{\Upsilon}}$, $\hat{b}_{\theta_{hs}^{\Upsilon} spike} = c_0^{\theta^{\Upsilon}}/d_0^{\theta^{\Upsilon}}$, and $\hat{a}^{\beta^{\theta^{\Upsilon}}} = c^{\theta^{\Upsilon}}$, $\hat{b}^{\beta^{\theta^{\Upsilon}}} = d^{\theta^{\Upsilon}}$, $\hat{a}_{\theta_{hs}^{\Upsilon} slab} = a^{\theta^{\Upsilon}}$, $\hat{b}_{\theta_{hs}^{\Upsilon} slab} = c^{\theta^{\Upsilon}}/d^{\theta^{\Upsilon}}$.

13. Initial assignments of θ_h^{Λ}, θ_h^F and θ_{hs}^{Υ} to spike or slab is done by randomly sampling from the probabilities of the slab, when the probabilities of the slab is set as $\alpha^{\tau^{\Lambda}}/(\alpha^{\tau^{\Lambda}} + H)$, $\alpha^{\tau^F}/(\alpha^{\tau^F} + H)$, and $\alpha^{\tau^{\Upsilon}}/(\alpha^{\tau^{\Upsilon}} + H_s)$, correspondingly.

14. Initial values for common factor loading covariance matrix $\boldsymbol{\Sigma}_j^{\lambda}$ are set as $\boldsymbol{\Sigma}_j^{\lambda} = (\boldsymbol{\Psi}_j^{init})^{-1}$, where $\boldsymbol{\Psi}_j^{init} = diag\left(\frac{\hat{a}_{\kappa^\Lambda}}{\hat{b}_{\kappa^\Lambda}}\frac{\hat{a}_{\phi}}{\hat{b}_{\phi_{j1}}}\mathbb{E}_{mix}\left(\frac{1}{\theta_1^\Lambda}\right), \ldots, \frac{\hat{a}_{\kappa^\Lambda}}{\hat{b}_{\kappa^\Lambda}}\frac{\hat{a}_{\phi}}{\hat{b}_{\phi_{jH}}}\mathbb{E}_{mix}\left(\frac{1}{\theta_H^\Lambda}\right)\right)$ with $\mathbb{E}_{mix}\left(\frac{1}{\theta_h^\Lambda}\right) = r_h^\Lambda \frac{\hat{a}_{\theta_h^\Lambda slab}}{\hat{b}_{\theta_h^\Lambda slab}} + (1 - r_h^\Lambda)\frac{\hat{a}_{\theta_h^\Lambda spike}}{\hat{b}_{\theta_h^\Lambda spike}}$, and the parameters are initiated as described above.

15. Initial values for subject-specific factor loading covariance matrix $\boldsymbol{\Sigma}_{js}^{\upsilon}$ are set as $\boldsymbol{\Sigma}_{js}^{\upsilon} = (\boldsymbol{\Psi}_{js}^{init})^{-1}$, where $\boldsymbol{\Psi}_{js}^{init} = diag\left(\mathbb{E}_{mix}\left(\frac{1}{\theta_{1s}^\Upsilon}\right), \ldots, \mathbb{E}_{mix}\left(\frac{1}{\theta_{Hs}^\Upsilon}\right)\right)$ with $\mathbb{E}_{mix}\left(\frac{1}{\theta_{hs}^\Upsilon}\right) = r_{hs}^\Upsilon \frac{\hat{a}_{\theta_{hs}^\Upsilon slab}}{\hat{b}_{\theta_{hs}^\Upsilon slab}} + (1 - r_{hs}^\Upsilon)\frac{\hat{a}_{\theta_{hs}^\Upsilon spike}}{\hat{b}_{\theta_{hs}^\Upsilon spike}}$, and the parameters are initiated as described above.

16. Subject-specific factor covariance matrix is initialised as

$$\boldsymbol{\Sigma}_{is}^{\eta} = \left(\boldsymbol{I}_{H_s} + \sum_{j=1}^{p} \tilde{\xi}_{js}^2 \left(\boldsymbol{\mu}_{js}^{\lambda}\left[\boldsymbol{\mu}_{js}^{\lambda}\right]^\top + \boldsymbol{\Sigma}_{js}^{\lambda}\right)\right)^{-1}.$$

17. Common factor covariance matrix is initialised as $\boldsymbol{\Sigma}_{is}^{f} = \left(\boldsymbol{\Omega}_{is}^{-1} + \sum_{j=1}^{p} \tilde{\xi}_{js}^2 \left(\boldsymbol{\mu}_{j}^{\lambda}\left[\boldsymbol{\mu}_{j}^{\lambda}\right]^\top + \boldsymbol{\Sigma}_{j}^{\lambda}\right)\right)^{-1}$, where $\boldsymbol{\Omega}_{is}^{-1} = diag\left(\frac{\hat{a}_{\kappa^F}}{\hat{b}_{\kappa^F}}\frac{\hat{a}_{\phi^F}}{\hat{b}_{\phi_{1is}^F}}\mathbb{E}_{mix}\left(\frac{1}{\theta_1^F}\right), \ldots, \frac{\hat{a}_{\kappa^F}}{\hat{b}_{\kappa^F}}\frac{\hat{a}_{\phi^F}}{\hat{b}_{\phi_{His}^F}}\mathbb{E}_{mix}\left(\frac{1}{\theta_H^F}\right)\right)$ with $\mathbb{E}_{mix}\left(\frac{1}{\theta_h^F}\right) = r_h^F \frac{\hat{a}_{\theta_h^F slab}}{\hat{b}_{\theta_h^F slab}} + (1 - r_h^F)\frac{\hat{a}_{\theta_h^F spike}}{\hat{b}_{\theta_h^F spike}}$, and the parameters are initialised as described above.

B.2 Hyperparameter specification

We specify the hyperparameters of the sparse exchangeable shrinkage process prior (ESP) to induce structured sparsity in both the common and study-specific components while allowing sufficient flexibility for dense signals. The slab inclusion probabilities for the common factor loadings and factor scores are governed by Beta distributions with parameters $(a^{\tau^\Lambda}, b^{\tau^\Lambda}) = (10, 2)$ and $(a^{\tau^F}, b^{\tau^F}) = (10, 2)$, favouring inclusion and thus allowing for a moderate number of common factors. In contrast, the study-specific factor loadings are assigned $(a^{\tau^\Upsilon}, b^{\tau^\Upsilon}) = (1, 4)$, encouraging smaller number of study-specific effects.

Global sparsity for common loadings and scores is controlled via inverse-Gamma hyperpriors with parameters $(a^{\kappa^\Lambda}, b^{\kappa^\Lambda}) = (2, 1)$ and $(a^{\kappa^F}, b^{\kappa^F}) = (2, 1)$, which favour moderate shrinkage while still allowing flexibility when supported by the data. Element-wise sparsity is further regulated through hierarchical inverse-Gamma distributions on local shrinkage parameters, with parameters $(a^{\phi^\Lambda}, c^{\phi^\Lambda}, d^{\phi^\Lambda}) = (7, 0.2, 0.5)$ for common loadings, $(a^{\phi^\Upsilon}, c^{\phi^\Upsilon}, d^{\phi^\Upsilon}) = (3, 0.5, 1)$ for study-specific loadings, and $(a^{\phi^F}, c^{\phi^F}, d^{\phi^F}) = (7, 0.2, 0.5)$ for common factor scores, promoting a balance between sparse and dense patterns.

The spike-and-slab components are specified through inverse-Gamma priors on the column-specific variance parameters, with separate Gamma hyperpriors on their rate parameters. For the common loading variances, the slab component is assigned shape

parameter $a^{\theta^\Lambda} = 3$, with rate parameter β^{θ^Λ} given the hyperprior $\beta^{\theta^\Lambda} \sim \mathcal{G}(c^{\theta^\Lambda}, d^{\theta^\Lambda})$ with $(c^{\theta^\Lambda}, d^{\theta^\Lambda}) = (2, 1)$. We use the same values for the corresponding common factor-score parameters: $a^{\theta^F} = 3$ and $\beta^{\theta^F} \sim \mathcal{G}(c^{\theta^F}, c^{\theta^F})$, with $(c^{\theta^F}, d^{\theta^F}) = (2, 1)$. For the study-specific loading column-level variance parameters, we use $a^{\theta^\Upsilon} = 2$ and $\beta^{\theta^\Upsilon} \sim \mathcal{G}(c^{\theta^\Upsilon}, d^{\theta^\Upsilon})$, with $(c^{\theta^\Upsilon}, d^{\theta^\Upsilon}) = (2, 1)$. The spike components are specified analogously, but with larger inverse-Gamma shape parameters (please see (Grushanina and Frühwirth-Schnatter, 2025) for the reasoning behind this choice), $a_0^{\theta^\Lambda} = 31$, $a_0^{\theta^F} = 31$, and $a_0^{\theta^\Upsilon} = 21$, together with Gamma hyperpriors on the corresponding spike rate parameters: $\beta_0^{\theta^\Lambda} \sim \mathcal{G}(c_0^{\theta^\Lambda}, d_0^{\theta^\Lambda})$ with $(c_0^{\theta^\Lambda}, d_0^{\theta^\Lambda}) = (1, 1)$, $\beta_0^{\theta^F} \sim \mathcal{G}(c_0^{\theta^F}, d_0^{\theta^F})$ with $(c_0^{\theta^F}, d_0^{\theta^F}) = (1, 1)$, and $\beta_0^{\theta^\Upsilon} \sim \mathcal{G}(c_0^{\theta^\Upsilon}, d_0^{\theta^\Upsilon})$ with $(c_0^{\theta^\Upsilon}, d_0^{\theta^\Upsilon}) = (1, 12)$. This choice yields clearly separated spike and slab regimes, with the spike components favouring substantially stronger shrinkage of inactive factors.

Overall, this specification encourages a decomposition with flexible common structure and moderately dense study-specific contributions, while allowing the data to determine the degree of sparsity at both global and local levels. The hyperparameter values used for estimatons in Sections 6 and 7 are presented in Table B.2.

B.3 Convergence

Convergence is monitored using blockwise relative Frobenius distances between successive iterates of the variational parameters. For each parameter block, the Frobenius norm of the difference between iterations is scaled by the norm of the previous iterate. For Bernoulli parameters (responsibilities), distances are evaluated on the logit scale. The algorithm is deemed converged when the maximum relative Frobenius distance across all parameter blocks falls below a fixed tolerance level for several consecutive iterations.

C Proof of Proposition 1

Proof. Take $(j, i) \in U_h$. By construction, $(j, i) \in B_h$, therefore both $\lambda_{jh} \neq 0$ and $f_{hi} \neq 0$, so factor h contributes nontrivially to the data entry (j, i). Since $(j, i) \in U_h$, we have $(j, i) \notin B_l$ for all $l \neq h$ and $(j, i) \notin B_r^s$ for all study-specific factors r and studies s. Therefore no other common or study-specific factor is active at (j, i). It follows that the signal at entry (j, i) is generated exclusively by factor h. Consequently, the contribution of factor h at (j, i) cannot be reproduced or absorbed by any other common or study-specific factor, since none of these components has support at (j, i). Therefore no observationally equivalent decomposition can reallocate the contribution of factor h to the remaining factors. Hence factor h is identified with respect to information switching. □

D Postprocessing

D.1 Post hoc identification procedure

In this section, we describe the procedure used to check identifiability of latent factors based on their unique support, as outlined in Section 4.4 of the main text.

Table B.2: *Hyperparameter specifications for the MSFRB model. The table is divided according to the different components of the model and prior hierarchy. In the prior for ξ_{js}^2, R_{js} denotes the range of the date in dimension i in study s.*

Variable(s)	Hyperparameter(s)	Hyperparameter value(s)
	Common factor loadings	
$\theta_h^\Lambda \mid \gamma_h^\Lambda = 1$	$a^{\theta^\Lambda},\ c^{\theta^\Lambda},\ d^{\theta^\Lambda}$	$3,\ 2,\ 1$
$\theta_h^\Lambda \mid \gamma_h^\Lambda = 0$	$a_0^{\theta^\Lambda},\ c_0^{\theta^\Lambda},\ d_0^{\theta^\Lambda}$	$31,\ 1,\ 1$
α^{τ^Λ}	$a^{\tau^\Lambda},\ b^{\tau^\Lambda}$	$10,\ 2$
ϕ_{jh}^Λ	$a^{\phi^\Lambda},\ c^{\phi^\Lambda},\ d^{\phi^\Lambda}$	$7,\ 0.2,\ 0.5$
κ^Λ	$a^{\kappa^\Lambda},\ b^{\kappa^\Lambda}$	$2,\ 1$
	Common factor scores	
$\theta_h^F \mid \gamma_h^F = 1$	$a^{\theta^F},\ c^{\theta^F},\ d^{\theta^F}$	$3,\ 2,\ 1$
$\theta_h^F \mid \gamma_h^F = 0$	$a_0^{\theta^F},\ c_0^{\theta^F},\ d_0^{\theta^F}$	$31,\ 1,\ 1$
α^{τ^F}	$a^{\tau^F},\ b^{\tau^F}$	$10,\ 2$
ϕ_{his}^F	$a^{\phi^F},\ c^{\phi^F},\ d^{\phi^F}$	$7,\ 0.2,\ 0.5$
κ^F	$a^{\kappa^F},\ b^{\kappa^F}$	$2,\ 1$
	Study-specific factor loadings	
$\theta_{hs}^\Upsilon \mid \gamma_{hs}^\Upsilon = 1$	$a^{\theta^\Upsilon},\ c^{\theta^\Upsilon},\ d^{\theta^\Upsilon}$	$2,\ 2,\ 1$
$\theta_{hs}^\Upsilon \mid \gamma_{hs}^\Upsilon = 0$	$a_0^{\theta^\Upsilon},\ c_0^{\theta^\Upsilon},\ d_0^{\theta^\Upsilon}$	$21,\ 1,\ 12$
α^{τ^Υ}	$a^{\tau^\Upsilon},\ b^{\tau^\Upsilon}$	$1,\ 4$
ϕ_{jhs}^Υ	$a^{\phi^\Upsilon},\ c^{\phi^\Upsilon},\ d^{\phi^\Upsilon}$	$3,\ 0.5,\ 1$
	Covariate coefficients	
β_{jl}^X	q_{jl}	1
	Idiosyncratic variances	
ξ_{js}^2	$a_\xi,\ a^{\beta^\xi},\ b_{js}^{\beta^\xi}$	$2,\ 1,\ 100/R_{js}^2$

Step 1: *Compute activity energy for each factor*

In variational inference, the ESP prior does not yield a sharp separation between active and inactive factors, as the corresponding $0/1$ indicators are replaced by continuous

responsibilities. While these quantities encode relative shrinkage, they are not guaranteed to concentrate at the extremes and thus may not provide a stable criterion for factor selection. This motivates the use of an alternative criterion for identifying active factors.

We identify active factors using posterior summaries of factor loadings and factor scores. We define activity energy for each factor as the sum of posterior second moments, combining posterior means and variances. This quantity corresponds to the expected squared magnitude of the factor under the variational posterior and is closely related to measures of factor strength used in sparse factor models (e.g., Hochreiter et al. (2010), Bhattacharya and Dunson (2011)).

For common factors, we compute activity energy separately for factor loadings and factor scores. The factor loading energy for factor h is defined as

$$A_h^{(\lambda)} = \sum_{j=1}^{p} \left(\left[\mu_{jh}^{\lambda}\right]^2 + \mathbf{\Sigma}_{j[h,h]}^{\lambda} \right),$$

and the factor score energy as

$$A_h^{(f)} = \sum_{s=1}^{S} \sum_{i=1}^{n_s} \left(\left[\mu_{his}^{f}\right]^2 + \mathbf{\Sigma}_{ns[h,h]}^{f} \right).$$

Activity energy for study-specific factors is computed separately for each study s and factor r in the following way:

$$A_r^{(\lambda_s)} = \sum_{j=1}^{p} \left(\left[\mu_{irs}^{\lambda}\right]^2 + \mathbf{\Sigma}_{js[r,r]}^{\lambda} \right),$$

$$A_r^{(\eta_s)} = \frac{1}{n_s} \sum_{i=1}^{n_s} \left([\mu_{ris}^{\eta}]^2 + \mathbf{\Sigma}_{is[r,r]}^{\eta} \right).$$

The averaging by n_s in $A_r^{(\eta_s)}$ ensures that the study-specific energy is not driven purely by study size. We then combine both metrics into an overall energy measure for study-specific factor r in study s: $A_r^{(s)} = A_r^{(\lambda_s)} A_r^{(\eta_s)}$.

Step 2: *Select active factors by relative thresholding*

As the next step, we identify active factors by comparing the activity energy of each factor to its largest value in the corresponding set.

For the common component, define the active factor loading columns and the active factor score rows as

$$\mathcal{H}_{\lambda}^{+} = \left\{ h : A_h^{(\lambda)} > \delta \max_l A_l^{(\lambda)} \right\} \quad \text{and} \quad \mathcal{H}_{f}^{+} = \left\{ h : A_h^{(f)} > \delta \max_l A_l^{(f)} \right\},$$

where $h, l \in \{1, \ldots, H\}$.

The set of active common factors is then defined as $\mathcal{H}^{+} = \mathcal{H}_{\lambda}^{+} \cap \mathcal{H}_{f}^{+}$. Thus, a common factor is retained as active only if it is active both in its loading column and in its factor score row.

For study-specific factors in study s we define the set of active factors as

$$\mathcal{H}_{s}^{+} = \left\{ r : A_r^{(s)} > \delta \max_l A_l^{(\lambda_s)} \right\},$$

where $r, l \in \{1, \ldots, H_s\}$. In our simulation studies and application to real data we used $\delta = 0.1$.

Step 3: *Estimate supports of active factors*

After selecting the active factors, we recover their supports by thresholding posterior means in the following way. For each active factor $h \in \mathcal{H}^+$, define its feature support as

$$G_h = \left\{ j \in 1, \ldots, p : |\mu^{\lambda}_{jh}| > c_\lambda \max_{1 \leq g \leq p} |\mu^{\lambda}_{gh}| \right\},$$

and its sample support as

$$T_h = \left\{ i \in 1, \ldots, n : |\mu^{f}_{hi}| > c_f \max_{1 \leq m \leq n} |\mu^{f}_{hm}| \right\},$$

where $n = n_1 + \ldots + n_S$.

Thus, G_h contains the features with sufficiently large posterior mean loadings on factor h while T_h contains the samples for which factor h has sufficiently large posterior mean score. Then, for each active factor $h \in \mathcal{H}^+$ we define its total support as the Cartesian product of its feature and sample supports $B_h = G_h \times T_h$. Thus, B_h represents the set of entries of the data matrix where both the loadings and scores of factor h are non-negligible.

Then, for each active study-specific factor $r \in \mathcal{H}^+_s$, define its feature support as

$$G^s_r = \left\{ j \in 1, \ldots, p : |\mu^{\lambda_s}_{jr}| > c_{\lambda_s} \max_{1 \leq g \leq p} |\mu^{\lambda_s}_{gr}| \right\}.$$

The sample support of the study-specific factors is fixed by the model structure and is given by $T^s_r = \mathcal{I}_s$, where $\mathcal{I}_s$ denotes the set of indices corresponding to the samples belonging to study s. This reflects the fact that the study-specific factors are, by modeling choice, active across all samples within their corresponding study. Then, for each active study-specific factor $r \in \mathcal{H}^+_s$ in study s we define its total support as $B^s_r = G^s_r \times \mathcal{I}_s$.

In our simulation studies and application to real data we used $c_\lambda = c_f = c_{\lambda_s} = 0.1$.

Step 4: *Calculate unique support for common factors*

For each active common factor h, the unique support is calculated by comparing its total support with the supports of all other active factors. Computationally, the total supports B_h and B^s_r can be represented as a binary mask of the data matrix, with entries equal to 1 if the factor is active at that location and 0 otherwise. With this in view, the unique support of common factor h can be computed as follows:

1. Create a binary mask for each common factor M_h such that $M_h(j, i) = 1$ if $(j, i) \in B_h$, otherwise $M_h(j, i) = 0$.

2. Create a binary mask for each study-specific factor in each study s M^s_r such that $M^s_r(j, i) = 1$ if $(j, i) \in B^s_r$, otherwise $M^s_r(j, i) = 0$, where $i = 1, \ldots, n_s$.

3. Construct the combined mask of all other factors by summing the binary masks of all other active common factors $l \neq h$ and all active study-specific factors in all studies $s = 1, \ldots, S$ as $M_{-h}(j, i) = \mathbf{1}\left\{ \sum_{l \neq h} M_l(j, i) + \sum_{s=1}^{S} \sum_{r \in \mathcal{H}^+_s} M^s_r(j, i) > 0 \right\}$.

4. Compute the unique support mask of common factor h as $U_h = M_h \odot (1 - M_{-h})$, where $\odot$ denotes element-wise multiplication.

5. Calculate the size of the unique support as $|U_h| = \sum_{j,i} U_h(j,i)$.

A common factor h is considered identified with respect to information switching if $|U_h| > 0$, that is, if its unique support is non-empty. We restrict the construction of unique supports to common factors, as study-specific factors are usually less sparse and, by design, active across all samples within a study, making the uniqueness criterion less likely to be satisfied. Moreover, our primary interest lies in the identification of common factors.

E Performance metrics

We use several complementary metrics to evaluate parameter, latent structure and biclustering recovery in the simulation studies.

Relative Frobenius distance. The relative Frobenius distance quantifies the reconstruction error relative to the overall magnitude of the true signal. For a true matrix A and its estimate B, it is defined as

$$d_F(A, B) = \frac{\|A - B\|_F}{\|A\|_F},$$

where

$$\|A - B\|_F = \sqrt{\sum_{j=1}^{p} \sum_{i=1}^{n} (A_{ji} - B_{ji})^2}.$$

Smaller values indicate better recovery, with a value of zero corresponding to perfect reconstruction.

RV coefficient. The RV coefficient (Abdi, 2007) measures the structural similarity between two matrices. For matrices A and B, it is defined as

$$\mathrm{RV}(A, B) = \frac{\mathrm{tr}\left(AA^\top BB^\top\right)}{\sqrt{\mathrm{tr}\left(AA^\top AA^\top\right)\mathrm{tr}\left(BB^\top BB^\top\right)}}.$$

The RV coefficient takes values in $[0, 1]$, with values close to zero indicating low structural similarity and values close to one indicating high structural similarity.

Bicluster similarity. Following Moran et al. (2021), similarity between biclusters is quantified using the Jaccard similarity coefficient. In the multi-study setting, each common bicluster is represented by the Cartesian product of its feature support and the pooled sample support across all studies. Specifically, for a bicluster $B_h = (G_h, T_h)$, where G_h denotes the set of active features and T_h the set of active samples, its support is

$$B_h = G_h \times T_h.$$

For two biclusters B_1 and B_2, their Jaccard similarity is

$$J(B_1, B_2) = \frac{|B_1 \cap B_2|}{|B_1 \cup B_2|}.$$

The Jaccard similarity takes values in $[0, 1]$, with one indicating identical bicluster supports.

Let $\mathcal{M}_t = \{B_1, \ldots, B_{K_t}\}$ denote the set of true biclusters and $\mathcal{M}_e = \{\widehat{B}_1, \ldots, \widehat{B}_{K_e}\}$ the set of estimated biclusters.

Relevance. The relevance score (Prelić et al., 2006) measures the average quality of the estimated biclusters and is defined as

$$\text{Relevance} = \frac{1}{K_e} \sum_{\widehat{B} \in \mathcal{M}_e} \max_{B \in \mathcal{M}_t} J(\widehat{B}, B).$$

Each estimated bicluster is compared with all true biclusters, and only the best matching true bicluster is retained. The resulting similarities are then averaged across estimated biclusters. High relevance therefore indicates that the estimated biclusters contain few false positives.

Recovery. The recovery score evaluates how well the true biclusters are recovered and is defined as

$$\text{Recovery} = \frac{1}{K_t} \sum_{B \in \mathcal{M}_t} \max_{\widehat{B} \in \mathcal{M}_e} J(B, \widehat{B}).$$

For each true bicluster, only the best matching estimated bicluster contributes to the score. Recovery therefore measures the proportion of the true underlying bicluster structure that has been successfully identified and is primarily affected by missed biclusters or incomplete recovery.

Consensus. While relevance and recovery do not penalise duplicated estimated biclusters, the consensus score of Hochreiter et al. (2010) explicitly accounts for overestimation of the number of biclusters. It is computed by first obtaining an optimal one-to-one matching between the true and estimated biclusters using the Hungarian assignment algorithm and then averaging the corresponding Jaccard similarities,

$$\text{Consensus} = \frac{1}{\max(K_t, K_e)} \sum_{(B, \widehat{B}) \in \mathcal{A}} J(B, \widehat{B}),$$

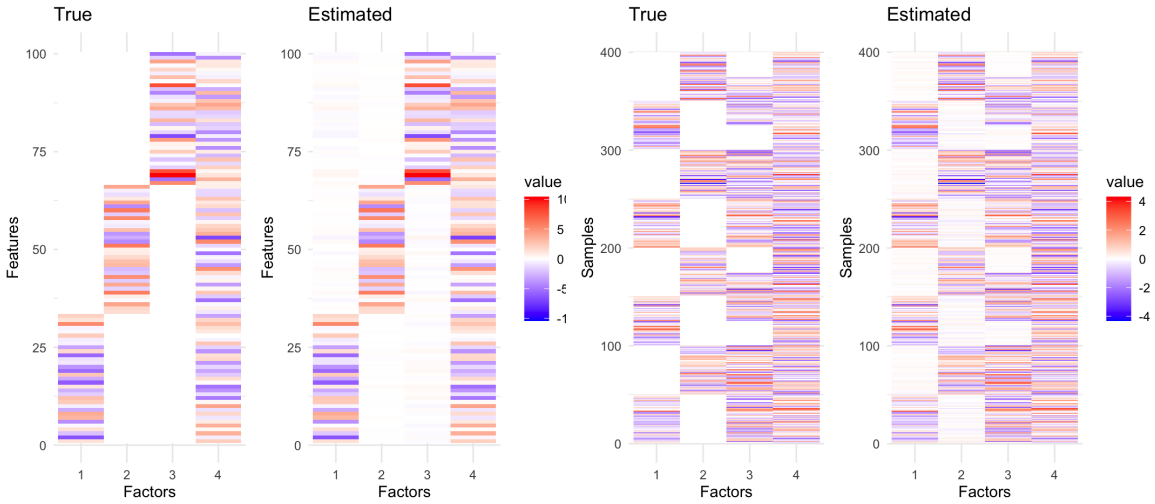
where $\mathcal{A}$ denotes the optimal assignment. Consequently, duplicated or spurious biclusters reduce the consensus score even if they have high individual Jaccard similarities.

The relevance, recovery and consensus measures all take values in $[0, 1]$, with larger values indicating better recovery of the true bicluster structure.

F Simulation Study 2: illustration of model parameter recovery

To provide a visual illustration of the ability of the MSFRB model to recover the individual components of the model as shown in Section 6.2, we generated an additional dataset using the same data-generating mechanism as in Section 6.2, but with $p = 100$ features and $n_s = 100$ samples per study, while keeping the number of studies fixed at $S = 4$. All other aspects of the simulation design were kept unchanged.

Figures F.1–F.3 compare the simulated true parameter values with their posterior mean estimates for one randomly selected replicate dataset. Since factor loadings and factor scores are identified only up to sign switching and column permutations, the estimated matrices were aligned with the corresponding true matrices before plotting. For visual clarity, only columns corresponding to active estimated factors are displayed.



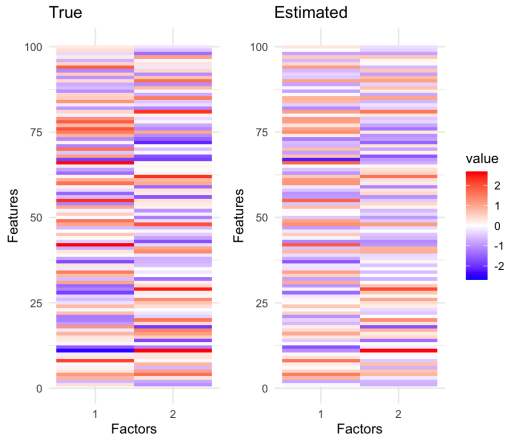
(a) *Common factor loadings* (b) *Common factor scores*

Figure F.1: *True and estimated common factor loadings and factor scores for one randomly selected replicate dataset.*

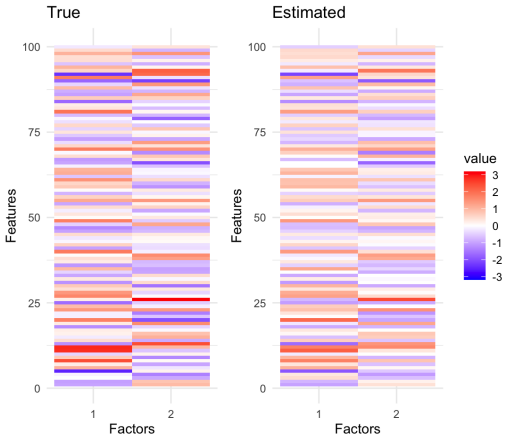
Figure F.1 shows that MSFRB accurately recovers both the sparse and dense common components, including their underlying support structure.

As shown in Figure F.2, the study-specific factor loadings are also recovered accurately. These factors capture variation unique to individual studies, which in transcriptomic applications may arise from differences in experimental platforms, laboratory protocols, batch effects, and other study-specific sources of variation.

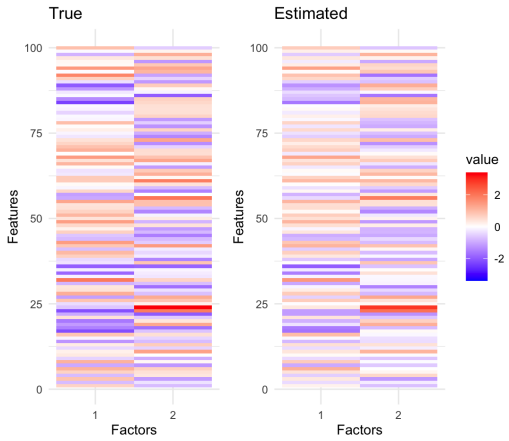
Finally, Figure F.3 compares the simulated and estimated covariate coefficients. The estimated coefficient matrix includes an intercept term, which is inferred jointly with the remaining model parameters. The intercept is necessary because, contrary to the usual practice in factor models, MSFRB is fitted directly to the original data without prior centering or scaling. This avoids removing variation that may be associated with the observed covariates and allows covariate effects to be estimated jointly with the latent structure. The figure shows that the covariate effects are also recovered with a high



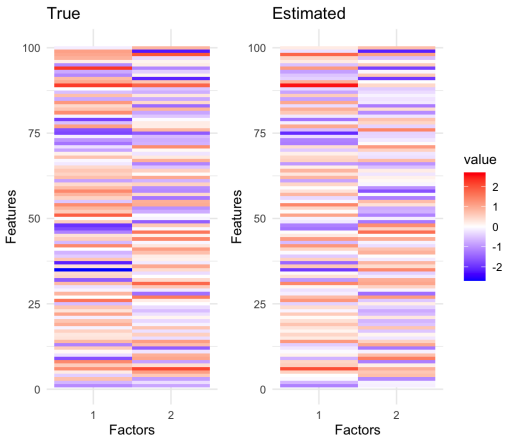
(a) *Factor loadings: study 1*



(b) *Factor loadings: study 2*



(c) *Factor loadings: study 3*



(d) *Factor loadings: study 4*

Figure F.2: *True and estimated study-specific factor loadings for one randomly selected replicate dataset.*

degree of accuracy.

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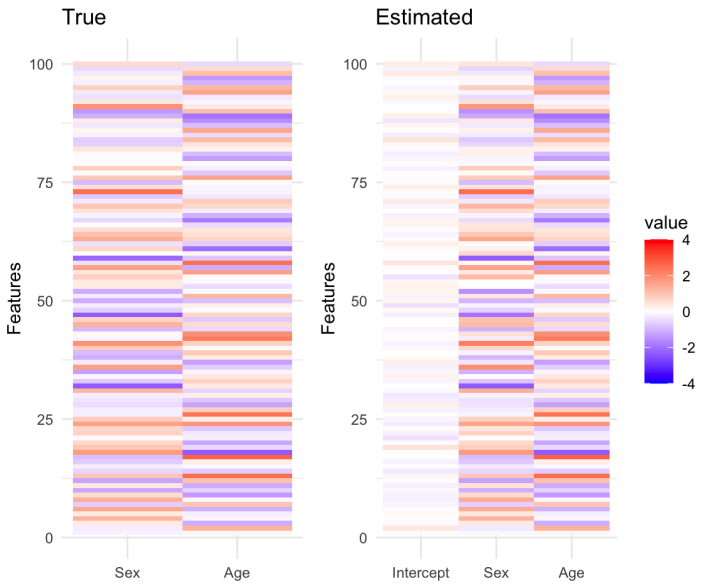


Figure F.3: *True and estimated covariate coefficients for one randomly selected replicate dataset.*

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